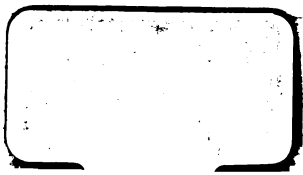


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OF BALLIOL COLLEGE.





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THE

COMPLETE
DISTILLER.

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Fig. 9.

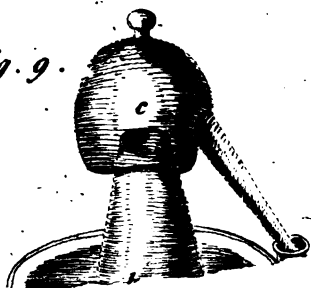
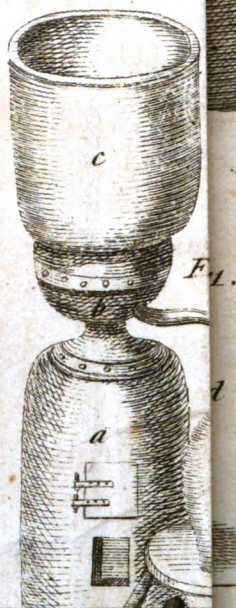
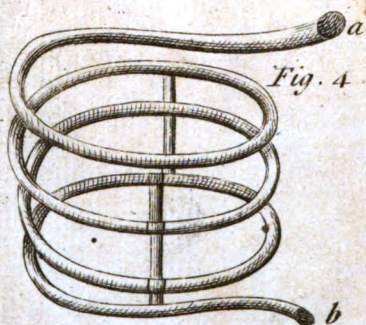
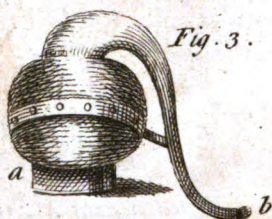
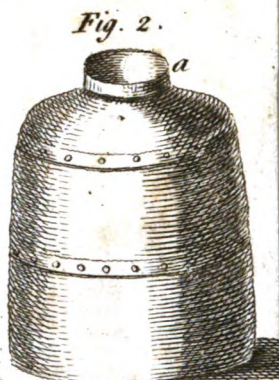


Fig. 8.



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THE COMPLETE DISTILLER:

CONTAINING,

- | | |
|--|---|
| I. The Method of performing the various Processes of Distillation, with Descriptions of the several Instruments: The whole Doctrine of Fermentation: The manner of drawing Spirits from Malt, Raisins, Molasses, Sugar, &c. and of rectifying them: With Instructions for imitating to the greatest Perfection | both the Colour and Flavour of <i>French</i> Brandies.
II. The manner of distilling all Kinds of Simple Waters from Plants, Flowers, &c.
III. The Method of making all the compound Waters and rich Cordials so largely imported from <i>France</i> and <i>Italy</i> ; as likewise all those now made in <i>Great Britain</i> . |
|--|---|

To which are added,

Accurate Descriptions of the several Drugs, Plants, Flowers, Fruits, &c. used by Distillers, and Instructions for chusing the best of each Kind.

The Whole delivered in the plainest manner, for the Use both of *Distillers* and *Private Families*.

By *A. COOPER*, DISTILLER.

L O N D O N :

Printed for P. VAILLANT in the *Strand*; and R. GRIFFITHS in *Pater-Noster-Row*.

M.DCC.LVII.





THE P R E F A C E.

IT is now some Years since I first formed a Design of compiling a complete System of Distillation; and accordingly read most of the Treatises on that Subject, and extracted from each what I thought necessary for my Purpose, proposing to supply the Defects from my own Experience. It is, however, more than probable, that this Design had never been executed, had not a French Treatise of Distillation* fell
A 3 into

* This Treatise is intitl'd *Traité Raisonné de la Distillation; ou La Distillation réduite en Principes: Avec un Traité des Odeurs.* Par M. DE' JEAN, Distillateur. Printed at Paris, in the Year, M.DCC.LIII.

P R E F A C E.

into my Hands ; but finding in that Book many useful Observations, and a great Number of Recipes for making various Sorts of compound Waters and Cordials, I determined to finish the Work I had begun, being now enabled to render it much more useful than it was possible for me otherwise to have done. What I have translated from this Author, will, I dare say, be kindly received by our Distillers, as the manner of making many of the foreign Compound Waters, &c. has never before been published in the English Language. And I flatter myself, if the several Hints interspersed through this Treatise are carefully adverted to, Distillation may be carried to a much greater Degree of Perfection than it is at present ; and the celebrated Compound Waters and Cordials of the French and Italians, imported at so great an Expence, and such Detriment to the Trade of this Nation,

P R E F A C E.

*Nation, may be made in England,
equal to those manufactured a-
broad.*

*My principal Intention being to
render this Treatise useful to all,
I have endeavoured to deliver every
thing in the plainest and most intel-
ligible Manner. Beauty of Stile
is not, indeed, to be expected in a
Work of this Nature; and there-
fore if Perspicuity be not wanting,
I presume the Reader will forgive
me, if he meets with some Passages
that might have been delivered in a
more elegant Manner. I have al-
so, for the same Reason, avoided, as
much as possible, Terms of Art, and
given all the Recipes in Words at
length.*

*Distillation, tho' long practised,
has not been carried to the Degree
of Perfection that might reason-
ably have been expected. Nor will*

P R E F A C E.

this appear surprizing, if it be considered, that the Generality of Distillers proceed in the same beaten Track, without hardly suspecting their Art capable of Improvements; or giving themselves any Trouble to enquire into the Rationale of the several Processes they daily perform. They imagine, that the Theory of Distillation is very abstruse, and above the Reach of common Capacities; or, at least, that it requires a long and very assiduous Study to comprehend it; and, therefore, content themselves with repeating the Processes, without the least Variation. This Opinion, however ridiculous it may appear to those not acquainted with the present Practice of Distillers, has, I am satisfied, been the principal Cause why Distillation has not been carried to the Height it would otherwise have been. I have therefore endeavoured in the following Treatise to destroy this idle Opinion, and shew the

P R E F A C E.

the Distiller how he may proceed on rational Principles, and direct his Enquiries in such a manner as cannot fail of leading him to such Discoveries in his Profession, as will be attended with Advantage both to himself and his Country.

But it is not to those only who make Distillation their Profession, that I have laboured to render this Treatise useful; I have also endeavoured to extend its Utility to those who distil Simple and Compound Waters for their own Use, or to distribute to their indigent Neighbours. And for this Reason I have adapted most of the Recipes to small Quantities, and briefly enumerated the Virtues and Uses of each Composition.

The short Descriptions of the most capital Ingredients, and the Directions for chusing the best of each Kind,

P R E F A C E.

Kind, I flatter myself will not be considered as improper : Because the Goodness of every Composition, must, in a great Measure, depend on the Goodness of the Ingredients.

As Typographical Errors are almost impossible to be avoided, the Reader will, I hope, pardon any he may chance to meet with in the ensuing Treatise ; and the rather as, I dare say, there are none but what he may himself very easily correct.



T H B



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T H E



A

Complete System

O F

DISTILLATION.

DISTILLATION is the Art of separating, or drawing off the spirituous, aqueous, and oleaginous Parts of a mixt Body from the grosser, and more terrestrial Parts, by means of Fire, and condensing them again by Cold.

We shall therefore divide this Treatise into three Parts ; in the first, we shall explain the Method of distilling Spirits from various Substances ; in the second, the manner of drawing simple Waters ; and in the third, the best Methods of making cordials or compound Waters.

B

PART

P A R T I.

Of the Distillation of Spirits.

BY the Distillation of Spirits is to be understood the Art by which all inflammable Spirits, Brandies, Rums, Arracks, and the like, are procured from vegetable Substances, by the means of a previous Fermentation, and a subsequent Treatment of the fermented Liquor by the Alembic, or hot Still, with its proper Worm and Refrigeratory.

But as it is impossible to extract vinous Spirits from any vegetable Subject without Fermentation, and previous to this Brewing is often necessary, it will be requisite first to consider these Operations.

C H A P. I.

Of Brewing, in order to the Production of inflammable Spirits.

BY Brewing, we mean the extracting a Tincture from some vegetable Substance, or dissolving it in hot Water, by which means it becomes proper for a vinous Fermentation.

A

of Distillation.

A Solution, or fermentable Tincture of this kind, may be procured, with proper Management, from any vegetable Substance, but the more readily and totally it dissolves in the Fluid, the better it is fitted for Fermentation, and the larger its Produce of Spirits. All inspissated vegetable Juices therefore, as Sugar, Honey, Treacle, Manna, &c. are very proper for this Use, as they totally dissolve in Water, forming a clear and uniform Solution ; but Malt, for its Cheapness, is generally preferred in *England*, though it but imperfectly dissolves in hot Water. The worst sort is commonly chosen for this Purpose ; and the Tincture, without the Addition of Hops, or Trouble of boiling it, is directly cooled and fermented.

But in order to brew with Malt to the greatest Advantage, the three following Particulars should be carefully attended to :
1. the Subject should be well prepared ; that is, it should be justly malted, and well ground : For if it be too little malted, it will prove hard and flinty ; and consequently, only a small Part of it dissolve in the Water : And, on the other hand, if too much malted, a great Part of the finer Particles, or fermentable Matter, will be lost in the Operation. With regard to grinding, the Malt
B 2 should

should be reduced to a kind of coarse Meal; for Experience has shewn, that by this means, the whole Substance of the Malt may, through the whole Process, continue mixed with the Tincture, and be distilled with it; whereby a larger Quantity of Spirit will be obtained, and also great Part of the Trouble, Time and Expence in Brewing saved. This Secret depends upon thoroughly mixing, or briskly agitating the Meal, first in cold Water, and then in hot; and repeating this Agitation after the Fermentation is finished: When the thick turbid Wash must be immediately committed to the Still. And thus the two Operations of Brewing and Fermenting may very commodiously be reduced to one, to the no small Profit and Advantage of the Distiller.

The second Particular to be attended to, is, that the Water be good, and properly applied. Rain Water is the best adapted to Brewing; for it not only extracts the Tincture of the Malt better than any other; but also abounds in fermentable Parts, whereby the Operation is quickened, and the Yield of the Spirit increased. The next to that of Rain, is the Water of Rivers and Lakes, particularly such as wash any large Tract of a fertile Country, or receive the Sullage of populous Towns. But
whatever

whatever Water is used, it must stand in a hot State upon the prepared Malt, especially if a clear Tincture be desired; but the greatest Care must be taken to prevent the Malt from running into Lumps or Clods; and, indeed, the best Way to prevent this, is to put a small Quantity of cold Water to the Malt first, and mix them well together; after which the remaining Quantity of Water may be added in a State of boiling, without the least Danger of coagulating the Malt, or what the Distillers call, making a Padding.

It has been found by Experience, that a certain Degree of Heat is necessary to extract the whole Virtue of the Malt: This Degree may, by the above Method, be determined to the greatest Exactness, as the Heat of boiling Water may at once be lessened to any assigned Degree of Warmth, by a proper Addition of cold Water; due Regard being had to the Season of the Year, and the Temperature of the Air. This Improvement, with that mentioned above, of reducing the two Operations of Brewing and Fermentation to one, will be attended with considerable Advantage.

With regard to the proper Quantity of Water, it must be observed, that if too little be used, a viscid clammy Mixture will be produced,

produced, little disposed to ferment, nor capable of extracting all the soluble Parts of the Malt. On the other hand, too much Water renders the Tincture thin and aqueous, and by that means increases the Trouble and Expence in all Parts of the Operation. A due Medium, therefore, should be chosen; and Experience has shewn, that a Wash about the Goodness of that designed by the *London* Brewers for Ten Shilling Beer, will best answer the Distiller's Purpose. When a proper Quantity of Water is mixed with the Malt, the whole Mass must be well agitated, that all the soluble Parts of the Malt may often come in contact with the aqueous Fluid, which being well saturated after standing a proper time, must be drawn off, fresh Water poured on, and the Agitations repeated, till at last the whole Virtue, or saccharine Sweetness of the Malt is extracted, and only a fixed husky Matter remains, incapable of being dissolved by either hot or cold Water.

The third requisite Particular is, that some certain Additions be used, or Alterations made according to the Season of the Year, or the Intention of the Operator. The Season of the Year is very necessary to be considered. In the Summer, the Water applied to the Malt must be colder than in the Winter; and in hot sultry Weather, the

the Tincture must be suddenly cooled, otherwise it will turn eager ; and, in order to check the too great Tendency it has to Fermentation, when the Air is hot, it will be necessary to add a proper Quantity of unmalted Meal, which being much less disposed to Fermentation than Malt, will greatly moderate its Impetuosity, and render the Operation suitable to the Production of Spirits, which, by a too violent Fermentation, would, in a great Measure, be dissipated and lost.

CHAP. II.

Of Fermentation.

THE Tincture, or, as the Distillers call it, the Wash, being prepared, as in the foregoing Chapter, it is next to be fermented ; for, without this Operation, no vinous Spirit can be produced.

By Fermentation is meant that intestine Motion performed by the instrumental Efficacy of Water, whereby the Salt, Oil and Earth of a fermentable Subject, are separated, attenuated, transposed, and again collected, and recomposed in a particular Manner.

The Doctrine of Fermentation, is of the greatest Use, and should be well understood by every Distiller, as it is the very Basis of the Art; and, perhaps, if more attended to, a much purer Spirit, as well as a greater Quantity of it, might be procured from the same Materials than at present. We shall therefore lay down a concise Theory of Fermentation, before we proceed to deliver the Practice.

Every fermentable Subject is composed of Salt, Oil, and a subtile Earth; but these Particles are so small, that, when asunder, they are imperceptable to the Senses; and, therefore, when mixed with an aqueous Fluid, they leave it transparent; neither have fermentable Bodies any Taste, except that of Sweetness.

These Particles are each composed of Salt, Oil and Earth, intimately mixed in an actual Cohesion, Connexion, and Union; and, therefore, when any one of those Principles too much abounds in any Subject, so that an intimate Union is prevented, the whole Efficacy of the Fermentation is either stopped or impaired, or at least limited to one certain Species.

This

This equal Connexion of Salt, Oil and Earth into a single compound Particle, forms a Corpuscle soluble in Water ; or, to speak more philosophically, this compound Corpuscle is, by means of its saline Particles, connected with the aqueous Corpuscles, and moved up and down with them. But where these Corpuscles are not thus connected with the Water, a Number of them join together, and form either a gross, or a loose, chaffy, and spongy Matter.

When these compound Particles are diluted with a small Quantity of an aqueous Fluid, they feel slippery, clammy, and unctuous to the Touch, and affect the Taste with a kind of rosy Sweetness. And when a proper Quantity of the Fluid is added, a Commotion is presently excited, and afterwards a subtile Separation,

This Commotion and Separation first begins in the whole Substance ; for before the Addition of Water, the Subject may remain in dry, solid, and large Pieces, as in Malt, Sugar, &c. which being reduced to Powder, each Grain thereof is an Aggregate of many smaller compound Corpuscles ; these being put into Water, dissolve, and separately float therein, till at length, they become

10 *A Complete System*

come so small as to be invifible, and only thicken the Confiftence of the Liquor.

These Corpuscles being thus separated from one another, there next enfues a Separation of their component Particles ; that is, the Salt, the Oil, and the Earth, are divided by the Interposition of the aqueous Particles.

The first Commotion is no more than a bare Solution ; for the faline Particles being easily diffolvable in Water, they are immediately laid hold of by the aqueous Particles, and carried about with them. But the fucceeding Separation, or fermentative Motion, is a very different thing ; for by this the faline Particles are divided from thofe of Oil and Earth, partly by the Impulfe of the others in their Motion, and partly by the Force of the aqueous Particles, which are now continually meeting and dafhing againft them.

This Motion is performed by the Water, as a Fluid, or Aggragate of an infinite Number of Particles, in actual and perpetual Motion ; their Smallnefs being proportionable to that of the fermenting Corpuscles, and their Motion, or constant Susceptibility of Motion, by Warmth, and the Motion of the Air, difpofing them to move
other

other subtile moveable Corpuscles also. The certain Agreement of Figure, or Size between the aqueous Particles, and those of the Salt in the fermentable Subject, tends greatly to increase this Commotion ; for, by this means, they are readily and very closely connected together ; and therefore move almost like one and the same compound Corpuscle ; whilst the Water is not at all disposed to cohere immediately with either the Oil or Earth. And thus an unequal Concussion is excited in the compound Corpuscles of the fermentable Subject ; which Concussion at length strikes out the saline Particle, loosens the others, and finally produces a Separation of the original Connexion of the Subject.

An aqueous Fluid, therefore, is the true, and indeed the only, Instrument for procuring a fermentable Motion in these compound Corpuscles of the Subject : For were an oily Fluid poured upon any fermentable Subject, no vinous Fermentation would ensue ; as the Oil could neither give a sufficient Impulse on the compound Corpuscles, which are grosser than its own constituent Particles, nor divide the oily or saline Particles of the Subject from their Connexion with the others, which detain, and, as it were, envelope, or defend them from its Action.

The

The compound Corpufcles of the fermentable Subject being affected by the perpetual Motion of the Particles of the aqueous Fluid, a proper Degree of Motion is necessary, or that the Particles move with a proper Degree of Velocity, which principally depends on external Heat. A considerable Degree of Cold, indeed, will not absolutely prevent Fermentation, though it will greatly retard it; and a boiling Heat will prevent it still more. A tepid, or middle Degree of Heat between Freezing or Boiling, is therefore the most proper for promoting and quickening the Operation.

The Admission of Air, also, though not of absolute Necessity, yet greatly promotes and quickens the Action, as being a capital Instrument in putting in a proper Degree of Motion the oily Particles of the Subject. But whilst the Air thus contributes to hasten the Effect, it causes at the same time by its Activity some remarkable Alterations in the oily Particles; for it not only moves, but absolutely dissolves and displaces them from their original Connexions; and thus carries them off with itself from the whole Mass. And, therefore, though the Consideration of the Air does not so properly belong to Fermentation in the general, yet it does in particular; as having an accidental Power to

to alter every Species of this Operation :
Consequently its Agency ought to be well
understood, either to procure Alterations at
pleasure in the fermenting Mass, or to pre-
vent and correct impending Dangers.

The oily Particles thus separated and dis-
solved by the Air, are also elastic, though
they probably derive that Property from
their Intercourse with the Air itself, and
their being rendered extremely minute.

When, therefore, an aqueous Fluid is
added to a fermentable Subject exposed to
a temperate Heat, a fermentative Struggle
immediately arises, the saline Part of the
compound Particles being dissolved by the
continual intestine Motion of the Water,
and carried up and down with it in all Di-
rections, amidst an infinite Number of other
Particles, as well fermentable, as aqueous
ones ; whence, by this Collision and At-
trition, the saline Particles are dissolved,
and separated from their Connexion with
the oily and earthy. And as the oily Par-
ticles are the most subtle and elastic, they
would, by this means, be thrown up to the
Surface of the Liquor, and carried off by the
Air, were they not closely connected with
the earthy ones, whose Gravity prevents
their Evaporation, and, by coming in con-
tact with others of the same kind, form
Aggregations,

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Aggregations, and sink down, with the oily Particles, to the Bottom. But before these can form a Bulk too large to be supported by the Water, many of the oily Particles are, by their frequent Collisions with the aqueous Fluid, separated from the earthy ones; and, by Degrees, more strongly connected again with the saline ones; whilst, on the other hand, the same saline Particles imbibe some of the earthy ones, which being left single, upon their Separation from the oily Particles, floated about separately in the Fluid.

And hence proceed the several different Consequences of Fermentation; viz. 1. From the Separation of the saline Particles of the fermentable Subject proceeds the tart, saline, or acid Taste of the Liquor; which is more sensible at first, before the Liquor is duly composed and settled, or the due Arrangement and Connection of the saline Particles with those of the oily and earthy Kinds, completed: After which the Liquor proves milder, softer, or less pungent. 2. From the oily Particles being set at liberty, proceeds the strong Smell of the Liquor, and the Head or shining Skin upon the Surface. 3. The earthy Particles collecting together in Clusters, cause the Fluid to appear turbid, and afterwards a visible earthy, or clay-like Matter to be precipitated: And some

some of the earthy Parts, in their Motion, arriving at the Head, or oily Skin on the Surface, cause it to thicken ; and afterwards taking it down along with it, thus constitute the Lees which abound in Oil. 4. From this new Struggle or Collision, which is productive both of Solution, and a new Connection in the saline and earthy Corpuscles, proceeds the Ebullition in Fermentation. And, lastly, by the same repeated Coalition of the oily with the aqueous and saline Particles, the inflammable Spirit is produced.

Having thus laid down a concise Theory of Fermentation, we shall now proceed to the Practice.

The Wash being brought to a tepid, or lukewarm State in the Backs, a proper Quantity of a good-conditioned Ferment is added ; but if the Ferment be solid, it should be previously broke into small Pieces, and gently thinned either with the Hand, Whisp, &c. in a little of the tepid Liquor. A complete and uniform Solution, however, should not be attempted, because that would greatly weaken the Power of the Ferment, or destroy its future Efficacy. The whole intended Quantity, therefore, being thus loosely mixed with a moderate Parcel of the Liquor, and kept in a tepid State, either by

by setting it near the Fire, or otherwise, and free from the too rude Commerce of the external Air ; more of the insensibly warm Liquor ought to be added, at proper Intervals, till, at length, the whole Quantity is properly set to working together. And, thus, by dividing the Business into Parts, it may much more speedily and effectually be performed, than by attempting it all at once.

The whole Quantity of Liquor being thus set to work, secured in a proper Degree of Warmth, and defended from a too free Intercourse of the external Air, Nature itself, as it were, finishes the Process, and renders the Liquor fit for the Still.

By Ferments, we mean any Substance, which, being added to any rightly disposed fermentable Liquor, will cause it to ferment much sooner and faster than it would of itself ; and, consequently, render the Operation shorter ; in contradiction to those abusively called so, which only correct some Fault in the Liquor, or give it some Flavour. Hence we see, that the principal Use of Ferments is to save Time, and make Dispatch in Business ; whilst they only occasionally, and, as it were, by Accident, give a Flavour, and increase the Quantity of Spirit. And, accordingly, any fermentable
Liquor,

Liquor, may, without the Addition of any Ferment, by a proper Management of Heat alone, be brought to ferment, and even more perfectly, though much slower, than with their Assistance.

These Ferments are, in general, the Flowers and Faces of all fermentable Liquors, generated and thrown to the Surface, or deposited at the Bottom, either during the Act of Fermentation, or after the Operation is finished.

Two of these are procurable in large Quantities, and at a small Expence; we mean, Beer-Yeast and Wine-Lees; a prudent and artificial Management, or Use of which, might render the Business of Distillation much more facile, certain and advantageous.

It has been esteemed very difficult, and a great Discouragement, in the Business of Distillation, to procure a sufficient Stock of these Materials, and preserve them at all times ready for use. The whole Secret consists in dexterously freeing the Matter from its superfluous Moisture; because in its fluid State, it is subject to a farther Fermentation, which is productive of Corruption; in which State it becomes intollerably fœtid and cadaverous.

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The Method of exposing it to the Air till it has required a proper Consistence, is subject to great Inconveniencies; and so peculiar and careful a Management necessary, that it rarely succeeds.

The best Way, therefore, is to press it very slowly and gradually, in a thick, close, and strong Canvas Bag, after the manner of Wine Lees, by the Tail press, till it becomes a kind of Cake; which, though soft, will easily snap, or break dry and brittle between the Fingers. Being reduced to that Consistence, and closely packed up in a tight Cask, it will remain a long Time uncorrupted, preserve its Fragrancy, and consequently, fit to be used for fermenting the finest Liquor.

The same Method is also practicable, and to the same Advantage, in the Flowers or Yeast of Wine; which may be thus commodiously imported from abroad: Or, if these cannot be procured, others of equal Efficacy may be procured from fresh Wine Lees, by barely mixing and stirring them into a proper warm Liquor; whence the lighter, or more volatile and active Parts of the Lees, will be thrown to the Surface, and may easily be taken off, and preserved by the above-mentioned Method, in any desired

desired Quantity. And hence, by a very easy Process, an inexhaustible Supply of the most useful Ferments may be readily and successively procured, so as to prevent for the future all Occasion of Complaint for want of them, in the Diffiller's Business.

Experience has demonstrated, that all Ferments abound much more in essential Oil, than the Liquor which produced them; and consequently they retain, in a very high Degree, the Smell and Flavour of the Subject. It is therefore requisite, before the Ferment is applied, to consider what Flavour is intended to be introduced, or what Species of Ferment is most proper for the Liquor.

The Alteration thus caused by Ferments is so considerable; as to render any neutral fermentable Liquor, of the same Flavour with that which yielded the Ferment. This Observation is of much greater Moment than will presently be conceived; for a new Scene is hereby opened, both in the Business of Diffillation, and others depending upon Fermentation. It must, however, be observed, that its Benefit does not extend to Malt, treated in the common Method; nor to any other Subject but what affords a Spirit tolerably pure and tasteless: For, otherwise, instead of producing a simple,

pure, and uniform Flavour, it causes a compound, mixed, and unnatural one. How far the fine Stiller may profit by it, well deserves his Attention ; and whether our native Cyder Spirit, Crab Spirit, &c. which have very little Flavour of their own, may not, by this Artifice, be brought nearly, if not intirely, into the State of some foreign Brandies, so highly esteemed, is recommended to Experience.

It is common with Distillers, in order to increase the Quantity of Spirit, give it a particular Flavour, or improve its Vinosity, to add several things to the Liquor, during the Time it is in a State of Fermentation; and these Additions may properly be reduced to Salts, Acids, Aromatics, and Oils.

All rich vegetable Juices, as Treacle, Honey, &c. which either want a natural Acid, have been deprived of it, or contain it in too small a Quantity, will be greatly improved by adding, at the Beginning of the Operation, a small Quantity of the vegetable or fine mineral Acids ; as Oil of Sulphur, Glauber's Spirit of Salt, Juice of Lemons, or an aqueous Solution of Tartar. These Additions will either give, or greatly improve the vinous Acidity of the Subject, but not increase the Quantity of the Spirit,
that

of Distillation.

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that Intention being performed by Aromatics and Oils.

All pungent Aromatics have a surprising Quality of increasing the Quantity of the Spirit, as well as in altering, or improving the Flavour; but their Use requires that the Fermentation should be performed in close Vessels. And if a large Quantity be intended to be added, Care must be taken not to do it all at once, lest the Oiliness of the Ingredients should check the Operation. But if the Flavour be the principal Intention, they should not be added till the Operation is nearly finished. After the same Manner a very considerable Quantity of any essential vegetable Oil may be converted into a surprisingly large Quantity of inflammable Spirit; but great Caution is here also necessary not to drop it too fast, or add too large a Quantity at a time, which would damp the Fermentation; it being the surest Method of checking, or totally stopping this Operation, at any Point of Time required. The best Method, therefore, of adding the Oil, so as to avoid all Inconveniencies, is to rub the Oil in a Mortar with Sugar, which the Chemists call making an *Oleosaccharum*, by which Means the Tenacity of the Oil will be destroyed, and the whole readily mix with the Liquor, and immediately ferment with it. The Distiller would do well

to consider these Observations attentively, as he may thence form an advantageous Method of increasing the Quantity of Spirits, and at the same Time greatly improve their Quality and Flavour.

But in order to put these Observations in practice, particular Regard must be had to the containing Vessel in which the Fermentation is performed, the Exclusion of the Air, and the Degree of the external Heat or Cold.

With regard to the containing Vessel ; its Purity, and the Provision for rendering it occasionally close, are chiefly to be considered. In cleansing it, no Soap, or other unctuous Body should be used, for fear of checking the Fermentation ; and, for the same Reason, all strong alkaline Lixiviums should be avoided. Lime-water, or a turbid Solution of quick Lime may be employed for this Purpose, without producing any ill Effect ; it will also be of great Service in destroying a prevailing acetous Salt, which is apt to generate in the Vessels when the warm Air has free Access to them ; and tends to pervert the Order of Fermentation, and, instead of a Wine or Wash, produce a Vinegar. Special Care must also be had, that no Remains of Yeast, or cadaverous Remains of former fermented Matters, hang about

about the Vessels, which would infect whatever should be afterwards put into them; and cannot, without the utmost Difficulty, be perfectly cured and sweetened.

The occasional Closeness of the Vessels may in the large way, be provided for by Covers properly adapted; and, in the small way, by Valves, placed in light Casks. These Valves will occasionally give the necessary Vent to preserve the Vessel, during the Height of the Fermentation; the Vessel otherwise remaining perfectly close, and impervious to the Air.

It is a Mistake of a very prejudicial Nature, in the Business of Fermentation, to suppose, that there is an absolute Necessity for a free Admission of the external Air. The express contrary is the Truth, and very great Advantages will be found by practising according to this Supposition. A constant Influx of the external Air, if it does not carry off some Part of the Spirit already generated, yet certainly catches up and dissipates the fine, subtile, or oleaginous and saline Particles, whereof the Spirit is made, and thus considerably lessens the Quantity. By a close Fermentation this Inconveniency is avoided; all Air, except that included in the Vessel, being excluded. The whole Secret consists in leaving a moderate Space for

the Air at the Top of the Vessel, unpossessed by the Liquor. When the Liquor is once fairly at work to bung it down close, and thus suffer it to finish the Fermentation, without opening or giving it any more Vent than that afforded it by a proper Valve placed in the Cask ; which, however is not of absolute Necessity, when the empty Space, or rather that possessed by the Air, is about one tenth of the Gage ; the artificial Air, generated in the Operation being then seldom sufficient to open a strong Valve, or at most not to endanger the Cask.

This Method may be practised to good Advantage by those whose Business is not very large ; but it requires too much Time to be used by the large Dealers, who are in a manner forced to admit the free Air, and thus sustain a considerable Loss in their Quantity of Spirit, that the Fermentation may be finished in the small Time allowed for that Purpose. It may, however, be said, that the silent, slow, and almost imperceptible vinous Fermentation, is universally the most perfect and advantageous.

During the whole Course of this Operation, the Vessel should be kept from all external Cold, or considerable Heat, in an equal, uniform, and moderate Temperature. In the Winter, a Stove Room, such as is common

mon in *Germany*, would be very convenient for this Purpose ; the Vessel being placed at a proper Distance from the Stove : But at other Seasons no particular Apparatus is necessary with us in *England*, if the Place allotted for the Business be but well defended from the Summer's Heat, and the ill Effects of cold bleak northern Winds.

The Operation is known to be perfected when the hissing, or small bubbling Noise can be no longer heard, upon applying the Ear to the Vessel ; and also by the Liquor itself appearing clear to Eye, and having a pungent Sharpness on the Tongue. And that it may fully obtain these Properties, and be well fitted to yield a pure and perfectly vinous Spirit by Distillation, it should be suffered to stand at rest in a somewhat cooler Place, if practicable, than that in which it was fermented ; till it has thoroughly deposited and cleansed itself of the gross Lee, and become perfectly transparent, vinous and fragrant ; in which State it should be committed to the Still, and the Spirit obtained will not only exceed that obtained in the common Way in Quantity, but also in Fragrance, Pungency, and Vinosity.

C H A P. III.

Of Distillation in general.

HAVING in the two preceding Chapters laid down the best Methods of Brewing and Fermentation, we shall now proceed to the Method of Distillation.

And in order to lead our Readers methodically through the Path which lies before them, we shall begin with explaining the Principles of Distillation; or, the Method of extracting the spirituous Parts of Bodies.

To extract the Spirits is to cause such an Action by Heat, as to cause them to ascend in Vapour from the Bodies which detain them.

If this Heat be natural to Bodies, so that the Separation be made without any adventitious Means, it is called Fermentation, which we have already explained.

If it be produced by Fire, or other heating Power, in which the Alembic is placed, it is called Digestion, or Distillation: Digestion, if the Heat only prepares the Materials for the Distillation of their Spirits; and Distillation, where the Action is of sufficient

sufficient Efficacy to cause them to ascend in Vapour, and distil.

This Heat is that which puts the insensible Parts of a Body, whatever it be, into Motion, divides them, and causes a Passage for the Spirits inclosed herein, by disengaging them from the Phlegm and the earthy Particles by which they are inclosed.

Distillation considered in this Light, is not unworthy the Attention and Countenance of the Learned. This Art is of infinite Extent; whatever the whole Earth produces, Flowers, Fruits, Seeds, Spices, aromatic and vulnerary Plants, odoriferous Drugs, &c. are its Objects, and come under its Cognizance; but we generally confine it to Liquids of Taste and Smell; and to the simple and spirituous Waters of aromatic and vulnerary Plants. With regard to its Utility, we shall omit saying any thing here, as we shall give sufficient Proofs of it in the Sequel.

CHAP. IV.

Of particular Distillation.

Distillation is generally divided into three Kinds; the first is called Distillation *per ascensum*, which is when the Fire, or other

other Heat, applied to the Alembic, containing the Materials, causes the Spirits to ascend. This is the most common, and indeed almost the only kind used by Distillers.

The second is called Distillation *per descensum*; which is, when the Fire being placed upon the Vessel precipitates, or causes the Spirit to descend. This Kind is hardly ever used by Distillers, but to obtain the Effence or Oil of Cloves.

The third is termed Distillation *per latus*, or oblique Distillation; but this being used only by the Chemists we shall say nothing farther of it here.

With regard to the different Methods of Distillation, occasioned by the different Vessels, or Materials made use of to excite Heat, improperly called Distillation; they are of various Kinds, and shall be explained as they occur in the Work.

There are various Kinds of Distillation, some of which arise from the different Constructions of Alembics; such are the Distillation by the common Alembic, with a Refrigeratory, the Glass Alembic, the serpentine Alembic, and the Retort: Others are produced from the Heat surrounding the

the Alembic ; such as the Distillation in *Balneum Mariæ*, the Vapor, the Sand, the Dung, and the Lime Baths.

These different Methods of Distilling, we shall explain in enumerating the Operations in which they are most proper ; and proceed to treat of the different Forms of Alembics and ther Constructions.

CHAP. V.

Of ALEMBICS, and their different Constructions.

THE Alembic is a Vessel usually of Copper tined, which serves for, and is essential to all Operations in the Distillery.

There are several Sorts of Alembics, all different, either with regard to Matter or Form. As, the common Alembic with a Refrigeratory, the earthen and the glass Alembic, the *Balneum Mariæ*, and the Vapour-Bath Alembic.

Every one of these being of a different Construction, are also used in different Operations,

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The common Alembic consists principally of two Parts, the lower Part called the Body, and the upper termed the Head.

The Body consists of two Pieces, the lower called the Cucurbit, and the upper the Crown. The Cucurbit, or lower Part of the Body, is a kind of Receptacle proportioned to the Size of the Alembic, in which the Bodies to be distilled are placed.

The Crown, or upper Part of the Body, is also another Part of the Alembic ; and is that Part of the Body to which the Head is immediately luted. But an Idea of these several Alembics will be much better attained from the following Figures, which represent them much stronger to the Imagination than is possible to be done by Words.

Fig. 1. Is a common Alembic, as it appears before it is placed in a Furnace, where *a* is the Bottom, *b* the Crown, *c* the Head.

Fig. 2. Is the Body without the Head ; *a* the Rim or Top of the Crown where the Head is luted.

Fig. 3.

Fig. 3. The Head ; *a* the Rim where it is to be luted to the Body ; *b* the Nose, or End which is luted into the Worm.

Fig. 4. The Worm, as it appears when out of the Tub in which it is fixed when in use ; *a* the End into which the Still Head is inserted, *b* that which conveys the Liquor into the Receiver.

Fig 5. Two Stills at work in one Refrigeratory ; *a, b* the two Still Heads, *c, d* the Bodies inclosed in the Brick-Work ; *e, e* the two Fire-Places ; *f, f* the two Ash-Holes ; *g* a Common Receiver ; *h* a Spout Receiver, called by Chemists a Separating-Glass, used in the Distillation of Herbs, in order to extract their essential Oil ; *i* a Crane for drawing the Water out of the Refrigeratory.

Fig. 6. A small Still with a Refrigeratory ; *a* the Body, *b* the Head, *c* the Refrigeratory filled with Water, *d* the Receiver, luted to the Bec. of the Alembic.

Fig. 7. A Glass Alembic to be used as a *Balneum Mariæ* ; *a* the Body, *b* the Head, *c* the Bec, which is to be luted to the Receiver, *d* a Trivet on which it is standing in the Water.

Fig.

Fig. 8. A proper Receiver for the Glass Alembic, called by Chemists a Bolt-Head, or Matrafs.

Fig 9. The Glass Alembic placed in a Copper Vessel ; *a* the Copper Vessel filled with Water, *b* the Body of the Glass Alembic, *c* the Head, *d* the Receiver luted at *e* to the Bec of the Alembic.

Fig. 10. A cold Still for distilling simple Waters ; *a* the Head, *b* the Bec, or Nose, *c* the Receiver, *d* the Plate on which Herbs are laid.

Fig. 11. A Vessel for Digestion, called by Chemists a Pelican or circulatory Vessel ; *a* the Body, *b* the Head, *c, c* two Tubes, luted at *d, d*, by which the Liquor returns from the Head into the Body ; *e* a Furnace on which it is placed, *f* the Fire-place, *g* the Ash-hole.

Fig. 12. Another Receiver, used when it is necessary to lute it to the End of the Worm, in order to prevent the most volatile Parts from being evaporated, and lost.

C H A P. VI.

Of the ACCIDENTS that too often happen in performing the Processes of DISTILLATION.

AMONG the Accidents which frequently happen in Distilling, the least of all is for the Operation to miscarry and the Ingredients to be lost.

And this being a Subject of the greatest Importance we shall treat it with all possible Accuracy.

All Accidents are occasioned by Fire, their primary Cause; by want of Attention they get too much Head, and Fear often suffers them to become irremediable.

The first Accident which may happen by the Fire, is when a Distiller, by too great a Heat, causes the Ingredients to be burnt at the Bottom of the Still; by this Means his Liquor is spoiled by an emperumatic Taste, and the Tin is melted off from the Alembic. An Empereuma resembles the Smell of burnt Tobacco, and is produced in Liquors by too great a Degree of Heat. To illustrate this, distil any Fruit, Flowers, or any Aromatic whatever; but especially

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Something whose Smell is very volatile, draw off only the best, unlute the Alembic, and what remains in the Still, will be found to have a very disagreeable Smell; whence it follows, that if a little more had been drawn off, it would have spoiled what was before obtained.

If the Fire be too violent, the extraordinary Ebullition of the Contents causes them to ascend into the Head; and, if a Glass Alembic, they fall ignited into the Recipient; the Heat breaks it, the Spirits are dissipated, and often take Fire from the Heat of the Furnace.

If the Fire be too strong, the Bottom of the Still becomes red hot, the Materials inflamed, and consequently the Fire reaches the Recipient.

When an earthen Alembic is used, the closest Attention is requisite to keep the Fire from burning the Materials at the Bottom. The Head, which is always of Glass, bursts, and the Spirits are spilt, and often catch fire. And the Remedy becomes the more difficult, as Earth retains the Fire much longer than a common Alembic.

If the Alembic be not firmly fixed, it is soon put out of Order, falls down and unlutes

lutes itself; thus the Liquor is spilt, and the Vapour sets the Spirits on fire.

If all the Joints be not carefully luted, the Spirits at their first Effort issue through the least Aperture, run into the Fire, which is propagated into the Alembic by the Vapour.

In Distillations where the Phlegm ascends first, its Humidity penetrates the Lute, and loosens it, so that when the spirituous Vapours ascend, they are exposed to the same Accident.

Lastly, when the Recipient is unluted, especially if near full, without the greatest Circumspection the Spirits will be spilt, and so catch Fire.

Hitherto I have only given a simple Account of what daily happens to Distillers; but the Consequences of these Accidents are infinitely more terrible than the Accidents themselves; for an Artist to lose his Time, his Labour and Goods, is no small Matter; but it follows from what we have premised, that both his Life and Fortune are in danger from these Conflagrations. Instances of the former are too common, as well as those of the latter, relating to the

Danger to which the Operator is exposed. They are evident, and we have seen very lately three Instances sufficient to intimidate the most sanguine. The Spirits catch, the Alembic and Recipient fly, and the inflamed Vapour becomes present Death to all who breathe it.

The Rectifiers, who perform the most dangerous Operations of Distillery, are particularly exposed to these terrible Accidents ; the Fineness of the Spirit at the same time that it renders it more inflammable, also causes the Fire to spread with the greater Rapidity. And when their Store-houses are once on fire, they are seldom or never saved.

Possibly I may be censured for my Conscience on this Head ; indeed the Importance of it requires the most particular Discussion ; but intending to speak of the Methods proper to prevent these Accidents, I shall close this Chapter, with recommending the Subject of it to the serious Reflection of all concerned in Distillation. And it being hitherto omitted, though of all others it requires the Attention of the Distiller, I shall further observe, that these Operations should never be left to Servants. What can be expected from ignorant Persons ? Fear will
seize

seize them, when the greatest Presence of Mind is requisite. — Let us now proceed to the Methods of preventing, or at least lessening their Effects.

CHAP. VII.

Of the Methods of preventing Accidents.

TO have informed the Reader of the Accidents which happen in Distilling, would have been of little Consequence, without shewing, at the same Time, the Methods of preventing them. In order therefore to fortify him against the Terror, which the foregoing Chapter may have excited, we will here point out the Remedies for all the Cases before specified.

To prevent Accidents, two Things especially must be known, and adverted to.

1. The Knowledge of the Fire, which depends on the Fuel, whether Wood or Coal.

2. The Manner of luting so as to prevent the Vapours from escaping through it, and by that Means of setting the whole on fire.

The hardest Wood generally makes the quickest Fire, such as Beech, Oak, Holm, Elm, &c. The white Woods, as the Ash, the Poplar, the Willow, and the Birch, make a milder Fire. This holds good also of the Coal made of these two kinds of Wood ; and, consequently, the Nature of the Wood or Coals must determine the Fire, and the Action of this must be proportioned to the Effect intended to be produced by it. That is, the Capacity of the Alembic, the Matters to be distilled, and their Quantity. The same may also be said of Pit Coal, which is generally used in *England*.

It is evident, that the larger the Alembic, the more Fire is necessary. What has not been digested, also, requires more Fire than that which has been prepared by that Operation. Spices require a stronger Fire than Flowers ; a Distillation of Simple Waters more than that of spirituous Liquors.

The surest Way of ascertaining the necessary Degree of Fire, is to regulate it by the Materials, as they are more or less disposed to yield them Spirits, &c. and this is done as follows. The Operator must not leave the Alembic, but attentively listen to what passes within, when the Fire begins to heat it. When the Ebu-
lition

lition becomes too vehement, the Fire must be lessened, either by taking out some of the Fuel, or covering it with Ashes or Sand.

It requires a long Experience in the several Cases, before a Distiller can acquire a competent Knowledge in this important Point. Nor is it possible to determine the Degree of Fire from the Quantity of Fuel ; Judgment, assisted by Experience, must supply this Defect.

Every thing being determined with regard to the Degree of Fire, we shall now proceed to explain the Method of luting Alembics.

By the Term luting an Alembic, we mean, the closing the Joints through which the Spirits might transpire.

Lute is a Composition of common Ashes, well sifted, and soaked in Water ; Clay, and a kind of Paste made of Meal or Starch are also used for this Purpose ; which, as I before observed, is to close all the Joints, &c. in order to confine the Spirits from transpiring.

Good Luting is one of the surest Methods of preventing Accidents. An Alembic,

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where all Transpiration is prevented, having nothing to fear but the too great Fierceness of the Fire ; and that may be regulated by the Rules already laid down.

The refrigerating Alembic is mostly used. The Body and the Head are joined to each other ; but notwithstanding the greatest Care be taken in luting the Juncture, there will still be some imperceptable Interstice for Transpiration ; and the least being of the greatest Consequence, a Piece of strong Paper, should be pasted over the Joint, and the Alembic never left, till the Spirits begin to flow into the Receiver, in order to apply fresh Paper, if the former should contract any Moisture. The Master himself should carefully attend to this, and whatever Precaution may have been previously used, the Eye must be constantly upon it.

The Alembic, when vinous Spirits are distilled, should be luted with Clay, carefully spread round the Junctures, in order to prevent all Transpiration ; because the Consequences here are terrible ; for when the Fire catches a large Quantity, it is often irremediable. Besides, as this Earth cracks in drying, it must be often moistened, and fresh applied, on the first Appearance of any Occasion for it.

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The Retort is also luted with Clay ; but as glass Retorts are also used, they are often coated with the same Clay, to prevent their melting by the Intensity of the Fire.

Lastly, the earthen and glass Alembics are luted with Paper and Paste as above.— Having thus explained the great Consequence of Circumspection with regard to Luting, and the Degree of Fire, we shall now proceed to a third Method of preventing them, and close this Chapter with a short Observation on portable Furnaces ; which is, That Alembics being never thoroughly secure on this kind of Furnaces, a Hook should be fastened to the Refrigerant for fixing it to the Wall.

C H A P. VIII.

Of the Remedies for Accidents, when they happen.

NOtwithstanding the best of Rules, and the strictest Observation, it is impossible entirely to prevent Accidents, and therefore it is of no less Importance to point out the Remedies on those Occasions.

The most essential, are Courage and Presence of Mind ; Fear only increasing the Misfortune.

I. If

1. If the Fire be too violent it must be covered, but not so as totally to prevent its Action, as by that Means the Process of the Distillation would be interrupted, and render it more difficult and less perfect.

2. When the Ingredients burn, which you will soon discover by the Smell, the Fire must be immediately put out, in order to prevent the whole Charge of the Still being entirely spoiled, which would otherwise inevitably be the Consequence.

3. If the Spirits should catch fire, the first care is to unlute immediately the Receiver, and stop both the End of the Beak and Mouth of the Receiver with wet Clothes.

The Fire must then be put out, and if the Flame issued through the Luting, the Joints must be closed with a wet Cloth, which, together with Water, should never be wanting in a Distil-house.

4. If the Alembic be of Earth, and the Contents burn at the Bottom, the Fire must immediately be put out, the Alembic removed, and Water thrown upon it, till the Danger is over ; and, for farther Security, covered with a wet Cloth.

5. If

5. If after all your Care in closing the Junctures to prevent Transpiration, you perceive any thing amiss, while the Spirits are ascending, apply Clay, or any other Composition, in order to stop the Aperture, and have always a wet Cloth ready to stifle the Flame, if the Spirits should take fire.

6. If the Heat detaches the Lute, or it becomes moist, immediately apply another, having always ready what is necessary for performing it. Should the Transpiration be so violent, that you cannot immediately apply a fresh Lute, clap a wet Cloth round the Joint, and keep it on firm and tight, till the Spirits have taken their Course. But if notwithstanding all your Efforts the Transpiration should increase, so that you fear a Conflagration, remove the Receiver as soon as possible from the Fire, and afterwards your Alembic, if portable; but if otherwise, put out the Fire immediately.

7. The Charge being worked off, be cautious in luting the Receiver, that nothing be spilt on the Furnace, and carry it to some Distance from it, that the Spirits exhaling may not take fire.

8. Lastly observe, that wherever a Remedy is required, there must be no Candle used ;

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used ; for the spirituous Vapours easily take fire, and propagate the Flame to the Vessels from whence they issue.

All that has been hitherto said concerns only the Management of the Alembic ; but what remains is still more interesting, and relates to those who work it, that they may not, by conquering the Accident, destroy themselves.

On discovering any of the above Accidents, when the Flame has not yet reached the Spirits, let the Remedies already mentioned be applied, either with regard to the Lute, or the Violence of the Fire.

But if the Flame has reached the Alembic, the following Precautions are to be used.

The Operator must not approach the Alembic without a wet Cloth over his Mouth and Nostrils, it being immediate Death to inhale the inflamed Vapour.

In hastening to stop any Accident, be careful to approach the Side opposite to that whither the Air impels the Flame ; for, without this Precaution you would be involved in it, and could not, without the utmost Difficulty, extricate yourself from it.

If

If notwithstanding this Precaution, the Eddy of the Air should force the Flame to your Side, quit the Place immediately, and do not return till its Direction be changed, always taking care to have a wet linen Cloth before your Nose and Mouth, and keep yourself on the Side opposite to the Direction of the Flame : And also to have another such Cloth, in order to smother the Flame, and close the Crevice through which the Spirits issue.

Should it be your Misfortune to be covered with inflamed Spirits, wrap yourself in a wet Sheet, which should be always ready for that Purpose. Self-Preservation is of too great Importance that any of these Precautions should be omitted in such Variety of Dangers.

If the Fire has acquired such a Head that it cannot be stopt, the Receiver must be broke, and the Alembic, if portable, thrown down ; but no Person must be suffered to go near them, especially those who are Strangers to the Business.

In a desperate Case, like that of a large Quantity of rectified Spirit taking Fire, if Time permit, the Communication of the Beak of the Alembic with the Recipient, which is usually a Cask, must be cut off, by
closely

closely stopping the Bung ; and be sure no Candle come near the Receiver, leaving the rest, as the Danger would be too great to expose ones self to the Flames of a large Charge, and the Distiller's Safety should be principally considered.

I thought it my Duty to give my Reader these Informations, and hope that in the Practice of Distillation, he will find them of great Advantage.

C H A P. IX.

On the Necessity of often cooling the Alembic, as another Means of preventing Accidents.

THE Refrigerant is so essential a Part of the Alembic, that for want of it several other Expedients are made use of to perform its Office, for cooling those whose Capacity, Brittleness, or lastly the Construction, will not admit of their having any.

The Refrigerant is usually in proportion to the Capacity of the Alembic, for which the following may serve as a Rule, that the Capacity of the Refrigerant should be to that of the Alembic, as 14 to 8.

The Necessity of cooling the Head of the Alembic is self-evident to all who have the

the least Knowledge of Distillation, as it condenses the Spirits, cools them, and causes them to flow into the Receiver, which, if of Glafs, would otherwise be broken by the Heat; and consequently serves to prevent Conflagrations.

The Alembics of the *Balneum Mariae*, and the Vapour Bath, ought also to have Refrigerants, like the common Alembic, unless they are of Glafs.

Those of Earth and Glafs are cooled, as we have already observed, with a wet Cloth, which is also used to cool the Head of other kinds of Alembics. But it is not difficult to contrive one which may be placed in a Refrigerant; such as the following.

To a common small Still apply and lute a Worm, or long tin or pewter Tube, forming several Circumvolutions, of the same Circumference with the Body, in order to give it some Elevation, place this Worm in a Refrigerant, proportioned to the Alembic. If the Capacity of this Alembic should make it bear too much on the Neck of the Matrafs, it may be supported by a Trevit of the same Circumference as the Body itself: The Extremity of the Worm may have a Beak projecting beyond the Side of the Refrigerant, for conveying the Spirits into the Receiver.

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This Apparatus will be attended with little Expence, will save the Distiller the Trouble of being perpetually cooling the Head of the Alembic, and is such a Safeguard against Accidents, that if the Worm be well luted, nothing need be apprehended but from the Violence of the Fire.

This Method of Practice, therefore, is productive of three valuable Particulars: The first is, that by cooling the Spirits it preserves the Receiver, and obviates the Accidents arising from their Heat. The second is, that the Spirits being kept in a moderate Heat, the Transpiration is less, and consequently the Spirits procured by the Operation have more Taste, Smell and Frangency than they would otherwise have had.

Experience demonstrates, that when the Spirits flow hot into the Receiver, however attentive the Distiller may be to lute the Junctures of the Alembic, there will be a very sensible Evaporation, which even in simple Waters greatly depreciates the Goodness of the Liquor.

Lastly, the third is, that the Cooling of Alembics is what principally contributes to the Perfection of the Operation; because the

the Coolness of the Head precipitates the Phlegm, and in the Case of too great a Degree of Fire, and where the Ebullition is too vehement, if after taking away Part of the Fire, or covering it, the Ebullition should continue, the Head may be cooled with a wet Cloth, till the Ebullition is reduced.

As there is a Necessity of cooling the Alembic; so what we have said cannot be too carefully observed. In fine, the Contrast of Cold and Heat, equally concurring, but by Methods directly opposite, to the same Process, and the Perfection of the Distillation, is a Phænomenon, which deserves the Attention of all who study the Operations of Nature.

C H A P. X.

Of the Necessity of putting Water into the Alembic, for several Distillations.

TWO principal Advantages attend putting Water into the Alembic. The first is, to prevent the Loss the Distiller would incur without that Precaution, and so prevent any Alteration in the Liquor procured by Distillation. This we shall illustrate by an Example. Suppose a Distiller should attempt to rectify Spirits of Wine, without putting Water in the Alembic. It
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is evident, that the Fire will consume Part of it, which is entirely loss, because the same Quantity of Spirit cannot be procured from it, which might, had there been any thing to moderate the Action of the Fire, which now preyed upon it.

Secondly, If Liquors are impregnated with strong Ingredients, especially Seeds, and the Quantity be sufficient to absorb all the Phlegm, a great Quantity of Spirit must be left in the Still, or the Ingredients will burn, and the Spirits contract an empyreumatic Taste, which is the more detrimental to the Spirit, as it is increased by Age.

Thirdly, If no Water be put into the Alembic with the Ingredients, the Spirit will be rendered finer by them, and the Fire, if ever so little too strong, will cause the Ingredients to burn, and the Spirits to contract an Empyreuma ; a Misfortune easily prevented by this Precaution.

Thus it is a Safe-guard against Accidents : But besides, Water being mixed with the Ingredients, they are at once prevented from burning, and the Spirit not weakened ; for no sooner are the Ingredients put in Motion by the Fire, than the Spirits immediately ascend, and the Liquor loses nothing of its Quality, provided the Receiver be removed as soon as the Phelegm begins to ascend.

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The Water therefore prevents the Waste of the Spirits, and thus the Distiller loses nothing of his Goods ; whereas, without Water, the Spirits by impregnating the Materials, their Quantity must be less. With regard to the Phlegm, there is no Difficulty in finding when it begins to ascend, the first Drop being cloudy, and when it has continued dropping for some time, it is perceived by a milky Cast at the Bottom of the Receiver.

Lastly, The Distiller is no Loser with regard to the Quality of his Liquor, which is not at all weakened thereby. Thus it is attended with the two capital Advantages, the Profit of the Distiller, and the Perfection of the Liquor. Let us now proceed to the different Manners of Distillation.

CH A P. XI.

Of the particular Advantages attending every kind of Distillation.

I N the third Chapter we mentioned the several kinds of Distillation, we shall here enlarge on the particular Advantages of each, and in what Circumstances each is to be used.

In order for Distillation, the Alembic must be charged with Materials, and placed on a Fire, or Substances capable of producing the same Effect.

The Method of Distilling with the common refrigerant Alembic.

This Method of Distilling is the most generally used, being one of the most speedy and profitable, as it requires fewer Preparatives, and less Time.

To distil with the common Alembic, the Body of it must be thoroughly cleansed, that no Taste or Smell of any preceding Materials may remain. The Materials are then to be put into the Alembic; but care must be taken that the Alembic be not above half full, in order that the Materials may have sufficient Room to move, without choaking the Neck of the Alembic. The same Care must be taken with regard to the Head, it must be thoroughly cleansed and dried; for it often happens that some small Quantity of Water is left in the Rim, which renders the first Spirits foul, and, by endeavouring to separate it from the other, some, and that the most volatile Part of the Spirit, will be lost.

After

After this the two Parts of the Alembic are to be carefully luted with strong brown Paper, well pasted on, and the Nose of the Alembic luted to the Worm; after which the Fire should be immediately made under the Still, lest too long an Infusion should prejudice the Liquor.

This Alembic being worked on an open Fire, the Operation is quicker than any other; but the Degree of Fire requires a very close Attention; as a different Management is necessary to different Materials. The Water of the Refrigeratory must be changed from time to time, and if the Case requires it, the whole Head, but especially the Bec, must be kept cold.

Of Distillation in Sand, and in what Cases it should be used.

This Species of Distillation is performed in two different Manners. First, by covering the Fire with Sand or Ashes, and placing the Alembic upon it. This Method is very necessary in Digestion, and for the perfect Rectification of Spirits. Sand is absolutely necessary for moderating the Action of the Fire, when there is Reason to fear the Matter contained in the Bottom of the Alembic will burn.

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The second Method of Sand Distillation, is to take the finest River Sand, and after thoroughly washing it, put into the Alembic a Quantity sufficient to cover it three Fingers deep; after which the Still is to be charged with the Ingredients to be distilled. This serves instead of Water in certain Cases, where the Use of it would prejudice the Ingredients; as in the fine spirituous Waters impregnated with the aromatic Parts of Flowers; the Sand preventing the Ingredients from burning. It is also necessary in distilling rectified Spirits from Seeds.

This Operation being finished, the Alembic must be thoroughly cleansed from the Sand, that the Taste or Smell contained therein, be not communicated to any other Charge of different Ingredients.

Of Distilling in Balneum Mariæ, and its Advantages.

This Method of Distillation is of great Use in several Cases. Its Operation is more perfect, and is subject to few, if any of those Accidents attending Distillations on an open Fire.

In distilling sweet-scented Waters from Flowers, aromatic Plants, and others of that

that kind, where neither Water, nor Spirit ought to be mixed with them, there is an absolute Necessity for using the *Balneum Mariæ*; as by every other Distillation, on an open Fire, the Ingredients would infallibly burn.

If Sand should be made use of, the Fire would melt the Tin from the Alembic, and the Contents be in the utmost Danger of being burnt.

In distilling in *Balneum Mariæ*, a glass Alembic is generally used. This Alembic is to be placed in a Copper Vessel filled with Water. This Vessel ought at least to be of half the Height of the Alembic: at the Bottom of the Copper Vessel must be a Trivet on which the Alembic is to be placed, that it may not touch the Bottom of the Copper, because when the Water begins to boil, it disperses itself towards the Sides, and leaving the Bottom dry, the Ingredients would be in danger of burning.

The Use of the *Balneum Mariæ* is excellent for those Ingredients which require little Spirit; but if a Copper Alembic be used, be sure to place Sand at the Bottom, that the distilled Liquor may not contract any ill Taste or Smell. This Method is also adviseable in the Rectification of Spirits,

on Account of the Danger attending this Operation when performed on a naked Fire.

Were this Method of Distillation as expeditious as that performed on a naked Fire, no other ought to be used; because it is subject to no Accidents, and at the same Time the Spirit, &c. distilled is much more fragrant and grateful.

In what Cases glass, or earthen Alembics are to be used; their Advantages and Disadvantages.

In the Chapter relating to Accidents, we have mentioned the earthen Alembic; we must now add, that it ought never to be used, except the Matter to be distilled have a strong and bad Smell, and then seldom above once, unless it be for Ingredients of the same or similar Qualities.

This Alembic being very difficult to be managed, we can only recommend it in the Case above-mentioned.

As a naked Fire is generally applied to this Alembic, it requires a Furnace where the Fire may be gradually increased, on account of the Accidents to which it is liable.

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The glass Alembic is more easily managed, as it is generally placed in a *Balneum Mariæ*. Its principal Use is for distilling Waters from Flowers, and making Quintessences; and were it not for the Length of the Operation, it would be preferable to any other Method.

This Alembic hardly admitting of a Refrigerant, a wet linen Cloth must be placed on the Head, and often changed.

The Receiver of this Alembic must not be very large, because of the Fragility of the Bec; but if it were ever so little bent into a Curve, the Largeness of the Receiver would be of no Prejudice; because then its whole Weight would be supported by its Stand.

Advantages of Distillation performed by the Vapour Bath.

This Method differs very little from the *Balneum Mariæ*, and is used nearly in the same Circumstances; but has greatly the Advantage of the *Balneum Mariæ* in the Quickness of the Operation. And *Lemery*, in the first Part of his Course of Chemistry, affirms its Operation to be more perfect.

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However that be, its Use is equal to that of the *Balneum Mariæ*; but in distilling sweet-scented Waters, or Flowers, Sand must be placed at the Bottom, that the Liquor may not contract a Taste from the Copper.

Cases where Dung, Husks of Grapes, and Lime, are to be used.

These Substances are rarely used except in Digestions; and therefore of no great Use to Distillers, they using only hot Ashes, or a Fire well covered for that Purpose.

If Dung be used it must be of the hottest kind, viz. that of the Horse or Sheep, and the Quantity proportioned to the Heat intended. The Lime must be quick; and if the Heat required be moderate, Lime which has lain some time in the Air must be used. The same is to be observed with regard to the Husks of Grapes. But in whatever manner these are used, the Digestions must be performed in a close covered Vessel.

CHAP.

CHAP. XII.

Of Bodies proper for Distillation.

THIS Chapter alone might make a Volume, were we to make a particular Enumeration of all its Parts ; but, as we have already observed, we shall confine ourselves to the Distillation of simple and compound Waters, &c.

If we acquit ourselves to the Satisfaction of the Public, we shall enjoy the Pleasure of having treated of one Part entirely new ; and, indeed, the only one that has been overlooked.

The Bodies proper for Distillation, are Flowers, Fruits, Seeds, Spices and aromatic Plants.

By Distillation and Digestion, we extract the Colour and Smell of Flowers, in simple Waters and Essences.

We extract from Fruits, at least from some, Colour, Taste, &c.

From aromatic Plants, the Distiller draws Spirits, Essences, simple and compound Waters.

From

60. *A Complete System*

From Spices are procured Essences, or in the Language of the Chemists, Oils, and Perfumes, and also pure Spirits.

From Seeds or Berries are drawn simple Waters, pure Spirits; and from some, as those of Anise, Fennel, and Juniper, Oil.

The Colour of Flowers is extracted by Infusion, and likewise by Digestion in Brandy or Spirit of Wine: The Smell is extracted by Distillation; the simple Water with Brandy or Spirit of Wine.

What is extracted of the Colour of Flowers, by Infusion in Water by a gentle Heat, or by Digestion in Brandy or Spirits of Wine, is called, in the Distiller's Phrase, Tincture of Flowers.

The Colour of Fruits is extracted in the same manner, either by Infusion or Digestion: Their Taste is also procured by the same Processes. But let it be observed, that the Time of these Operations must be limited; for otherwise the Fruit, after Fermentation, would render it acid. The Taste is also extracted by Distillation in Spirit of Wine.

From aromatic Plants are extracted by the Alembic pure Spirits, Odours, and simple

ple Waters. But these require different Methods of Distillation. The first by Water or Brandy only, the second by rectified Spirit, which will give them the greatest Excellency they are capable of.

The Plants themselves with their Flowers may also be distilled, which is still better.

From Spices are drawn Spirits, and oily or spirituous Quintessences. The Spirits are drawn by Brandy, or Spirit of Wine, with very little Water : The Oils are distilled *per Descensum*; and the spirituous Quintessences by pounding the Spices, and after infusing them in Spirit of Wine, decanting it gently by Inclination.

From Seeds are extracted simple Waters, Spirits and Oils. Very few of the first and last, Spirits being what is generally extracted from Seeds and Berries.

Some Distillers, through a Notion of Frugality, distil Seeds with Water ; but their Liquors are not to be compared with those which are distilled with Spirits. When Oils are drawn from Seeds, the Operation is performed either by the *Balneum Mariæ*, or the Vapour Bath.

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We only deliver in this Place, the first Elements of each of these Operations, which will be farther illustrated in the Sequel, when we treat more particularly of these Subjects.

C H A P. XIII.

Of what is procured by Distillation.

BY Distillation are procured Spirit, Essence, simple Waters and Phlegm.

Spirits are very difficult to be defined. I consider them as the most subtil and volatile Parts of a Body.

All Bodies without Exception have Spirits more or less.

These Parts are an ignited Substance, and consequently by their own Nature disposed to a violent Motion.

These volatile Particles are more or less disposed to separate themselves, as the Bodies are more or less porous, or abound with a greater or lesser Quantity of Oil.

By the Term *Essence*, we understand the oleaginous Parts of a Body. An essential
Oil

Oil is found in all Bodies, being one of their constituent Principles. I have observed in all my Distillations, Spirit of Wine excepted, a soft unctuous Substance floating on the Phlegm; and this Substance is Oil, which we call Essence; and this is what we endeavour to extract.

Simple Waters are those distilled from Plants, Flowers, &c. without the Help of Water, Brandy, or Spirit of Wine. These Waters are commonly odoriferous, containing the Odour of the Body from whence it is extracted, and even exceeds in Smell the Body itself.

Phlegm is the aqueous Particles of Bodies; but whether an active or passive Principle, we shall leave to the Decision of Chemists.

It is of the last Importance to a Distiller to be well acquainted with its Nature; many mistaking for Phlegm several white and clouded Drops, which first fall into the Receiver, when the Still begins to work. These, however, are often the most spirituous Particles of the Matter in the Alembic, and consequently ought to be preserved. What has given occasion to this Mistake, is some Humidity remaining in the Head, &c. of the Alembic. And had it been thoroughly wiped, the first Drops would have been.

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been equally bright with any during the whole Operation.

The following Remark deserves Attention. In Bodies that have been digested the Spirits ascend first; whereas in Charges not digested, the Phlegm ascends before the Spirits. The Reason of this is very plain and natural.

In Substances previously digested, the Action of the Fire no sooner causes the Matter in the Alembic to boil, than the Spirits, being the most volatile Parts, detach themselves, and ascend into the Head of the Alembic. But when the Matter to be distilled has not undergone a proper Digestion, the Spirits being intangled in the Phlegm, are less disposed to ascend, till the Phlegm itself separates, and gives them room to fly upward. The Phlegm being aqueous rises first: This is more particularly observable in Spices. I am, however, inclined to believe, that were the Operation performed in an Alembic, whose Head was at a great Distance from the Surface of the Charge, they would not ascend high enough to come over the Helm, but fall back again by their own Gravity, and by that means leave the Spirits at Liberty to ascend. But in the common Refrigeratory Alembic this always happens.

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If this Observation be not readily admitted, I appeal to Experience, which I desire may be the Test of every thing I shall advance.

Another Observation, which has verified the above Assertion by innumerable Instances, is, that in an extraordinary Run of Business, when I had not time sufficient to digest the Substances, I used to bruise them in a Mortar ; but notwithstanding the Trituration, the Phlegm first came over, and afterwards the Spirits. But I desire to be understood, that I speak here only of the volatile Parts of the Plants not drawn with vinous Spirits, but contained in a simple Water.

Another Remark I must add, and which I hope will be acceptable to the Curious, as it has not yet been made public, though doubtless the Observation has often occurred to others ; it is this : That in mixed Charges, consisting of Flowers, Fruits, and aromatic Plants, put into the Alembic without a previous Digestion, the Spirits of the Flowers ascend first ; and notwithstanding the Mixture, they contracted nothing of the Smell or Taste of the Fruits and Plants. Next after the Spirits of the Flowers, those of the Fruits ascend, not in the least impregnated with the Smell or Taste of either of the

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Flowers

Flowers or Plants. And in the last Place the Spirits of the Plants distil no less neat than the former. Should this appear strange to any one, Experience will convince him of the Truth.

Another Observation I have made on aromatic Herbs, is, that whether they are, or are not digested ; whether the Spirits or Phlegm ascend first ; the Spirits contain very little of the Taste and Smell of the Plants from whence they were extracted ; and I have always been obliged to put to these Spirits a greater or lesser Quantity of the Phlegm, in order to give the Spirits I had drawn the Taste of an aromatic Odour of the Plants ; the Phlegm containing the greatest Quantity of both.

This Observation I insert, as of great Use to those who practise Distillation.

As the Term Digestion often occurs in this Essay, I cannot avoid pointing out its Advantages, and even shew the Necessity of using it in several Circumstances.

Substances are said to be in Digestion, when they are infused in a Menstruum, over a very slow Fire. This Preparation is often necessary in Distillation ; for it tends to open the Bodies, and thereby free the Spirits.

rits from their Confinements, whereby they are the better enabled to ascend.

Cold Digestions are the best; those made by Fire, or in hot Materials, diminish the Quality of the Goods, as some Part, as the most volatile, will be lost.

In order to procure Essences, the Bodies must be prepared by Digestion. It is even of absolute Necessity for extracting the Spirits and Essences of Spices.

C H A P. XIV.

Of the proper Season for Distilling.

FLOWERS of all Kinds must be distilled in their proper Seasons. To begin with the Violet. Its Colour and Smell can only be extracted when it is in its greatest Vigour, which is not at its first Appearance, nor when it begins to decay. *April* is the Month in which it is in its greatest Perfection; the Season being never so forward in *March*, as to give the Violet its whole Fragrancy.

The same must be observed of all other Flowers. And let them be gathered at the hottest Time of the Day; the Odour and

Fragrancy of Flowers being then in their greatest Perfection.

The same Observation holds good, with regard to Fruits; to which must be added; that they are the finest, and of the most beautiful Colour, especially those from whence Tinctures are drawn; they must be free from all Defects, as the Goods would by that Means be greatly detrimented.

Berries and Aromatics may be distilled at any Season, all that is necessary being a good Choice. But in this Distillers are sometimes mistaken, as may easily happen without a very accurate Knowledge. We shall therefore, in the Sequel, lay down more particular Directions for making a proper Choice of Materials.

C H A P. XV.

Of the Filtration of Liquors.

Filtration consists in passing Liquors thro' some porous Substance, in order to free them from those Particles which obscure their Brightness.

Nothing is finer than a Liquor newly distilled; but the Syrup and colouring Particles

ticles render it thick and opaque ; in order, therefore, to restore their Brightness they are filtrated, which is done by passing them through Sand, Paper, Cloth, &c.

All the Attention of the Distiller cannot in ordinary Operations always prevent some aqueous Particles from rising with the Spirits, either in the Beginning of the Process, in those Compositions where they ascend first, or at the Conclusion when they rise last. As this is almost unavoidable, so it is also sometimes necessary.

In distilling Flowers, or aromatic Plants, fresh gathered, the Phlegm rises first ; and this Part cannot be taken out of the Receiver without depriving the Spirits of a considerable Part of their Fragrancy.

In distilling Spices, their Odour being more entangled, will remain in the Alembic till Part of the Phlegm is drawn off. But when, instead of these Substances, their Quintessences are used, the Necessity ceases. But the Phlegm commonly causing a Cloudiness in the Liquor, it may be rendered tolerably fine, by pouring it gently off by Inclination, without the Trouble of Filtration, the aqueous Particles, by their Gravity, falling to the Bottom. But to render it entirely bright and fine, put some Cot-

ton in a Funnel, and pour the Liquor thro' it, by which means the aqueous Particles will be retained in the Cotton. You must however remember to cover the Top of the Funnel, to prevent the most volatile Parts of the Spirits from evaporating.

C H A P. XVI.

Of the Distillation of Malt Spirits.

THE Wash, or Liquor being prepared by Brewing and Fermentation, as directed in the first and second Chapters of this Treatise, the Still is to be charged with it, and worked off with a pretty brisk Fire. But it should be observed, that the only Apparatus used in this Process, is the Alembic with a Refrigeratory, as represented in *Fig. 1.*

The Wash being of a mucilaginous Nature, a particular Management is necessary to prevent its burning, and cause it to work kindly in the Still: If it should happen to be burnt in the Operation, the Spirit will have a most disagreeable Flavour, which can hardly ever be removed; and therefore to prevent this ill Effect, the Wash should be made dilute or thin, the Fire well regulated, and the whole kept in a continual Agitation during the whole Process. The most judicious

cious Distillers always take care to have their Wash sufficiently diluted, and constantly find their Spirit the purer for it. With regard to the Fire, it may be easily kept regular by a constant Attendance, and observing never to stir it hastily, or throw on fresh Fuel; and the stirring of the Liquor in the Still is to be effected by Means of a Paddle, or Bar kept in the Liquor till it just begins to boil, which is the Time for luting on the Head; and after which there is no great Danger, but from the improper Management of the Fire: This is the common Way; but it is no easy Matter to hit the exact Time, and the doing it either too late, or too soon, is attended with great Inconvenience, so that several have discovered other Methods; some put more solid Bodies into the the Still with the Wash; others place some proper Matter at the Bottom and Sides of the Still, which are the Places where the Fire acts with the greatest Force.

The Use of the Paddle would, however, answer better than either of these Methods, could it be continued during the whole Time the Still is working; and this may be done by the following Method: Let a short Tube of Iron or Copper be soldered in the Center of the Still-head, and let a cross Bar be placed below in the same Head, with a Hole in the Middle, corresponding to that

at the Top ; through both these let an iron Pipe be carried down in the Still, and let an iron Rod be passed through this with wooden Sweeps at its End ; this Rod may be continually worked by a Winch at the Still-head, and the Sweeps will continually keep the Bottom and Sides scraped clean, the Interstices of the Tube being all the time well crammed with Tow to prevent any Evaporation of the Spirit.

The same Effect may, in a great Measure, be produced by a less laborious Method, namely, by placing a Parcel of cylindrical Sticks lengthways, so as to cover the whole Bottom of the Still, or by throwing in a loose Parcel of Faggot Sticks at a Venture ; for the Action of the Fire below moving the Liquor, at the same time gives Motion to the Sticks, making them act continually like a Parcel of Stirrers upon the Bottom and Sides of the Still, which might, if necessary, be furnished with Buttons and Loops, to prevent them from starting. Some also use a Parcel of fine Hay laid upon the loose Sticks; and secured down by two cross Poles, laid from Side to Side, and in the same Manner fastened down with Loops. Care is to be taken in this Case not to press the Hay against the Sides of the Still ; for that would scorch nearly as soon as the Wash itself ; but the Sticks never will ; These are simple
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but effectual Contrivances, and in point of Elegance, they may be improved at Pleasure.

There is another Inconvenience attending the distilling of Malt Spirit, which is, when all the Bottoms, or gross mealy Fœculence is put into the Still along with the Liquor, the thinner Part of the Wash going off in Form of Spirit; the mealy Mass grows by Degrees more and more stiff, so as to scorch towards the latter Part of the Operation. The best Method of remedying this is to have a Pipe with a Stop-cock, leading from the upper Part of the Worm-tub into the Still; so that upon a half, or a quarter Turn, it may continually supply a little Stream of hot Water, in the same Proportion as the Spirit runs off, by which Means the Danger of scorching is avoided, and the Operation, at the same time, not in the least retarded.

In *Holland*, the Malt Distillers work all their Wash thick, with the whole Body of Meal among it; yet they are so careful in keeping their Stills clean, and so regular and nice in the Management of their Fires, that though they use no Artifice at all on this Head, only to charge the Still while it is hot and moist, they very rarely have the Misfortune to scorch, except now and then in

in the Depth of Winter. When such an Accident has once happened in a Still, they are extremely careful to scrape, scrub and scour off the Remains of the burnt Matter, otherwise they find the same Accident very liable to happen again in the same Place. But beyond all the other Methods in Use on this Occasion, would be the working the Stills not by a dry Heat, but in a *Balneum Mariæ*, which might possibly be so contrived by the Basen being large, and capable of working a great many Stills at once, as to be extremely worth the Proprietor's while in all respects.

Another Requisite to be observed is, that the Water in the Worm-tub be kept cool ; this may be affected, by placing in the middle of the Tub a wooden Pipe or Gutter, about three Inches square within, reaching from the Top almost to the Bottom ; by this Contrivance cold Water may, as often as necessary, be conveyed to the Bottom of the Worm-tub, and the hot Water at the Top forced either over the Sides of the Tub, or, which is better, through a leaden Pipe of moderate Size, called a Waste-pipe, soldered into the Top of the Tub, and extended to the Gutter formed to carry away the Water.

C H A P.

C H A P. XVII.

Of the Distillation of Molosses Spirits:

TH E Spirit distilled from Molosses or Treacle, is very clean or pure. It is made from common Treacle dissolved in Water, and fermented in the same Manner as the Wash for the common Malt Spirit.

But if some particular Art is not used in Distilling this Spirit, it will not prove so vinous as Malt Spirit, but more flat and less pungent and acid, though otherwise much cleaner tasted, as its essential Oil is of a less offensive Flavour. Therefore, if good fresh Wine-lees, abounding in Tartar, be added and duly fermented with the Molosses, the Spirit will acquire a much greater Vinosity and Briskness, and approach much nearer to the Nature of foreign Spirits.

Where the Molosses Spirit is brought to the common Proof Strength, if it be found not to have a sufficient Vinosity, it will be very proper to add some good dulcified Spirit of Nitre ; and if the Spirit be clean worked, it may, by this Addition only, be made to pass on ordinary Judges for *French Brandy*.

Great

Great Quantities of this Spirit are used in adulterating foreign Brandy, Rum and Arrac. Much of it is also used alone in making Cherry-Brandy, and other Drams by Infusion ; in all which many, and perhaps with Justice, prefer it to foreign Brandies.

Molosses, like other Spirits, is entirely colourless when first extracted ; but Distillers always give it, as nearly as possible, the Colour of foreign Spirits ; the Methods of performing which we shall explain in a subsequent Chapter.

C H A P. XVIII.

Of the Nature of Brandies, and Method of distilling them in France.

TH E general Method of distilling Brandies in *France* need not be formally described, as it differs in nothing from that commonly practised here in working from Wash or Molosses ; nor are they in the least more cleanly, or exact in the Operation.

They only observe more particularly to throw a little of the natural Lee into the Still, along with the Wine, as finding this gives

gives their Spirit the Flavour, for which it is generally admired abroad.

But though Brandy is extracted from Wine, Experience tells us, that there is a great Difference in Grapes from which the Wine is made. Every Soil, every Climate, every kind of Grapes varies with regard to the Quantity and Quality of Spirits extracted from them. There are some Grapes which are only fit for eating ; others for drying ; as those of *Damascus*, *Corinth*, *Provence*, and *Avignon* ; but not fit to make Wine.

Some Wines very proper for Distillation, others much less so. The Wines of *Languedoc* and *Provence* afford a great deal of Brandy by Distillation, when the Operation is made in their full Strength : The *Orleans* Wines, and those of *Blois* afford yet more ; but the best are those of the Territories of *Cogniac* and of *Andaye*, which are however in the Number of those the least drank in *France*. Whereas those of *Burgundy* and of *Champaign*, though of a very fine Flavour, are improper, because they yield but very little in Distillation.

It must also be farther observed, that all the Wines for Distillation, as those of *Spain*, the *Canaries*, of *Alicant*, of *Cyprus*, of *St. Perés*, of *Toquet*, of *Grave*, of *Hungary*, and

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and others of the same kind, yield very little Brandy by Distillation ; and consequently would cost the Distiller considerably more than he could sell it for. What is drawn from them is indeed very good, always retaining the saccharine Quality, and rich Flavour of the Wine from whence it is drawn ; but as it grows old, this Flavour often grows aromatic, and is not agreeable to all Palates.

Hence we see, that Brandies always differ, according as they are extracted from different Species of Grapes. Nor would there be so great a Similarity as there is between the different kinds of *French* Brandies, were the strongest Wines used for this Purpose : But this is rarely the Case, the weakest and lowest-flavoured Wines only are distilled for their Spirit, or such as prove absolutely unfit for any other Use.

A large Quantity of Brandies is distilled in *France* during the Time of the Vintage ; for all those poor Grapes that prove unfit for Wine, are usually first gathered, pressed, their Juice fermented, and directly distilled. This rids their Hands of their poor Wines at once, and leaves their Casks empty for the Reception of better. It is a general Rule with them not to distil any Wine, that will fetch any Price as Wine ; for, in this
State,

State, the Profits upon them are vastly greater than when reduced to Brandies. This large Stock of small Wines, with which they are almost over-run in *France*, sufficiently accounts for their making such vast Quantities of Brandy in *France*, more than other Countries, which lie in warmer Climates, and are much better adapted to the Production of Grapes.

Nor is this the only Fund of their Brandies; for all the Wine that turns eager, is also condemned to the Still; and, in short, all that they can neither export, nor consume at home, which amounts to a large Quantity; since much of the Wine, laid in for their Family Provision, is so poor, as not to keep during the Time in spending.

Hence many of our *English* Spirits, with proper Management, are convertible into Brandies, that shall hardly be distinguished from the foreign in many Respects, provided this Operation be neatly performed. And, in particular, how far a Cyder Spirit, and a Crab Spirit, may, even from the first Extraction, be made to resemble the fine and thin Brandies of *France*, we would recommend to those Distillers, whose Skill and Curiosity prompts them to Undertakings condemned by those who only work mechanically, and scorn

scorn to deviate from the beaten Tract, tho' they have the fairest Prospect of acquiring Profit to themselves, and a lasting Emolument to their Country.

C H A P. XIX.

Of the Distillation of Rum.

RUM differs from what we simply call Sugar Spirit, as it contains more of the natural Flavour, or essential Oil of the Sugar Cane ; a great deal of raw Juice, and even Parts of the Cane itself being often fermented in the Liquor, or Solution, of which the Rum is prepared.

Hence we see from whence Rum derives its Flavour ; namely, from the Cane itself. Some, indeed, are of Opinion, that the unctuous or oily Flavour of the Rum proceeds from the large Quantity of Fat used in boiling the Sugar. This Fat, indeed, if coarse, will give a stinking Flavour to the Spirit in our Distillations of the Sugar Liquor, or Wash, from our refining Sugar-houses ; but this is nothing like the Flavour of the Rum ; which, as we have already observed, is the Effect of the natural Flavour of the Cane.

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Great Quantities of Rum are made at *Jamaica, Barbadoes, Antigua*, and other Sugar Islands : The Method of making it is this :

When a sufficient Stock of the Materials is got together, they add Water to them, and ferment them in the common Method, though the Fermentation is always carried on very slowly at first ; because at the Beginning of the Season for making Rum in the Islands, they want Yeast, or some other Ferment to make it work ; but after this, they, by Degrees, procure a sufficient Quantity of the Ferment, which rises up as a Head to the Liquor in the Operation ; and thus they are able afterwards to ferment, and make their Rum with a great deal of Expedition, and in very large Quantities.

When the Wash is fully fermented, or to a due Degree of Acidity, the Distillation is carried on in the common Way, and the Spirit is made up Proof ; though sometimes it is reduced to a much greater Degree of Strength, nearly approaching to that of Alcohol, or Spirit of Wine ; and it is then called double distilled Rum.

It would be easy to rectify the Spirit, and bring it to a much greater Degree of Purity
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than we usually find it to be of ; for it brings over in the Distillation a large Quantity of the Oil ; and this is often so disagreeable, that the Rum must be suffered to lie by a long time to mellow before it can be used ; whereas, if well rectified, its Flavour would be much less, and consequently much more agreeable to the Palate.

The best State to keep Rum, both for Exportation, and other Uses, is doubtless that of Alcohol, or rectified Spirits. In this manner, it would be contained in half the Bulk it usually is, and might be let down to the common proof Strength with Water when necessary : For the common Use of making Punch, it would likewise serve much better in the State of Alcohol ; as the Taste would be cleaner, and the Strength might always be regulated to a much greater Degree of Exactness than in the ordinary Way.

If the Business of rectifying Rum was more nicely managed, it seems a very practicable Scheme to throw out so much of the Oil ; as to reduce it to the fine light State of a clear Spirit, but lightly impregnated with the Oil ; in this State it would nearly resemble Arrac, as is easily proved by mixing a very small Quantity of it with a tasteless Spirit ;
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for it then bears a very near Resemblance to Arrac in Flavour.

C H A P. XX.

Of Sugar-Spirit.

WE mean by a Sugar-Spirit, that extracted from the Washings, Scummings, Dross, and Waste of a Sugar-baker's Refining-house.

These recrementitious, or drossy Parts of the Sugar are to be diluted with Water, fermented in the same manner as Molasses or Wash, and then distilled in the common Method. And if the Operation be carefully performed, and the Spirit well rectified, it may be mixed with foreign Brandies, and even Arrac in a large Proportion, to great Advantage ; for this Spirit will be found superior to that extracted from Treacle, and consequently more proper for these Uses.

C H A P. XXI.

Of Raisin-Spirits.

BY Raisin-Spirits, we understand, that extracted from Raisins, after a proper Fermentation.

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In order to extract this Spirit, the Raisins must be infused in a proper Quantity of Water, and fermented in the manner described in the Chapter on Fermentation. When the Fermentation is completed, the whole is to be thrown into the Still, and the Spirit extracted by a strong Fire.

The Reason why we here direct a strong Fire, is, because by that Means a greater Quantity of the essential Oil will come over the Helm with the Spirit, which will render it much fitter for the Distiller's Purpose; for this Spirit is generally used to mix with common Malt Goods; and it is surprizing how far it will go in this Respect, ten Gallons of it being often sufficient to give a determining Flavour, and agreeable Vinosity to a whole Piece of Malt Spirits.

It is therefore well worth the Distiller's while to endeavour at improving the common Method of extracting Spirits from Raisins; and perhaps the following Hint may merit Attention.

When the Fermentation is completed, and the Still charged with fermented Liquor, as above directed, let the whole be drawn off with as brisk a Fire as possible; but instead of the Cask or Can, generally
used

used by our *English* Distillers for a Receiver, let a large Glass, called by Chemists, a Separating-Glass, be placed under the Nose of the Worm, and a common Receiver applied to the Spout of the Separating-Glass; by this means the essential Oil will swim upon the Top of the Spirit, or rather low Wine, in the Separating-Glass, and may be easily preserved at the End of the Operation.

The Use of this limpid essential Oil is well known to Distillers; for in this resides the whole Flavour, and consequently may be used to the greatest Advantage in giving that distinguishing Taste, and true Vinosity, to the common Malt-Spirits.

After the Oil is separated from the low Wine, the Liquor may be rectified in *Bal-neum Mariae* into a pure and almost tasteless Spirit, and therefore well adapted to make the finest compound Cordials, or to imitate or mix with the finest *French* Brandies, Arracs, &c.

In the same Manner a Spirit may be obtained from Cyder. But as its particular Flavour is not so desirable as that obtained from Raisins, it should be distilled in a more gentle Manner, and carefully rectified in the Manner we shall shew in the Chapter on

Rectification ; by which Means a very pure and almost insipid Spirit will be obtained, which may be used to very great Advantage in imitating the best Brandies of *France*, or in making the finest compound Waters or Cordials.

C H A P. XXII.

Of Arracs.

WHAT is properly meant by the Term Arracs, are Spirits extracted from the fermented Juice of certain Trees common in the *East-Indies*, particularly those of the Cocoa, or Palm-tree. The whole Process of making Arrac, is performed in the following Manner.

In order to procure the vegetable Juice for this Operation, the Person provides himself with a sufficient Number of small earthen Pots, with Bellies and Necks, resembling our common glass Bottles ; a Number of these he fastens to his Girdle, or to a Belt across his Shoulders, and climbs up the tall Trunk of the Cocoa tree : Having reached the Boughs of the Tree, he cuts off with a Knife certain small Buds, or Buttons, applying immediately to the Wound one of his Bottles, and fastens it with a String to the Bough. In this Manner he proceeds

proceeds till he has fixed his whole Number of Bottles, which serve as Receivers to the Juice distilling from the Wounds. This Operation is generally performed in the Evening, a greater Quantity of Juice flowing from the Tree in the Night than in the Day. The Bottles are next Morning taken off, and the Liquor emptied with a proper Vessel, where it spontaneously ferments. As soon as the Fermentation is completed, the Liquor is thrown into the Still, and drawn down to a low Wine ; but so very poor and dilute, that they are obliged to rectify it in another Still, to that weak kind of Proof Spirit, we generally see it ; for though it appears Bubble-Proof, it rarely contains more than a sixth, and sometimes only an eighth of Alcohol, all the rest being no more than an acidulated Water, which might be supplied from any common Spring. Why Arrac appears Bubble-Proof, when in reality so far below what we mean by Proof, is not so great a Mystery, as at first Sight it appears to be ; for this kind of Proof is entirely owing to a certain Tenacity of the Parts of the Liquor, or to the particular Property of the Oil incorporated in the Spirit ; as we shall abundantly shew in a subsequent Chapter.

From this Account of Arrac, it should seem no very difficult Matter to imitate it

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here. And, perhaps, the whole Difficulty lies in procuring a pure and insipid Spirit ; for it is ridiculous to attempt it with our common Malt-Spirit. With regard to the Flavour of the Arrac, it may be effectually imitated by some essential Oils easily procurable.

Hence we see of what prodigious Advantage a pure and insipid Spirit would be of to Distillers, and consequently the great Encouragement there is to attempt the Discovery. Perhaps a Spirit of this kind may be extracted from Sugar properly refined. The Hint is worth prosecuting ; and the Writer of this Essay, from repeated Experiments, is abundantly convinced that the Thing is practicable. Had he entirely succeeded, he would readily have communicated the Whole for the Benefit of his Country ; but is now obliged to defer, to some future Opportunity, the Result of his Enquiries. In the mean Time, he would recommend the Prosecution of this Hint to those Distillers, who endeavour to improve their Art, and advance it nearer to Perfection.

Since Arrac is a Spirit extracted from the Juice of the Cocoa tree, it might perhaps be worth enquiring how nearly it might be imitated by fermenting and distilling the
Juices

Juices of the Birch and Sycamore-trees. We should by this Means obtain an *English* Arrac; and, perhaps, a Spirit equal in Flavour to that imported from *Batavia*.

When the Cask, in which the Arrac is imported happens to be decayed; or the Liquor touches any Nails, or other Iron, it dissolves Part of it, and at the same time extracts the resinous Parts of the Oak, by which means the whole Liquor in the Cask acquires an inky Colour. In order to whiten and clarify Arrac, which has contracted this Colour, a large Quantity of new or skimmed Milk must be put into the Cask, and the whole beat together, as Vintners do to whiten their brown Wines; by this means the inky Colour will be absorbed by the Milk, and fall with it to the Bottom, so that the greatest Part of the Arrac may be drawn off fine; and the Remainder procured in the same Condition by being filtrated through a conical Flannel Bag.

C H A P. XXIII.

Of Rectification.

TH E R E are several Methods of performing this Operation; though some, and indeed those in general practised by our Distillers, hardly deserve the Name; because, instead

instead of rectifying, that is freeing the Spirit from its essential Oil and Phlegm, they alter the natural Flavour of the Spirit that comes over in the Operation.

The principal Business of Rectification is to separate the Spirit from the essential Oil of the Ingredient, which is very apt to adhere strongly to the Spirit. And in order to this, Care should be taken in the first Distillation ; that is, the Spirit, especially that from Malt, should be drawn by a gentle Fire, by which means great Part of the essential Oil will be kept from mixing with the Spirit ; for Experience has abundantly proved, that it is much easier to keep asunder, than to separate them when once mixed.

But as it is almost impossible to draw low Wines without the Spirit being in some Measure impregnated with the essential Oil, it is absolutely necessary to be acquainted with some Methods of separating the Spirit from the Oil, and also of freeing it from its Phlegm. The best Methods of doing this to Perfection, are Re-distillation and Percolation.

In order to rectify low Wines, they should be put into a tall Body or Alembic, and gently distilled in *Balneum Mariæ* ; by this means

means a large Proportion, both of the Oil and Phlegm will remain in the Body. But if the Spirit should be found, after this Operation, to contain some of the essential Oil, it must be let down with fair Water, and re-distilled in the same gentle Manner. And thus it may be brought to any Degree of Purity; especially if in the working the Spirit be suffered to fall into a proper Quantity of clear Water, and the Spirit afterwards rectified to the Height proposed. The same Method should be used in freeing Proof-Spirit, or even Alcohol, from this Oil; namely, by letting it down with clean Water to the Strength of low Wines, and re-distilling it in *Balneum Mariæ*. But it must be remembered, that it is much more difficult to cleanse Alcohol, or Proof-Spirit than low Wines, because the Oil is more intimately mixed with the two former than with the latter. This Oil may however be separated from Proof-Spirit, &c. by the Method already proposed, especially if it be previously filtrated through Paper, thick Flannel, Sand, Stone, &c.

But this Method, though it effectually answers the Intention, is generally rejected by our Distillers, because of the Slowness of the Operation; and others substituted in its stead, though instead of freeing the Spirit from the Oil, they only abolish the natural Flavour

Flavour of the Spirit, and make a more intimate Mixture between the Particles of the Spirit, and those of the essential Oil.

It is impossible to enumerate all the Methods practised by Distillers, as almost every one pretends to have a secret Nostrum for this Purpose. The principal Methods in use for rectifying Malt-Spirits, are however reducible to three, namely, by fixed alkaline Salts, by acid Spirits mixed with alkaline Salts, and by saline Bodies, and flavouring Additions.

The Method of rectifying by alkaline Salts is thus performed. To every Piece of Proof-Spirit, add fourteen Pounds of dry Salt of Tartar, fixed Nitre, or calcined Tartar; lute on the Head, and distil, by a gentle Heat, but be very careful to leave out the Faints. By this Method a large Proportion of the foetid Oil will be left in the Still; and what comes over with the Spirit will be greatly attenuated. But this Operation is generally performed in a very different manner; for, instead of distilling the the Spirit in a gentle and equable manner, the Still is worked in its full Force; by which means the Oil, which should have remained in the Still, is driven over, and intimately mixed with the Spirit; and, consequently, the whole Operation frustrated, and

and the Spirit rendered much harder to cleanse than it was before.

But even when the Operation is performed according to the Rules of Art, it is far from being perfect ; for it is well known, that Part of the fixed Salts become volatile in the Operation, pass over the Helm, and intimately mixes with the essential Oil still contained in the Spirits : by this means the Oil becomes more perfectly united with the Spirits, and consequently much harder to be separated by repeated Distillations. Nor is this all, for the Still being worked in its full Force, the bitter Oil of the Malt, formed into a kind of liquid Soap in the Still, by means of the alkaline Salt, is brought over the Helm with the Faints, and suffered to mix with the Spirit, whereby it is rendered almost as nauseous and ill-tasted as before the Operation. Besides, if this Operation were performed in its utmost Perfection, it would never answer the Intention ; for the alkaline Salt destroys the Vinosity of the Spirit ; and consequently deprives it of one of its most valuable Properties. Our Distillers are well acquainted with this Defect in the Operation, and endeavour to supply it by an Addition of Acids. This is what we call the second Method by Alcalies and Acids.

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The Operation of rectifying by the Method of fixed Alcalies and Acids is the same as that above described ; the Spirit is drawn over from fixed Alcalies as before ; but in order to mortify the Alkali in the Spirit, and restore its Vinosity, a proper Quantity of some acid Spirit is added. Various kinds of Acids are used on this Occasion ; but principally those of the mineral Kind, because of their Cheapness ; as Oil of Vitriol, Spirit of Nitre, Oil of Sulphur, and the like. We would, however, caution a young Distiller from being too busy with these corrosive Acids, the sulphurous Spirit of Vitriol, dulcified Spirit of Nitre, or Mr. *Boyle's* acid Spirit of Wine well rectified, will much better answer his Purpose.

The third Method of Rectification is that by saline Bodies, and flavouring Ingredients. There is no Difference in the Operation between this and the two foregoing Methods ; fixed alkaline Salts, common Salt decrepitated or dried, calcined Vitriol, Sandiver, Alum, &c. is put into the Still with the low Wines, and the Spirit drawn over as before. When the Quantity is drawn off, the flavouring Ingredients are added to give the Spirit the Flavour intended. But as the Spirit is not by this means rendered sufficiently pure, the disagreeable Flavour of
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the Spirit generally overpowers that of the Ingredients, whereby the whole Intention is either destroyed, or a compound Flavour produced, very different from that intended.

Some Distillers, instead of alkaline Salts, use quick Lime in rectifying their Malt Spirit ; this Ingredient cleanses and dephlegmates the Spirit considerably ; but like that rectified from alkaline Salts, it requires an alkaline Disposition, and also an nidorous Flavour. Acids, therefore, are as necessary to be mixed with those Spirits rectified with quick Lime, as with those rectified with an alkaline Salt. If Chalk, calcined and well purified animal Bones, &c. were used instead of quick Lime, the Spirit would have a much less alkaline or nidorous Flavour ; and, consequently, the flavouring Ingredients might be added to it with more Success than can be expected from a Spirit rectified from alkaline Salts.

But, perhaps, if neutral Salts were used instead of the alkaline ones, the Spirit might be rendered pure, without contracting an alkaline Flavour ; soluble Tartar might be used for this Purpose, though the Spirit acquires from hence a little saponaceous Flavour. Dr. Cox has mentioned another Method

thod for this Purpose, namely, to deprive the volatile Salts of their Oil, by rendering them neutral with Spirit of Salt, and afterwards subliming them with Salt of Tartar : The Acid may be varied if the Spirit of Salt should not be found so well adapted to the Purpose as could be wished : But fine dry Sugar seems the best adapted to the Purpose of rectifying these Spirits ; as it readily unites with the essential Oil, detains and fixes it, without imparting any urinous, alkaline, or other nauseous Flavour to the Spirits rectified upon it.

Thus have I considered the principal Methods used by our Distillers in rectifying their Spirits ; and shall conclude this Chapter with remarking, that there is no other Way of rectifying to Perfection besides what we first laid down, namely, by gentle Distillation. But then it must be remembered, that the whole Process must be of a Piece : We mean, that the first Distillation from the Wash must be performed in a gentle manner ; for otherwise the essential Oil will be so intimately blended with the Spirit, as not to be easily separated by Re-distillation. Another good Property attending this Method is its Universality ; all kinds of Spirits, from whatever Ingredients extracted, require Rectification ; and this is adapted to all kinds.

C H A P.

C H A P. XXIV.

Of the Flavouring of Spirits.

WE have observed in the preceding Chapter, that the common Method of rectifying Spirits from alkaline Salts, destroys their Vinosity, and in its stead introduces an urinous or lixivious Taste. But as it is absolutely necessary to restore, or at least to substitute in its room some Degree of Vinosity, several Methods have been proposed, and a Multitude of Experiments performed, in order to discover this great *Desideratum*: But none has succeeded equal to the Spirit of Nitre; and accordingly this Spirit, either strong or dulcified, has been used by most Distillers to give an agreeable Vinosity to their Spirits.

Several Difficulties however occur in the Method of using it; the principal of which is, its being apt to quit the Liquor in a short Time, and consequently depriving the Liquor of that Vinosity it was intended to give. In order to remove this Difficulty, and prevent the Vinosity from quitting the Goods, the dulcified Spirit of Nitre, which is much better than the strong Spirit, should be prepared by a previous Digestion continued for some Time with Alcohol; the

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longer

perhaps a preparation of Ether would answer that end best.

longer the Digestion is continued the more intimately will they be blended, and the Compound rendered the milder and softer.

After a proper Digestion, the dulcified Spirit should be mixed with the Brandy, by which Means the Vinosity will be intimately blended with the Goods, and disposed not to fly off for a very considerable Time.

No general Rule can be given for the Quantity of this mineral Acid requisite to be employed; because different Proportions of it are necessary in different Spirits. It should, however, be carefully adverted to, that though a small Quantity of it will undoubtedly give an agreeable Vinosity resembling that naturally found in the fine subtile Spirits drawn from Wines, yet an over large Dose of it will not only cause a disagreeable Flavour, but also render the whole Design abortive, by discovering the Imposition. Those, therefore, who endeavour to cover a foul Taste in Goods by large Doses of dulcified Spirit of Nitre, will find themselves deceived.

But the best, and indeed the only Method of imitating *French* Brandies to Perfection, is by an essential Oil of Wine; this being the very thing that gives the *French* Brandies their Flavour. It must, however,

however, be remembered, that in order to use even this Ingredient to Advantage, a pure, tasteless Spirit must be first procured ; for it is ridiculous to expect that this essential Oil should be able to give the agreeable Flavour of *French* Brandies, to our fulsome Malt Spirit, already loaded with its own nauseous Oil, or strongly impregnated with a lixivious Taste from the alkaline Salts used in Rectification. How a pure insipid Spirit may be obtained has been already considered in some of the preceding Chapters ; it only therefore remains to shew the Method of procuring this essential Oil of Wine, which is this :

Take some Cakes of dry Wine Lees, such as are used by our Hatters, dissolve them in six or eight times their Weight of Water, distil the Liquor with a slow Fire, and separate the Oil by the Separating Glass ; reserving for the nicest Uses that only which comes over first, the succeeding Oil being coarser and more resinous.

Having procured this fine Oil of Wine, it may be mixed into a Quintessence with pure Alcohol ; by which Means it may be preserved a long time fully possessed of all its Flavour and Virtues ; but without such Management, it will soon grow resinous and rancid.

When a fine essential Oil of Wine is thus procured, and also a pure and insipid Spirit, *French* Brandies may be imitated to Perfection with regard to the Flavour. It must, however, be remembered, and carefully adverted to, that the essential Oil be drawn from the same sort of Lees, as the Brandy to be imitated was procured from; we mean, in order to imitate *Coniac* Brandy, it will be necessary to distil the essential Oil from *Coniac* Lees; and the same for any other kind of Brandy. For as different Brandies have different Flavours; and as these Flavours are owing entirely to the essential Oil of the Grape, it would be preposterous to endeavour to imitate the Flavour of *Coniac* Brandy, with an essential Oil procured from the Lees of *Bordeaux* Wine.

When the Flavour of the Brandy is well imitated by a proper Dose of the essential Oil, and the Whole reduced into one simple and homogeneous Fluid, other Difficulties are still behind: The Flavour, though the essential Part, is not however the only one; the Colour, the Proof and the Softness must be also regarded, before a Spirit, that perfectly resembles Brandy, can be procured. With regard to the Proof, it may be easily hit, by using a Spirit rectified

tified above Proof ; which, after being intimately mixed with the essential Oil of Wine, may be let down to a proper Standard by fair Water. And the Softness may in a great Measure be obtained by distilling and rectifying the Spirit with a gentle Fire; and what is wanting of this Criterion in the Liquor, when first made, will be supplied by Time ; for it must be remembered, that it is Time alone that gives this Property to *French* Brandies ; they being at first, like our Spirits, acrid, foul, and fiery. But with regard to the Colour a particular Method is necessary to imitate it to Perfection : And how this may be done shall be considered in the next Chapter.

C H A P. XXV.

Of the Methods of colouring Spirits.

TH E Art of colouring Spirits owes its Rise to Observations on foreign Brandies. A Piece of *French* Brandy that has acquired by Age a great Degree of Softness and Ripeness is observed, at the same time, to have acquired a yellowish brown Colour ; and hence our Distillers have endeavoured to imitate this Colour in such Spirits as are intended to pass for *French* Brandy. And in order to this a great Variety of Experiments has been made on various Substances,

in order to discover a direct and sure Method of imitating this Colour to Perfection. But, in order to do this, it is necessary to know from whence the *French* Brandies themselves acquire their Colour ; for till we have made this Discovery, it will be in vain to attempt an Imitation ; because, if we should be able to imitate exactly the Colour, which is indeed no difficult Task, the Spirit will not stand the Test of different Experiments, unless the Colour in both be produced from the same Ingredient.

This being undeniably the Case, let us try if we cannot discover this mighty Secret ; the Ingredient from whence the *French* Brandy acquires its Colour.

We have already observed, that this Colour is only found in such Brandies as have acquired a mellow Ripeness by Age ; it is therefore not given it by the Distiller, but has gained it by lying long in the Cask. Consequently, the Ingredient from whence this Colour is extracted, is no other than the Wood of the Cask, and the Brandy in reality is become a dilute Tincture of Oak.

The common Experiment used to prove the Genuineness of *French* Brandy proves, that this Opinion is well founded. The
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Experiment is this : They pour into a Glass of Brandy a few Drops of a Solution of calcined Vitriol of Iron in a diluted Spirit of Sulphur, or any other mineral Acid, and the Whole turns of a blue Colour ; in the same Manner, as we make Ink of a Tincture of Galls and Vitriol.

Since, therefore, the Colour of *French* Brandies is acquired from the Oak of the Cask, it is no Difficulty to imitate it to Perfection. A small Quantity of the Extract of Oak, or the Shavings of that Wood properly digested, will furnish us with a Tincture capable of giving the Spirit any Degree of Colour required. But it must be remembered, that as the Tincture is extracted from the Cask by Brandy, that is Alcohol and Water, it is necessary to use both in extracting the Tincture ; for each of these Menstruums dissolves different Parts of the Wood. Let, therefore, a sufficient Quantity of Oak Shavings be digested in strong Spirit of Wine ; and also at the same Time other Oak Shavings be digested in Water : And when the Liquors have acquired a strong Tincture from the Oak, let both be poured off from the Shavings, into different Vessels, and both placed over a gentle Fire till reduced to the Consistence of Treacle. In this Condition, let the two

Extracts be intimately mixed together ; which may be done effectually by adding a small Quantity of Loaf Sugar, in fine Powder, and well rubbing the Whole together. By this Means a liquid essential Extract of Oak will be procured, and always ready to be used as Occasion shall require.

There are other Methods in Use for colouring Brandies ; but the best, besides the Extract of Oak above-mentioned, are common Treacle and burnt Sugar.

The Treacle gives the Spirits a fine Colour, nearly resembling that of *French Brandy* ; but as its Colour is but dilute, a large Quantity must be used ; this is not however attended with any bad Consequences ; for notwithstanding the Spirit is really weakened by this Addition, yet the bable Proof, the general Criterion of Spirits, is greatly mended by the Tenacity imparted to the Liquor by the Treacle. The Spirit also acquires from this Mixture a sweetish or luscious Taste, and a Fulness in the Mouth ; both which Properties render it very agreeable to the Palates of the common People, who are, in fact, the principle Consumers of these Spirits.

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A much smaller Quantity of burnt Sugar than of Treacle will be sufficient for colouring the same Quantity of Spirits; the Taste is also very different; for, instead of the Sweetness imparted by the Treacle, the Spirit acquires from the burnt Sugar an agreeable Bitterness, and by that Means recommends itself to nicer Palates, which are offended with a luscious Spirit. The burnt Sugar is prepared by dissolving a proper Quantity of Sugar in a little Water, and scorching it over the Fire till it acquires a black Colour.

Either of the above Ingredients, Treacle or burnt Sugar, will nearly imitate the genuine Colour of old *French* Brandy; but neither of them will succeed, when put to the Test of the vitriolic Solution.

Thus have I traced the Subject of Distillation from its Origin; shewn the Methods commonly made use of by Distillers, and pointed out various Improvements, that might be introduced into this Art with great Advantage; and shall conclude this Part with recommending the several Hints to those Distillers who are desirous of improving their Art, and proceeding on a rational Foundation, it being from such only that

that Improvements are to be expected ; for where the Operations are constantly carried on in the same beaten Tract, it is in vain to expect Improvements, unless Chance should be kind enough to throw that in their Way, which a rational Theory would have easily led them to discover.





A

Complete System O F DISTILLATION.

PART II.

*Containing the Method of distilling
Simple Waters.*

CHAP. I.

THE Instruments chiefly used in the Distillation of Simple Waters, are of two Kinds, commonly called the *Hot Still*, or *Alembic*, and the *Cold Still*; the former is represented in *Fig. 5.* and the latter in *Fig. 10.*

The

The Waters drawn by the cold Still from odoriferous Plants are much more fragrant, and more fully impregnated with their Virtues than those drawn by the hot Still, or Alembic; but the Operation is much more slow and tedious by the former than the latter, so that very few care to comply with it: And, therefore a Method has been invented, to avoid the Tediouſness of the one, and the Inconveniencies of the other. The Method is this:

A Pewter Body is suspended in the Body of the Alembic, and the Head of the Still fitted to the Pewter Body: Into this Body the Ingredients to be distilled are put, the Alembic filled with Water, the Still Head luted to the Pewter Body, and the Nose luted into the Worm of the Refrigeratory or Worm.

The same Intention will be answered, by putting the Ingredients into a Glass Alembic, and placing it in a Bath Heat, or *Balneum Mariæ*, as we have before directed, Chap. XI.

By either of these Means, the Ingredients have greater Heat given them than in the cold Still; and yet, by the Interposition of the Water, in which the Vessel, containing

containing them is placed, they are not so forcibly acted upon by the Fire, as in the common Way of the hot Still. So that all those Things which require a middle Way between the other ; that is, those Simples which are of a Texture between very volatile, and very fixed, are treated very properly by this Method ; but neither the very odoriferous Simples, nor those whose Parts are very heavy and fixed, can be treated this Way but to Disadvantage.

One of the greatest Advantages of this Contrivance is, that Waters so drawn come over much cooler than from the hot Still ; that is, they have not so much of the Fire in them, as the Distillers term it ; so that a hot spicy Water, thus ordered, will taste as cool on the Palate when just drawn, as it would, when drawn by the hot Still, after it had acquired a considerable Age.

C H A P. II.

Of Waters drawn by the cold Still.

THE cold Still is much best adapted to draw off the Virtues of Simples, which are valued for their fine Flavour when green, which is subject to be lost in drying. For when we want to extract from Plants a Spirit so light and volatile, as not
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to subsist in open Air any longer than while the Plant continues in its Growth, it is certainly the best Method to remove the Plant from its native Soil, into some proper Instrument, where, as it dries, these volatile Parts can be collected and preserved. And such an Instrument is what we call the cold Still, where the drying of the Plant or Flower, is only forwarded by a moderate Warmth, and all that rises is collected and preserved.

As the Method of performing the Operation by the cold Still, is the very same, whatever Plant or Flower is used, the following Instance of procuring a Water from Rosemary, will be abundantly sufficient to instruct the young Practitioner in the manner of conducting the Process in all Cases whatever.

Take Rosemary, fresh gathered, in its Perfection, with the Morning Dew upon it, and lay it lightly and unbruised upon the Plate, or Bottom of the Still. Cover the Plate with its conical Head, and apply a Glass Receiver to the Nose of it. Make a small Fire of Charcoal under the Plate, continuing it as long as any Liquor comes over into the Receiver. When nothing more comes over, take off the Still Head, and remove the Plant, putting fresh in its
stead,

stead, and proceed as before ; continue to repeat the Operation successively, till a sufficient Quantity of Water is procured. Let this distilled Water be kept at Rest, in clean Bottles close stopped, for some Days in a cold Place ; by this Means it will become limpid, and powerfully impregnated with the Taste and Smell of the Plant.

In this Water are contained the Liquor of Dew, consisting of its own proper Parts, which are not without Difficulty separated from the Plant, and cleave to it even in the drying. This Dew, also, by sticking to the Outside, receives the liquid Parts of the Plant, which being elaborated the Day before, and exhaling in the Night, are hereby detained ; so that they concrete together into one external Liquid, which is often viscid, as appears in Manna, Honey, &c. This Water also contains the Fluid, which exhales from the Vessels of the Rosemary, and which principally consists of simple Water, as appears upon long standing in an open Vessel, when the Taste and Odour vanishing, leave an insipid Water behind. Another Part of this Water is that subtle, volatile Substance, which give the Plant its peculiar Taste and Odour ; for this the Senses discover in it ; but what remains after the Process is finished, scarce afford any thing thereof. The same Water seems also

also to contain Seeds, or other little Bodies, which in a certain Time usually grows into a kind of thin, whitish Weed, suspended in the middle of the Water; and daily increasing or spreading itself, becomes a Mucilage, which did not appear at first.

I have kept these Waters undisturbed in separate well closed Vessels, and observed that in a Year's Time, they began to appear thick, which Thickness gradually increased every Year, till at length the Liquor grew ropy and mucilaginous. Hence we see, that this Water contains the elementary Water, and presiding Spirit of the Plant; a Spirit small in Bulk, but rich in Virtues, and exhibiting the specific Smell and Taste of the Subject. This Water, therefore, in exhaling, proves a Vehicle to that Spirit, which contains in a small, subtile, extremely volatile, and thence easily separable Substance; the particular Virtue of the Plant, leaving the Remainder exhausted in this Respect: and hence proceeds the medicinal Virtues of these Waters, which principally depend upon their native Spirit. For this Spirit, in most Plants, having a brisk Mobility, affects the Nerves, and raises the Spirits in case of their Depression.

If the Vessel be close stopped, and set in a cool Place, the Waters drawn by the cold
Still

still will retain their Virtues for a Year; but if negligently kept, or any Crack should happen in the Glâs, their extremely volatile Spirit secretly flies off, and leaves the Water vapid.

Hence we learn what it is that Plants lose by being dried in the Summer-time; namely, the Water and Spirit we have been describing. Hence we also know the Nature of that Fluid, which first rises from Plants in Distillation, and what that Matter properly is in Plants, that gives their peculiar Odour; that is, their presiding Spirit. Lastly, we hence learn, in some measure at least, what those *Effluvia* are, which principally in the Summer-Season, and in the open Air, exhale from Vegetables; for it is highly probable, that these constant Exhalations of Plants, especially in the Dry-time, have a near Agreement in their peculiar Nature, with the Liquor extracted by the cold Still, though differing in this, that the Exhalation made from the Parts is continually recruited by the Root; whilst by our Operation, those Parts alone are collected, which are driven off from the Plant, after being gathered, and no longer supplied with fresh Nourishment.

C H A P. II.

Of the Distilling Simple Waters by the Alembic.

TH E Plants designed for this Operation are to be gathered when their Leaves are at full Growth, and a little before the Flowers appear, or, at least, before the Seed comes on ; because the Virtue of the Simple expected in these Waters is often little, after the Seed or Fruit is formed, at which Time Plants begin to languish : The Morning is best to gather them in, because the volatile Parts are then condensed by the Coldness of the Night, and kept in by the Tenacity of the Dew, not yet exhaled by the Sun.

This is to be understood, when the Virtue of the distilled Water resides principally in the Leaves of Plants ; as it does in Mint, Marjoram, Pennyroyal, Rue, and many more ; but the Case differs when the aromatic Virtue is only found in the Flowers, as in Roses, Lillies of the Valley, &c. in which Case we choose their flowery Parts, whilst they smell the sweetest, and gather them before they are quite opened, or begin to shed, the morning Dew still hanging on them.

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In other Plants the Seeds are to be preferred, as in Anise, Caraway, Cumin, &c. where the Herb and the Flower are indolent, and the whole resides in the Seed alone, where it manifests itself by its remarkable Fragrance, and aromatic Taste. We find that Seeds are more fully possessed of this Virtue, when they arrive at perfect Maturity.

We must not omit that these desirable Properties are found only in the Roots of certain Plants, as appears in Avens and in Orpine, whose Roots smell like a Rose. Roots of this Kind should be gathered, for the present Purpose, at that Time when they are richest in these Virtues; which is generally at that Season of the Year, just before they begin to sprout, when they are to be dug up in a Morning.

If the Virtues here required be contained in the Barks or Woods of Vegetables, then these Parts must be chosen for the Purpose.

The Subject being chosen, let it be bruised, or cut, if there be occasion, and with it fill two thirds of a Still, leaving a third Part of it empty, without squeezing the Matter close; then pour as much Rain or River Water into the Still as will fill it to

the same Height ; that is, two thirds together with the Plant : Fit on the Head, luting the Juncture, so that no Vapour may pass through ; and also lute the Nose of the Still-head to the Worm. Apply a Receiver to the Bottom of the Worm, that no Vapour may fly off in the Distillation ; but that all the Vapour being condensed in the Worm, by cold Water in the Worm-tub, may be collected in the Receiver.

Let the Plant remain thus in the Still to digest for twenty-four Hours, with a small Degree of Heat. Afterwards raise the Fire, so as to make the Water in the Still boil ; which may be known by a certain hissing Noise, proceeding from the breaking Bubbles of the boiling Matter ; as also by the Pipe of the Still-head, or the upper-end of the Worm, becoming too hot to be handled ; or the smoaking of the Water in the Worm-tub heated by the Top of the Worm ; and, lastly, by the following of one Drop immediately after another, from the Nose of the Worm, so as to form an almost continual Stream. By all these Signs we know that the requisite Heat is given ; if it be less than a gentle Ebullition, the Virtues of the Simple, here expected, will not be raised : On the contrary, when the Fire is too strong, the Water hastily rises into the Still-head, and fouls both the Worm and the

the distilled Liquor ; and the Plant being also raised, it blocks up the Worm ; for which Reason it is no bad Caution to fasten a-piece of fine Linen before the Pipe of the Still-head ; that, in case of this Accident, the Plant may be kept from stopping up the Worm : But, notwithstanding this Precaution, if the Fire be too fierce, the Plant will stop up the Pipe of the Still-head ; and, consequently, the rising Vapour finding no Passage, will blow off the Still-head, and throw the boiling Liquor about the Still-house, so as to do a great deal of Mischief, and even suffocate the Operator, without a proper Caution ; and the more oily, tenacious, gummy, or resinous the Subject is, the greater the Danger, in case of this Accident ; because the Liquor is the more frothy and explosive.

Let the due Degree of Fire therefore be carefully observed, and equally kept up, as long as the Water, distilling into the Receiver, is white, thick, odorous, sapid, frothy and turbid ; for this Water must be carefully kept separate from that which follows it. The Receiver, therefore, should be often changed, that the Operator may be certain that nothing but this first Water comes over ; for there afterwards arises a Water that is transparent, thin, and without the peculiar Taste and Flavour of the

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Plant, but generally somewhat tartarish and limpid, though somewhat obscured and fouled by white dreggy Matter : And if the Head of the Still be of Copper, and not tinned, the Acidity of this last Water corrodes the Copper, so as to become green, nauseous, emetic and poisonous to those who use it, especially to Children, and Persons of weak Constitutions.

The first Water above-described, principally contains the Oil and presiding Spirit of the Plant; for the Fire by boiling the Subject, dissolves its Oil, and reduces it into small Particles, which are carried upwards by the Assistance of the Water, along with those Parts of the Plant that become volatile with their Motion. And, if the Vessels are exactly closed, all these being united together, will be discharged without Loss, and without much Alteration, into the Receiver; and, consequently, furnish us with a Water richly impregnated with the Smell, Taste, and particular Virtues of the volatile Parts of the Plants it was extracted from.

The Water of the second Running, wants the volatile Part above described, and has scarce any other Virtue than that of cooling.

And

And this is the best Method of preparing simple Waters, provided the two sorts be not mixed together, for both of them would be spoiled by such a Mixture.

Hence it plainly appears at what time, with the same Degree of Fire, quite contrary Virtues may arise from a Plant ; for so long as a milk Water continues to come over from such Plants as are aromatic, so long the Water remains warming and attenuating ; but when it comes to be thin and pellucid, it is acid and cooling.

Hence we may also learn the true Foundation for conducting of Distillation ; for if the Operation be stopped, as soon as ever the white Water ceases to come over, the Preparation will be valuable and perfect ; but if, through a Desire of increasing that Quantity, more be drawn off, and the latter acid Part suffered to mix with the first Running, the whole will be spoiled, or at least rendered greatly inferior to what it would otherwise have been.

Such is the general Method of procuring simple Waters, that shall contain the volatile Virtues of the Plants distilled ; some Rules are however necessary to render it

applicable to all Sorts of Plants ; these Rules are the following :

1. Let the aromatic, balsamic, oily, and strong-smelling Plants, which long retain their natural Fragrance, such as Balm, Hyssop, Juniper, Marjoram, Mint, Origanum, Pennyroyal, Rosemary, Lavender, Sage, &c. be gently dried a little in the Shade ; then digest them, in the same manner as already mentioned, for twenty-four Hours, in a close Vessel, with a small Degree of Heat, and afterwards distil in the manner above delivered, and thus they will afford excellent Waters.

2. When Waters are to be drawn from Barks, Roots, Seeds, or Woods that are very dense, ponderous, tough and resinous, let them be digested for three, four, or more Weeks, with a greater Degree of Heat, in a close Vessel, with a proper Quantity of Salt added, to open and prepare them the better for Distillation. The Quantity of Sea-salt is here added, partly to open the Subject the more, but chiefly to prevent Putrefaction, which otherwise would certainly happen in so long a Time, and with such a Heat as is necessary in this Case, and so destroy the Smell, Taste, and Virtues expected from the Process.

3. Those Plants which diffuse their Odour to some Distance from them, and thus soon lose it, should immediately be distilled after being gathered in a proper Season, without any previous Digestion; thus Borage, Bugloss, Jessamin, white Lilies, Lilies of the Valley, Roses, &c. are hurt by Heat, Digestion, or lying in the Air.

• C H A P. III.

Of increasing the Virtues of Simple Waters by means of Cohobation.

BY Cohobation is meant the returning the distilled Water procured in the manner described in the preceding Chapter, upon more of the fresh Plant. The Operation is performed in the following manner :

Take the Plant and Liquor remaining in the Still after the Operation described in the foregoing Chapter is performed; and press them strongly in a Bag for that Purpose, that all the Decoction may be obtained; and with this mix all the Water before drawn over. Return this Mixture into the Still, and a fresh Quantity of the same Plant, and if necessary, as much Water as will make the former Proportion to the

the Plant. Close all the Junctures exactly, and digest the whole in a gentle Degree of Heat for three Days and three Nights, that the Herb, being so long steeped in its own Liquor, may be opened, loosened, and disposed the easier to part with its Virtues. This Digestion is of great Service; but if protracted too long, introduces a Change tending to Putrefaction. Let the Water now be distilled off, in the same manner as before; only proceeding more cautiously, and somewhat more slowly at first; because the Liquor in the Still being now thicker, more impregnated with the Plant, and therefore more apt to smell upon feeling the Fire, it easily boils over; but after about half of the expected Water is come off, the Fire may be gradually raised.

By this Method, and carefully observing to change the Receiver, as soon as the first Water is all come over, a noble Liquor, highly impregnated with the Virtues of the Plant, will be obtained. And as this Operation may be repeated as often as desired, the Virtues of Plants may be thus exalted to any Degree the Artist shall think proper; which shews the extraordinary Power of Distillation. This Method I would particularly recommend for making the simple Water of Balm, Elder Flowers, Roses, and

and the like Simples, but sparingly furnished with an essential Oil.

C H A P. IV.

Of the Method of procuring a simple Water from Vegetables, by previously fermenting the Vegetable before Distillation.

BY this elegant Method we obtain the Virtues of Plants very little altered from what they naturally are, though rendered much more penetrating and volatile. The Operation is performed in the following manner.

Take a sufficient Quantity of any recent Plant, cut it, and bruise it if necessary; put it into a Cask, leaving a Space empty at top of about four Inches deep; then take as much Water as would, when added, fill the Cask to the same Height, including the Plant, and mix therein about an eighth Part of Honey, if it be cold Winter Weather; or a twelfth Part, if it be warm: In the Summer the like Quantity of coarse, unrefined Sugar might be added instead of Honey, or half an Ounce of Yeast to each Pint of Water will have the same Effect; though most prefer Honey for this Purpose. When the proper Quantity of Honey is added to the Water, let it be warmed and
poured

poured into the Cask, and set it in a warm Place to ferment for two or three Days; but the Herb must not be suffered to fall to the Bottom, nor the Fermentation above half finished. The Whole must then be immediately committed to the Still, and the Fire raised by Degrees; for the Liquor, containing much fermenting Spirit, easily rarefies with the Fire, froths, swells, and therefore becomes very subject to boil over; we ought therefore to work slower, especially at first.

By this Method there will come over at first, a limpid, unctuous, penetrating, odorous, sapid Liquor, which is to be kept separate: After this there follows a milky, opaque, turbid Liquor, still containing something of the same Taste and Odour; and at length comes one that is thin, acid, without either Smell, or scarce any Property of the Plant.

The first Water, or rather Spirit, may be kept several Years, in a close Vessel, without changing or growing ropy. It also excellently retains the Taste and Odour of the Plant, though a little altered; but if less Honey were added, less Heat employed, or the Fermentation continued for a smaller Time, the distilled Liquor of the first

first running would be white, thick, opaque, unctuous, frothy, and perfectly retain the Scent and Taste of the Plant, or much less altered than in the former Case ; though the Water will not be so sharp and penetrating. After this is drawn off, a tartish, limpid, inodorous Liquor will come over.

And thus may simple Waters be made fit for long keeping without spoiling ; the Proportion of inflammable Spirit generated in the Fermentation, serving excellently to preserve them.

CHAP. V.

Of the Simple Waters commonly in Use.

SIMPLE Waters are not so much used at present as they were formerly ; and perhaps one Reason for their being neglected, is the bad Methods used in distilling them ; the Process is carried on in the same manner with every Herb ; though some should be gently dried, and others distilled green ; some should be drawn with the cold, and others with the hot Still.

The general Rule that should be observed with regard to the hot Still is, that all Herbs should have twice their Weight of Water added

added to them in the Still ; and not above a fourth, or a sixth Part of it drawn off again ; for simple Waters have their Faints, if drawn too low, as well as those that are spirituous.

Some Plants, particularly Balm, require to have the Water drawn from them cohobated, or poured several times on a fresh Parcel of the Herb, in order to give it a proper Degree of Strength or Richness. Others, on the contrary, abound too much with an essential Oil that floats on the distilled Water ; in this case all the Oil should be carefully taken off. Lastly, those that contain a more fixed Oil, should be imperfectly fermented, in the manner laid down in the preceding Chapter, before they are distilled ; of this Kind are Carduus, Chamomile, &c.

The simple Waters now commonly made, are Orange-flower-water, Rose-water, Cinnamon-water, Fennel-water, Pepper-mint-water, Spear-mint-water, Balm-water, Pennyroyal-water, *Jamaica* Pepper-water, Castor-water, Simple-water of Orange-peel, and of Dill seed.

C H A P. VI.

Of Orange flower-water.

TH E Orange-tree grows plentifully in *Italy, Spain, and Portugal*, and bears Flowers and Fruit all the Year ; but the Fruit is gathered chiefly in *October and November*.

The Flowers grow on the younger Shoots among the Leaves : They are white, and consist of a single cup-fashioned Leaf, cut into five Parts, with several yellow Stamina in the middle, and of a fragrant odoriferous Smell.

Some Degree of Attention is requisite to draw a simple and odoriferous Water for the Orange Flowers ; the Fire must be carefully regulated ; for too small a Degree will not bring over the essential Oil of the Flowers, in which their odoriferous Flavour consists : and, on the contrary, too strong a Fire destroys the Fragrancy of the Water, and is very apt to scorch the Flowers, and give the Water an empyreumatic Smell. Care should also be taken to fasten the Receiver to the End of the Worm with a Bladder, to prevent the volatile Parts from evaporating. The Quantity of Water, also, should be carefully

carefully attended to, if you hope to succeed in the Operation. The following Receipts will answer the Intention.

Receipt for Orange-flower-water.

Take twelve Pounds of Orange-flowers, and twenty-four Quarts of Water, and draw over three Pints.

Or,

Take twelve Pounds of Orange Flowers, and sixteen Quarts of Water; draw over fifteen Quarts, carefully observing what has been observed at the beginning of the Chapter with regard to the Regulation of the Fire.

The Manner of making Double Orange-flower-water, and the essential Oil, or Quintessence of Orange Flowers.

Having shewn how to make simple Orange-flower-water, we shall now shew how to make double Orange-flower-water, and the essential Oil, or Quintessence of Orange Flowers.

Double Orange-flower-water is made, by distilling the Orange Flowers in a cold Still; in the Manner laid down in the first Chapter. The Water extracted in this manner will be very odoriferous and grateful;

ful ; being what is called Double Orange-flower-water. The same odoriferous Water will be obtained by distilling the Flowers in *Balneum Mariæ*, without any Water in the Still. If the cold Still be used put into it as many Flowers as the Head will well cover ; and then make a gentle Fire under the Plate, and as soon as you perceive the Still is beginning to work, fasten the Receiver to the Beak of the Still with a Bladder. The same Caution must be observed if the Flowers are distilled in *Balneum Mariæ*.

To make this Water to Perfection, the Flowers should be fresh gathered in the Morning with the Dew upon them, if possible ; and carefully picked from the Leaves. You should likewise make Choice of the largest Flowers, because these yield most in Distillation. The Fire must be brisk when the Flowers are distilled in *Balneum Mariæ* ; because the Operation is longer in performing than by the common Alembic, and the Flowers are not here in Danger of being burnt at the Bottom of the Cucurbit. If you would have your Water of a fine Smell, let it be cohobated on fresh Flowers.

With this double Water, the essential Oil or Quintessence will come over, and float on the Surface of the Water. But a
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much larger Quantity of it will be obtained, by cohobating the Water on fresh Flowers in *Balneum Mariae*. The essential Oil is at first of a green Colour, but after some Days it will turn reddish. The essential Oil is easily separated from the Water, by the separating Glass, in the following manner : Stop the Spout of the separating Glass with a Cork, and then fill it with the Orange-flower-water ; when it has stood a small time the Oil will float on the Surface. Then pull out the Cork, and let the Water run out at the Spout into another Receiver placed for that Purpose. As the Water runs out at the Spout of the Separating-glass, let it be supplied at the Mouth, that the Separating-glass may be always full of Water, till the whole is in this manner poured into it. Then by gently inclining the Glass, pour out all the Water in it through the Spout, and the Oil will remain in the Separating-glass, and may be poured into another Bottle, and kept separate from the Water. The double Orange-flower water is odoriferous ; but the essential Oil much more so.

Orange-flower-water is not at present so much used as formerly ; but as it is a very odoriferous Water, I thought the Method of making it would be not unacceptable to the young Distiller.

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The essential Oil, or Quintessence of Orange-flowers will make a very grateful Cordial, by mixing it with a clean proof Spirit: The Method of mixing it is this:

Take some fine Loaf Sugar, and drop on it the Quantity of Oil you intend to dissolve in the Spirit, and rub them well together in a Glass Mortar, which is what the Chemists call making an Oleosaccharum. Put this Oleosaccharum into the Spirit; mix them well together, and dulcify it with Sugar to your Taste. If the Spirit be too strong, it may be lowered with Water; but you must observe, that if you add Water enough to bring the Spirit considerably below Proof, it will turn milky; and in order to render it fine, you must filtrate it thro' thick Flannel, or thin Paper. Twenty Drops of the essential Oil will be sufficient for a Pint of Spirit, and the same Proportion to a larger Quantity.

CHAP. VII.

Of Rose-water.

THE Damask Rose is the Species intended to be used in this Operation; it is of a very fragrant Smell, and flowers in *June* and *July*. The Water may
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be made either by the hot Still, the cold Still, or the *Balneum Mariæ*. If the hot Still be used, the Leaves picked from the Stalks must be put into the Still with a sufficient Quantity of Water to prevent an Eum-pyreuma, and the Water drawn off by a gentle Fire. The Receiver must be luted with a Bladder to the Nose of the Worm, to prevent the finest and most volatile Parts from evaporating, which they would otherwise do, to the great Prejudice of the Water.

If the cold Still be used, the Rose Leaves either with the Dew on them, or sprinkled with Water, must be laid on the Iron Plate, and covered with the conical Head. A gentle Fire must then be made under the Plate, and a Receiver luted with a Bladder to the Nose of the Still. The Water will gradually distil into the Receiver, and be strongly impregnated with the odoriferous Parts of the Roses.

The same Method with regard to the *Balneum Mariæ* must be used in the Distillation of Roses as in that of Orange-flowers, and therefore need not be repeated here. We shall therefore only observe, that Rose-water drawn either by the cold Still, or the *Balneum Mariæ*, is much preferable to that drawn by the hot Still.

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The Essence, or essential Oil of Roses is looked upon as one of the most valuable Perfumes in the World; but at the same Time the most difficult to be procured in any Quantity. A small Quantity of it is made in *Italy*, but it has always been thought impossible to procure it here; and, therefore, a Method of acquiring this valuable Commodity will not, I presume, be disagreeable to the Reader.

Take a Quantity of Damask Rose Leaves, put them into a proper Vessel, with a sufficient Quantity of Water, adding some mineral Acid, as Spirit of Salt, Vitriol, &c. In this Menstruum let the Roses be digested for fifteen Days; after which put the Whole into an Alembic, and draw off the Water with a pretty brisk Fire. But, instead of the common Receiver, a Separating-Glass must be placed under the Nose of the Worm, and a Receiver added to the Tube of the Separating-Glass. By this Means all the Oil or Essence will float on the Surface of the Water in the Separating-Glass, and may easily be separated from it, when the Operation is finished.

C H A P. VIII.

Of Cinnamon-water.

Cinnamon is a thin fine Bark, rolled up in a sort of little Pipes, from the thickness of a Goose-quill, to that of a Man's Thumb, and sometimes more, and about two or three Feet long. Its Colour brownish, with a Mixture of red. It is of an extremely aromatic Smell, and of an acrid and pungent, but very agreeable Taste. It is the interior, or second Bark of a Tree that grows plentifully in *Ceylon*. The People who gather it take off the two Barks together, and immediately separating the outer one, which is rough, and has very little Fragrancy, they lay the other to dry in the Shade in an airy Place, where it rolls itself up into the Form wherein we see it.

The greatest Cheats in the Sale of Cinnamon, are the selling such as has already had its essential Oil distilled from it, and dried again, and the imposing Cassia Lignea in its Place. The first of these is discovered by the want of Pungency in the Cinnamon; the second by this, that the Cassia, when held a little Time in the Mouth, becomes mucilaginous, which the true Cinnamon never does. Cinnamon is a noble Drug, ended

duced with many capital Virtues; it strengthens the Viscera, assists Concoction, dispels Flatulencies, and is a pleasant Cardiac.

Recipe for one Gallon of simple Cinnamon-Water.

Take a Pound of the best Cinnamon grossly powdered, digest for twenty-four Hours, in two Gallons of Water; put the Whole into an Alembic, and draw over one Gallon with a pretty brisk Fire.

The Oil of Cinnamon, in which the specific Virtue of the Drug consists, is very ponderous, and therefore will not come over, the Helm unless the Fire be pretty brisk, especially with a simple Water. It will therefore be in vain to attempt distilling simple Cinnamon-water by the *Balneum Mariæ*.

C H A P. IX.

Of Fennel-water.

Fennel-water is extracted from a Seed larger and more beautiful than that produced by our common Fennel; it is called *Sweet Fennel seed*, being of a fragrant Smell, and aromatic sweet Taste, and is cultivated in *France* and *Italy*. It is to be chosen new, large and fair; but when damp or dusty, to be rejected.

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Recipe

Recipe for one Gallon of Fennel-water.

Take one Pound of Sweet Fennel-seeds, and two Gallons of Water ; put them into an Alembic, and draw off one Gallon with a gentle Fire.

C H A P. X.

Of Pepper-mint-water.

PEPPER-mint is a very celebrated Stomachic, and on that account greatly used at present, and its Simple Water often called for.

Recipe for a Gallon of Pepper-mint-water.

Take of the Leaves of dried Pepper-mint, one Pound and a half ; Water two Gallons and a half ; put all into an Alembic, and draw off one Gallon, with a gentle Fire.

The Water obtained from Pepper-mint by Distillation in *Balneum Mariæ*, is more fragrant and more fully impregnated with the Virtues of the Plant than that drawn by the Alembic. The same may be said with regard to that extracted by the cold Still ; when the cold Still is used the Plant must be green, and if possible committed to the still with the Morning Dew upon it.

C H A P.

C H A P. XI.

Of Spear-mint-water.

Spear-mint is also like Pepper-mint a great Stomachic, and therefore constantly used.

Recipe for one Gallon of Spear-mint-water.

Take of the Leaves of dried Spear-mint one Pound and a half; Water two Gallons and a half; draw off by a gentle Fire one Gallon.

This Water, like that drawn from Pepper-mint, will be more fragrant if distilled in *Balneum Mariæ*, or the cold Still; but if the latter be used, the same Caution must be observed of distilling the Plant green.

C H A P. XII.

Of Baum-water.

BAUM is a Plant well known in our Gardens. It flowers in *July*, and is of a fine cordial Flavour; but so weak, that it is soon dissipated and lost; nor is it easy to dry it so as to preserve its natural Scent.

Baum-

Baum-water, therefore should be drawn when the Plant is green; and in order to procure the Water in full Perfection, it should be cohobated, or returned, several times upon fresh Parcels of the Plant; by this means a Water may be procured from Baum extremely rich, and of considerable Use as a Cordial.

If the *Balneum Mariæ* be used, the Water is much better than that drawn by an Alembic. The Water drawn from this Plant by the cold Still will also be very fragrant, and highly impregnated with the Virtues of the Plant.

C H A P. XIII.

Of Penny-royal-water.

Penny-royal, a Plant very common in *England*, is very warm, and its Parts very subtile and penetrating: It is one of the first Plants in Esteem in the present Practice, as well as in former Ages, as an Attenuant and Uterine. It is good in Flatulences and Suppressions of Urine, and by many is greatly recommended in Dropsies, Jaundices, and other chronic Distempers. It communicates its Virtues to Water in Infusion, and its simple Water has, perhaps,
more

more Virtue than any other kept in the Shops. But as it is requisite in order to obtain a Water fully impregnated with the Virtues of Baum, to cohobate it on fresh Parcels of the Plant; the Water drawn from green Penny-royal, on the contrary; generally contains so large a Portion of the essential Oil, that it is necessary to separate what floats on the Surface of the Water, by the Separating-glass.

Recipe for one Gallon of Penny-royal-water.

Take of the dried Leaves of Penny-royal one Pound and a half, of Water three Gallons; draw off one Gallon with a gentle Fire.

The Water drawn from green Penny-royal by the cold Still, is very fragrant, and fully impregnated with the Virtues of the Plant.

C H A P. XIV.

Of Jamaica-Pepper-water.

J*amaica-Pepper*, or Pimento, is the Fruit of a tall Tree growing in the mountainous Parts of *Jamaica*, where it is much cultivated; because of the great Profit arising from the cured Fruit, sent in large Quantities annually into *Europe*.

it

It is gathered, when green, and exposed to the Sun for many Days on Cloths, and frequently shaken and turned, till thoroughly dry; great Care is taken during the Time of drying to defend the Fruit from the Morning and Evening Dews; when thoroughly dried it is sent over to us.

It is a very noble Aromatic, and deserves to be used more frequently than it is at present. The simple Water drawn from it is a better Carminative than any other simple Water at present in use.

Recipe for a Gallon of Jamaica Pepper-water.

Take of Jamaica Pepper half a Pound. Water two Gallons and a half; draw off one Gallon, with a pretty brisk Fire. The Oil of this Fruit is very ponderous, and therefore this Water is best made in an Alembic.

C H A P. XV.

Of Castor-water.

THIS Drug is brought to us in the Pods or Bags that naturally contained it, and these so much resemble the Testicles of

of Distillation.

141.

of an Animal, both in their dry State, and when on the Body of the Creatures, that it is no wonder People who did not examine their Situation on the Animal, really took them for such ; it is, however, a peculiar secreted Matter, contained in Bags destined to receive it.

Castor is an indurated Substance, formed of a Matter once fluid ; the thinner Part of which has been evaporated by drying. It is a light and friable Matter, of a moderately lax Texture, and of a deep dusky brown Colour. It is of a somewhat acrid and bitterish Taste, and of a strong fœtid Smell, which, to many, is very disagreeable.

The Animal that produces the Castor is by all Authors called Castor and Fiber, and by the Vulgar, the Beaver.

The Castor of several Parts of the World differs in Goodness, and in regard to the Care taken in the drying. The *Russian* Castor has long been the most esteemed, and the *New England* kind the least.

Castor-water is of great Use in hysteric Cases, and all Diseases of the Nerves ; in Epilepsies, Palsies, and all Complaints of that kind.

Recipe

Recipe for making one Gallon of Castor-water.

Take of *Russia* Castor an Ounce, of Water three Gallons; draw off one Gallon with a pretty brisk Fire.

C H A P. XVI.

Of Orange-peel-water.

THE Orange is a Fruit too well known to need a Description here. The Water is very grateful to the Taste, and often used in Fevers, &c.

Recipe for one Gallon of Orange-peel-water.

Take of the outward yellow Rind of *Seville* Oranges, four Ounces; Water three Gallons and a half; draw off one Gallon by the Alembic, with a pretty brisk Fire.

C H A P. XVII.

Of the Water of Dill-seed.

DILL greatly resembles Fennel both in Root, Stalk, or Leaf, but rarely grows so tall, or is so much branched; it bears the same kind of yellow Umbels of Flowers, after which come Seeds rounder, broader, and

and flatter than those of Fennel. The whole Plant is of a strong Scent, less pleasant than Fennel. It grows in Gardens, and flowers and seeds in *July* and *August*. The Water drawn from the Seeds is heating and carminative, good in Cholics, and all Disorders arising from Wind.

Recipe for making a Gallon of the Water of Dill-seed.

Take of Dill-seed one Pound, Water three Gallons; distil off by the Alembic one Gallon, with a pretty brisk Fire.

The Waters we have enumerated in this Part are those now commonly in use; though there are many other Herbs, from whence Waters of great Use may be drawn; but as the Method of Distillation is the same in all, it would be of no Use to extend these Instructions to a greater length; we shall therefore only observe, that when unfavourable Seasons have prevented the Herbs from attaining a proper Degree of Perfection, it will be necessary to increase their Proportion in extracting the several Waters ordered to be drawn by the Alembic.

A



A
Complete System
O F
DISTILLATION.

P A R T III.

*Of making compound Waters and
Cordials.*

THE Perfection of this grand Branch
of Distillery depends upon the Ob-
servation of the following general Rules;
easy to be observed and practised.

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I.

1. The Artist must always be careful to use a well cleansed Spirit, or one freed from its own essential Oil, as were before observed, Part I. Chap. xxiii. For as a compound Water is nothing more than a Spirit impregnated with the essential Oil of the Ingredients, it is necessary that the Spirit should have deposited its own.

2. Let the Time of previous Digestion be proportioned to the Tenacity of the Ingredients, or the Ponderosity of their Oil. Thus Cloves and Cinnamon require a longer Digestion before they are distilled than Calamus Aromaticus or Orange-peel. Sometimes Cohobation (explained in Part II. Chap. iii.) is necessary; for Instance, in making the strong Cinnamon-water; because the essential Oil of Cinnamon is so extremely ponderous, that it is difficult to bring it over the Helm with the Spirit without Cohobation.

3. Let the Strength of the Fire be proportioned to the Ponderosity of the Oil intended to be raised with the Spirit. Thus, for Instance, the strong Cinnamon Water requires a much greater Degree of Fire than that from lax Vegetables, as Mint, Baum, &c.

4. Let only a due Proportion of the finest Parts of the essential Oil be united with the Spirit; the grosser and less fragrant Parts of the Oil not giving the Spirit so agreeable a Flavour, and at the same Time renders it thick and unsightly. This may in a great Measure be effected by leaving out the Faints, and making up to Proof with fine soft Water in their stead.

These four Rules carefully observed will render this extensive Part of Distillation far more perfect than it is at present. Nor will there be any Occasion for the use of burnt Allum, White of Eggs, Isinglass, &c. to fine down Cordial Waters; for they will presently be fine, sweet and pleasant tasted; without any farther Trouble.

CHAP. I.

Of Strong Cinnamon Water.

WE have already (Chap. viii. Part II.) described this Drug, and given some Directions for chusing the best Sort, to which the Reader is referred.

*Capsicums are in general Recipe
used in Cinnamon Water & give a
much finer flavour —*

Recipe for sixteen Gallons of strong Cinnamon Water.

Take eight Pounds of fine Cinnamon bruised, seventeen Gallons of clean rectified Spirit, and two Gallons of Water. Put them into your Still, and digest them twenty-four Hours with a gentle Heat; after which draw off sixteen Gallons by a pretty strong Heat.

I have ordered a much larger Quantity of Cinnamon than is common among Distillers, because when made in the Manner above directed, it is justly looked upon as one of the noblest Cordial Waters of the Shops; but when made in the common Way of two Pounds to twenty Gallons of Spirit, as some have ordered, is only an Imposition on the Buyer. Some also, to render the Goods cheaper, use equal Quantities of Cinnamon and Cassia Lignea; but by this means the Cordial is rendered much worse; and, therefore, if you desire a fine Cinnamon-water the above Recipe will answer your Intention: But if a cheaper Sort be desired you may lessen the Quantity of Cinnamon, and add Cassia Lignea in its stead. If you would dulcify your Cinnamon-water, take double refined Sugar, what Quantity you please, the general Proportion

portion is, about two Pounds to a Gallon, and dissolve it in the Spirit after you have made it up Proof with clean Water. One general Caution is here necessary to be added; namely, that near the End of the Operation you carefully watch the Spirit as it runs into the Receiver, in order to prevent the Faints mixing with the Goods. This you may discover by often catching some of it, as it runs from the Worm in a Glass, and observing whether it is fine and transparent; for as soon as ever the Faints begin to rise, the Spirit will have an azure or bluish Cast. As soon therefore as you perceive this Alteration, change the Receiver immediately; for if you suffer the Faints to mix with your other Goods, the Value of the whole will be greatly lessened. With regard to the Faints, they are to be kept by themselves, and poured into the Still when a fresh Parcel of the same Goods is to be made.

It is also necessary to observe here, once for all, that the Distillers call all Goods made up Proof, *double Goods*; and those which are below Proof, *single*. This Observation will be alone sufficient to instruct the young Distiller, how he may at any Time turn his Proof or double Goods into single.

C H A P. II.

Of Clove Water.

CLOVES, from whence this Water takes its Name, are the Fruit of a Tree growing in the *Molucca* Islands. The Figure of this Fruit is oblong, and not very thick, resembling in some measure, a Nail. The Surface of it is rough, and the Colour a dusky brown with an admixture of reddish. The whole Fruit is of an extremely fragrant Smell, and of an acrid, pungent, and very aromatic Taste. Cloves are to be chosen the largest, fairest, darkest coloured, the heaviest and most unctuous on the Surface, when pressed between the Fingers. Cloves are carminative, and good against all Distempers of the Head arising from cold Causes. They strengthen the Sight, and are good against Faintings, Palpitations of the Heart, and Crudities in the Stomach.

Recipe for fifteen Gallons of Clove Water.

Take of Cloves bruised four Pounds, Pimento or All-spice half a Pound, clean Proof of Spirit sixteen Gallons; let it digest twelve Hours in a gentle Heat, and then draw

draw off fifteen Gallons with a pretty brisk Fire.

Or,

Take *Winter's Bark*, four Pounds, *Pimento* six Ounces, Cloves one Pound and a Quarter; clean Proof Spirits sixteen Gallons: Digest, and draw off as before.

The *Winter's Bark*, added in the second Recipe is the Bark of a large Tree, growing in several Parts of *America*, and has its name from its Discoverer Captain *Winter*. The outer Rind of it is of an uneven Surface, and of a loose Texture, very brittle, and easily powdered. The inner Part, in which the principal Virtue resides, is hard, and of a dusky reddish brown Colour. It is of an extremely fragrant and aromatic Smell, and of a sharp, pungent, and spicy Taste, much hotter than Cinnamon in the Mouth, and leaving in it a more lasting Flavour. It is to be chosen in Pieces not too large, having the inner or brown Part firm and sound, and of a very pungent Taste. It is apt to be worm-eaten; but in that Case it should be wholly rejected, as having lost the most essential Part of its Virtue.

If you desire to have your Clove Water red, it may be coloured either by a strong Tincture of Cochineal, Alkanet-root, or

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Corn-

Corn-poppy-flowers. The first gives the most elegant Colour, but it is not often used on Account of its Dearness.

You may dulcify it to your Palate, by dissolving in it double refined Sugar. Some for Cheapness use a coarser Kind of Sugar; but this renders the Goods foul and unsightly. Some also, to save Expences, make what they call Clove Water, with Cloves and Caraway-seeds; the Proportion they generally use is half an Ounce of Cloves, and two Drams of Carraway-seeds to a Gallon of Spirit.

C H A P. III.

Of Lemon Water.

THE Peel of the Lemon, the Part used in making this Water, is a very grateful bitter Aromatic, and on that Account very serviceable in repairing and strengthening the Stomach.

Recipe for ten Gallons of Lemon Water.

Take of dried Lemon-peel four Pounds, clean Proof Spirit ten Gallons and a Half, and one Gallon of Water. Draw off ten Gallons by a gentle Fire. Some dulcify
Lemon-

Lemon-water, but by that means its Virtues as a Stomachic, are greatly impaired,

CHAP. IV.

Of Hungary Water.

ROSEMARY, the principal Ingredient in Hungary Water, has always been a favourite Shrub in Medicine; it is full of volatile Parts, as appears by its Taste and Smell. It is a very valuable Cephalic, and is good in all Disorders of the Nerves; in Hysteric and Hypocodriac Cases, in Palsies, Apoplexies, and Vertigoes. Some suppose that the Flowers possess the Virtues of the whole Plant in a more exalted Degree than any other Part; but the flowery Tops, Leaves, and Husks, together with the Flowers themselves, are much fitter for all Purposes, than the Flowers alone.

Recipe for ten Gallons of Hungary Water.

Take of the flowery Tops, with the Leaves and Flowers of Rosemary fourteen Pounds, rectified Spirit eleven Gallons and a Half, Water one Gallon, distil off ten Gallons with a moderate Fire. If you perform this Operation in Balneum Mariæ, your Hungary Water will be much finer, than if drawn by the common Alembic.

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This is called Hungary Water, from its being first made for a Princess of that Kingdom. Some add Lavender-flowers, and others Florentine-orice-root; but what is most esteemed is made with Rosemary only.

C H A P. V.

Of Lavender Water.

TH E R E are two Sorts of Lavender Water, the Simple and Compound; the first is much used externally on Account of its Fragrancy, and cephalic Virtues; the latter internally in a great Number of Disorders.

Recipe for ten Gallons of simple Lavender Water.

Take fourteen Pounds of Lavender-flowers, ten Gallons and a Half of rectified Spirit of Wine, and one Gallon of Water; draw off ten Gallons with a gentle Fire; or, which is much better, in Balneum Mariæ.

Both the Hungary and Lavender Water, may be made at any Time of the Year without Distillation, by mixing the Oil of the Plant with highly rectified Spirit of Wine.

Wine. In order to this, when the Plant is in Perfection, you should distill a large Quantity of it in Water with a very brisk Fire; placing under the Nose of the Worm the Separating-glass (described Page 31. Part I. of this Treatise) by which means you will obtain the essential Oil of the Plant, in which both its Fragrancy and Virtues reside. Having procured the essential Oil of the Plant, the Water may readily be made in the following Manner. Put the rectified Spirit into the Receiver (described Page 32. Fig. xii.) and let an Assistant shake it with a quick Motion; whilst the Spirit is thus agitated, drop in leisurely the essential Oil, and it will mix without any Foulness or Milkiness. The Oils of Lavender and Rosemary are imported cheaper from abroad, than they can be made here; but these Oils will not mix with the Spirit, without rendering it foul and milky; and therefore if you propose making Hungary or Lavender Water in this Manner; it will be necessary to extract the Oil yourself.

Recipe for making three Gallons of compound Lavender Water.

Take of Lavender Water above described two Gallons, of Hungary Water one Gallon, Cinnamon and Nutmegs of each three Ounces, and of red Saunders one Ounce;
digest

digest the whole three Days in a gentle Heat, and then filter it for use. Some add Saffron, Musk, and Ambergrease of each half a Scruple ; but these are now generally omitted.

This compound Lavender Water has been long celebrated in all nervous Cases. In all Kinds of Palsies, and Loss of Memory it is of the greatest Service ; and has been so much remarked for its Efficacy in these Complaints, as almost universally to obtain the Name of *Palsy Drops*.

C H A P. VI.

Of Citron Water.

THE Citron is an agreeable Fruit resembling a Lemon in Colour, Smell and Taste. The Inside is white, fleshy and thick, containing but a small Quantity of Pulp, in Proportion to the Bigness of the Fruit.

Recipe for making ten Gallons of Citron Water.

Take of dry yellow Rinds of Citron four Pounds, clean Proof Spirit ten Gallons and a Half, Water one Gallon, digest the whole twenty-four Hours with a gently Heat ;
draw

draw off ten Gallons, with a gentle Fire ;
or, which is much better, in Balneum
Mariæ, and dulcify it with fine Sugar to
your Palate.

Or,

Take of dry yellow Rinds of Citrons
three Pounds, of Orange-peel two Pounds,
Nutmegs bruised three Quarters of a Pound ;
digest, draw off, and dulcify as before.

This is one of the most pleasant Cor-
dials we have ; and the Addition of the
Nutmegs, in the second Receipt, increases
its Virtue as a Cephalic and Stomachic.

C H A P. VII.

Of Aniseed Water.

ANISEED is a small Seed of an ob-
long Shape, each way ending in an ob-
tuse Point ; its Surface is very deeply striated,
and it is of a soft and lax Substance, very
light and easily broken. Its colour is a
Kind of pale olive, or greenish grey ; it
has a very strong and aromatic Smell, and
a sweetish but acrid Taste, but in the whole
not disagreeable. Aniseed should be
chosen large, fair, new, and clean, of a
good Smell, and acrid Taste. The Plant
that produces it is cultivated in many Parts
of *France* ; but the finest Seed comes from
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the Island of *Malta*, where it is raised for Sale, and whence a great Part of *Europe* is supplied.

Recipe for ten Gallons of Aniseed Water.

Take of Aniseed bruised two Pounds, Proof Spirits twelve Gallons and a Half, Water one Gallon; draw off ten Gallons with a moderate Fire.

Or,

Take of the Seeds of Anise and Angelica, each two Pounds, Proof Spirits twelve Gallons and a half; draw off as before.

Aniseed Water should never be reduced below Proof, because of the large Quantity of Oil with which the Spirit is impregnated, and which will render the Goods milky and foul, when brought down below Proof; but if there be a necessity for doing this, the Goods must be filtrated either thro' Paper or the filtrating Bag, which will restore their Transparency.

Aniseed Water is a good Carminative, and therefore in great Request among the common People against the Cholic.

CHAP.

C H A P. VIII.

Of Caraway Water.

CARAWAY-SEED is of an oblong and slender Figure, pointed at both Ends, and thickest in the Middle. It is striated on the Surface, considerably heavy, of a deep brown Colour, and somewhat bright or glossy. It is of a very penetrating Smell, not disagreeable, and of a hot, acrid and bitterish Taste. Caraway-seed is to be chosen large, new and of a good Colour, not dusty, and of an agreeable Smell. The Plant which produces the Caraway-seed grows wild in the Meadows of *France* and *Italy*, and in many other Places; but is sown in Fields for the sake of the Seeds in *Germany*, and many other Parts of *Europe*.

Recipe for making ten Gallons of Caraway Water.

Take of Caraway-seeds bruised three Pounds, Proof Spirit twelve Gallons, Water two Gallons; draw off ten Gallons, or till the Faints begin to rise; make the Goods up with clean Water, and dulcify with common Sugar to your Taste.

Or,

Or,

Take of Caraway-seed bruised two Pounds and a Half, Orange or Lemon-peel dried one Pound, Proof Spirit twelve Gallons, Water two Gallons ; draw off and dulcify as before.

Caraway Water, like that of Aniseed, is a good Carminative ; but not so much used, tho' much pleasanter.

C H A P. IX.

Of Cardamom Seed Water.

THE Seed from whence this Water takes its Name, is called by Botanists *Cardamum Minus*, or the lesser Cardamom ; to distinguish it from the *Cardamomum Majus*, or Grains of Paradise.

The lesser Cardamom is a small short Fruit, or membranaceous Capsule, of a trigonal Form, about a third of an Inch long, and swelling out thick about the Middle ; beginning small and narrow from the Stalk, and terminating in a small, but obtuse Point at the End. It is striated all over very deeply with longitudinal Furrows, and consists of a thin but very tough Membrane, of a fibrous Texture, and pale-brown

brown Colour, with a faint Cast of red. When the Fruit is thoroughly ripe, this Membrane opens at the three Edges all the way, and shews that it is internally divided by three thin Membranes into three Cells, in each of which is an Arrangement of Seeds, separately lodged in two Series. The Seeds are of an irregular angular Figure, rough, and of a dusky brown Colour on the Surface, with a Mixture of yellowish and reddish, and of white Colour within. They have not much Smell, unless first bruised, when they are much like Camphire under the Nose. They are of an acrid, aromatic and firey hot Taste. They should be chosen sound, close shut on all Sides, and full of Seeds, of a good Smell, and of an acrid aromatic Taste.

Recipe for ten Gallons of Cardamom Seed Water.

Take of the lesser Cardamom Seeds husked two Pounds and a Half, of clean Proof Spirit ten Gallons and a Half, and of Water one Gallon; draw off ten Gallons by a gentle Heat. You may either dulcify it or not with fine Sugar at pleasure.

This Water is Carminative, assists Digestion, and good to strengthen the Head and Stomach.

C H A P. X.

Of Aqua Mirabilis; or, the Wonderful-Water.

MOST of the Ingredients in this Composition have already been described, and an Account of the Nutmegs will be given in Chap. xxv. But the Cubebs and Ginger remain to be mentioned.

Cubebs are small dried Fruit resembling a Pepper-corn, but often somewhat longer; of a dark brown Colour, composed of a wrinkled external Bark; of an aromatic, tho' not very strong Smell, and of an acrid and pungent Taste, tho' less so than Pepper; but its Acrimony continues long on the Tongue, and draws forth a large Quantity of Saliva. We have two Kinds of Cubebs, which differ only in their Periods of gathering, both are produced from the same Plant. The unripe Cubebs are small, very wrinkled on the Surface, and their Nucleus, when broken, is flacid: But the ripe ones not so. Cubebs are brought from the Island *Java*, where they grow in great Abundance. They should be chosen large, fresh, and sound, and the heaviest possible. They are warm and carminative, and esteemed

steemed good in Vertigoes, Palsies, and Disorders of the Stomach.

Ginger is a Root too well known to need a long Description ; it is sufficient to observe that it is of a pale yellowish Colour when broken, of a fibrous Structure, and easily beat into a Sort of woolly or long thready Matter. It is of very hot, acrid, and very pungent Taste ; but Aromatic withal, and of a very agreeable Smell. We have it both from the *East* and *West Indies* ; but the oriental is much superior to the occidental in its Flavour, of a firmer Substance, and does not beat out so much into Threads. Ginger is an excellent ~~Carmenative~~ and Stomachic ; it assists Digestion, dispells Flatus's, and takes off Cholic Pains almost instantaneously.

There are several Receipts for making this celebrated Cordial ; but the following are allowed to be the best.

Take of Cinnamon one Pound and a Quarter, Rind of Lemon-peels ten Ounces, Cubebs one Ounce and a Quarter, Leaves of Baum one Pound ; bruise all these Ingredients, and pour on them eleven Gallons of clean Proof Spirit, and one Gallon of Water ; digest the whole twenty-four Hours, and distil off ten Gallons with a

M 2

pretty

pretty brisk Fire ; and dulcify it with fine Sugar.

Or,

Take of the lesser Cardamoms, Cloves, Cubebs, Galangal, Mace, Nutmeg, and Ginger, of each one Pound and three Quarters, of the yellow part of Citron-peel and Cinnamon of each three Pounds and a Half, of the Leaves of Balm one Pound ; bruise these Ingredients, and pour on them eleven Gallons of Spirit and one Gallon of Water ; digest, and draw off, &c. as before.

This Cordial has been long celebrated as a noble Stomachic, and therefore greatly called for.

Some instead of all the Ingredients enumerated in the above Receipts use only Pimento ; and this is the sort of Aqua Mirabilis which some sell so very cheap.

C H A P. XI.

Of Mint Water.

THE Mint intended in this Recipe is the common Spear-mint, an Account of which has already been given. Page 137.

Recipe

Recipe for ten Gallons of Mint Water.

Take of dry Spear-mint Leaves fourteen Pounds, Proof Spirit ten Gallons and a Half, Water two Gallons; draw off ten Gallons by a gentle Heat. You may dulcify it with Sugar if required.

Mint Water is greatly recommended by the learned *Boerhaave* and *Hoffman*, against Vomitting, Nauseas, and the Cholic.

C H A P. XII.

Of Pepper Mint Water.

THE Pepper-mint has been already described Page 136, to which the Reader is referred.

Recipe for ten Gallons of Pepper Mint Water.

Take of dry Pepper Mint Leaves fourteen Pounds, Proof Spirit ten Gallons and a Half, Water one Gallon; draw off ten Gallons by a gentle Fire. You may either dulcify it or not.

Pepper Mint Water is a noble Stomachic, good against Vomiting, Nauseas, Cholic,
M 3 and

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and other griping Pains in the Bowels, in all which Intentions it greatly exceeds the common Spear Mint Water.

C H A P. XIII.

Of Angelica Water.

TH E R E are two Sorts of Angelica Water, the Single and the Compound. I shall give Receipts for making both Kinds; and with regard to the Nature of Angelica, it is sufficient to observe, that it is an excellent Carminative.

Recipe for ten Gallons of single Angelica Water.

Take of the Roots and Seed of Angelica cut and bruised, of each one Pound and a Half, Proof Spirit eleven Gallons, Water two Gallons; draw off ten Gallons, or till the Faints begin to rise, with a gentle Fire; and dulcify it, if required, with lump Sugar.

This Angelica Water is a good Carminative, and therefore good against all Kinds of flatulent Cholics, and Gripings of the Bowels.

Recipe

Recipe for ten Gallons of Compound Angelica Water.

Take of the Roots and Seeds of Angelica, and of sweet Fennel-seeds of each one Pound and a Half, of the dried Leaves of Baum and Sage of each one Pound; slice the Roots and bruise the Seeds and Herbs, and add to them of Cinnamon one Ounce, of Cloves, Cubebs, Galangals, and Mace, of each three Quarters of an Ounce, of Nutmegs, the lesser Cardamom-seed, Pimento, and Saffron of each half an Ounce; infuse all these in twelve Gallons of clean Proof Spirit, and draw off ten Gallons, with a pretty brisk Fire. It may be dulcified or not at pleasure.

This is an excellent Composition, and a powerful Carminative; and good in all flatulent Cholics, and other griping Pains in the Bowels. It is also good in Nauseas, and other Disorders of the Stomach.

It may not be amiss to observe here, that in distilling this and several other Compositions, abounding with oily Seeds, the Operator should be careful not to let the Faints mix with the other Goods, as they would by that means be rendered nauseous and unsightly; he should therefore be care-

ful towards the latter End of the Operation, to catch some of the Spirit as it runs from the Worm in a Glass; and as soon as ever he perceives it the least cloudy, to remove the Receiver, and draw the Faints by themselves.

C H A P. XIV.

Of Orange Water.

THIS Water is made in the same Manner from the Peels of Oranges, as Citron Water; Chap. vi. is from the Peels of Citrons.

Recipe for ten Gallons of Orange Water.

Take of the yellow Part of fresh Orange-peels five Pounds, clean Proof Spirit ten Gallons and a Half, Water two Gallons; draw off ten Gallons with a gentle Fire.

This is a good Stomachic, and may also be used for making bitter Tinctures, as that called *Stoughton's Drops*.

CHAP.

C H A P. XV.

Of Plague Water.

TH E R E are several Receipts for making Plague Water ; but the following are much the best.

Recipe for ten Gallons of Plague Water.

Take of the Roots of Masterwort and Butter-burr, of each one Pound and a Quarter, *Virginia* Snake-root and Zeadory, of each ten Ounces, Angelica-seeds and Bay-seeds of each fourteen Ounces, and of the Leaves of Scordium one Pound and a Half. Cut or bruise these Ingredients and put them into the Still, with twelve Gallons of clean Proof Spirit, and two Gallons of Water ; digest the whole for twenty-four Hours, and draw off ten Gallons.

Or,

Take of the Leaves of Celandine, Rosemary, Rue, Sage, Roman Wormwood, Dragons Agrimony, Baum, Scordium, the lesser Centory, *Carduus Benedictus*, Betony, and Mint, of each twenty Handfuls ; of dried Angelica-root, Zeadory and Gentian, of each ten Ounces, and of *Virginia* Snake-root five Ounces ; digest these twenty-four Hours, in twelve Gallons of clean Proof Spirit,

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Spirit, and two Gallons of Water ; and then draw off ten Gallons as before.

Or,

Take of Rue, Rosemary, Baum, Carduus Benedictus, Scordium, Marigold-flowers, Dragons, Goat's-rue, and Mint, of each ten Handfuls ; Roots of Masterwort, Angelica, Butter-burr, and Peony, of each one Pound and a Quarter ; and of Viper-grass ten Ounces ; digest in twelve Gallons of Spirit, &c. as before.

Or,

Take of the Roots of Masterwort, Gentian, and Snake-root, of each seven Ounces ; green Walnuts bruised eighty ; Venice-treacle and Mithridate of each three Ounces ; Camphire six Drams ; of the Roots of Rue and Elecampane, of each three Ounces ; Hore-hound six Ounces ; Saffron six Drams ; Proof Spirit twelve Gallons ; digest, &c. as before.

Or,

Take Dragons, Rosemary, Wormwood, Sage, Scordium, Mugwort, Scabious, Baum, Carduus, Angelica, Marigold-flowers, Centory, Betony, Pimpernel, Celadine, Rue, and Agrimony, of each three Pounds ; of the Roots of Gentian, Zeadory, Liquorice, and Elecampane, of each twelve Ounces ; twelve Gallons of Spirits ; digest, &c. as before.

Or,

Or,

Take of green Walnuts, five Pounds ; of Angelica-root two Pounds ; of the Leaves of Angelica, Rue, Sage, and Scordium, of each ten Handfuls ; of Nutmegs, Long Pepper, Ginger, Camphire, and Gentian-root, of each five Ounces ; of Snake-root, Contrayerva, Elecampane, Zeadory, and Viper's Flesh, of each thirteen Ounces ; Venice Treacle and Mithridate of each thirteen Ounces ; White-wine Vinegar seven Pounds ; Proof Spirits twelve Gallons ; digest, &c. as before.

You may either dulcify your Plague-water, or not, as you see occasion.

All the above Receipts for making Plague Water are in use ; but the first the most elegant, containing nothing but what is proper in the Intention, and at the same time adapted to give its Virtues by Distillation ; which cannot be said of any of the rest, several of the Ingredients adding no Virtue at all to the Water. Of this Kind are the Celandine, Carduus, Centory, Gentian, Walnuts, &c.

Plague Water is a noble Alexipharmic, and a high Carminative Cordial in malignant Cases, and of great Use in Lowness of Spirits, and Depressions.

CHAP.

C H A P. XVI.

Of Dr. Stephens's Water.

THIS Water has its Name from its Inventor, a Physician of great Learning and Practice.

Recipe for ten Gallons of Dr. Stephens's Water.

Take of Cinnamon, Ginger, Galangal, Cloves, Nutmegs, Grains of Paradise, the Seeds of Anise, sweet Fennel, and Caraway, of each one Ounce ; of the Leaves of Thyme, Mother of Thyme, Mint, Sage, Penniroyal, Rosemary, Flowers of red Roses, Camomile, Origanum, and Lavender, of each eight Handfuls ; of clean Proof Spirit twelve Gallons, Water two Gallons ; digest all twenty-four Hours, and then draw off ten Gallons, or till the Faints begin to rise. Dulcify with fine Sugar to your Palate.

This is a noble Cephalic Cordial and Carminative ; and also in some Degree an Hysteric ; good in all cholic Pains in the Stomach and Bowels, and Diseases of the Nerves.

CHAP.

C H A P. XVII.

Of Surfeit Water.

T H E R E are two Kinds of Surfeit-water, one made by Distillation, and the other by Infusion, the former is generally called white *Surfeit Water*, and latter red *Surfeit Water*.

Recipe for ten Gallons of white Surfeit Water.

Take Marigold-flowers, Mint, Centory, Rosemary, Scordium, Mugwort, Carduus, Rue, St. John's-wort, Baum, and Dragons, of each seven Handfuls; of the Roots of Peony, Viper-grass, Butter-burr, and Angelica, of each one Pound and a Half; of Galangal, Calamus Aromaticus, and of the Seeds of Angelica and Caraway, of each four Ounces; of the Flowers of red Poppies ten Handfuls; Proof Spirit twelve Gallons, Water two Gallons; digest for twenty-four Hours, and then draw off ten Gallons, or till the Faints begin to rise; and dulcify with fine Sugar.

This is a good Cordial, but would not be the worse, if the Carduus, Mugwort, Rue and St. John's-wort, were omitted, as little
of

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of their Virtues can be obtained by Distillation. It is, however, a good Alexipharmic, Carminative, and Stomachic; and therefore good in all flatulent Pains in the Stomach and Bowels, in Nauseas and Surfeits, from whence it had its Name.

Recipe for making ten Gallons of red Surfeit Water.

Take of the Flowers of red Poppies, two Bushels, eleven Gallons of clean Proof Spirit, and digest them with a gentle Heat for three Days, or till the Spirit has extracted all the Colour of the Flowers: Then press out the Liquor from the Flowers, and add to the Tincture of the Seeds of Caraway and Coriander, and Liquorice-root sliced, of each ten Ounces; of Cardamoms and Cubebs of each four Ounces; of Raisins stoned five Pounds; of Cinnamon five Ounces; of Nutmegs, Mace, and Ginger, of each three Ounces; of Cloves two Drams; of Juniper-berries three Ounces; let the whole be digested three Days, then press-out the Liquor adding to it a Gallon of Rose-water; and then strain or filter the whole through a Flannel Bag.

This Water is much Superior to the preceding, as all the Ingredients will give their Virtues to the Tincture, tho' they will not rise in Distillation. It is a noble Alexipharmic,

pharmic, it strengthens the Stomach, and greatly assists Digestion ; it is also an excellent Carminative and good against the Cholic and Gripes : Its Cordial Virtues renders it serviceable in all Tremblings of the Nerves, and Depressions of the Spirits.

C H A P. XVIII.

Of Wormwood Water.

THERE are two Sorts of Wormwood Water, distinguished by the Epithets of *greater* and *lesser*.

Recipe for making ten Gallons of the lesser Composition of Wormwood Water.

Take of the Leaves of dried Wormwood five Pounds ; of the lesser Cardamom-seeds five Ounces ; of Coriander-seeds one Pound ; of clean Proof Spirit eleven Gallons ; Water one Gallon ; draw off ten Gallons, or till the Faints begin to rise, with a gentle Fire. It may be dulcified with Sugar, or not, at pleasure.

This is a good Stomachic and Carminative ; and on that Account often called for.

Recipe

Recipe for ten Gallons of the greater Composition of Wormwood Water.

Take of the common and Sea Wormwood, dried, of each ten Pounds; of Sage, Mint, and Baum dried, of each twenty Handfuls; of the Roots of Galangal, Ginger, Calamus Aromaticus, and Elecampane; of the Seed of sweet Fennel and Coriander, of each three Ounces; of Cinnamon, Cloves, and Nutmegs, the lesser Cardamoms and Cubebs, of each two Ounces. Cut and bruise the Ingredients as they require; digest them twenty-four Hours, in eleven Gallons of fine Proof Spirit, and two Gallons of Water, and draw off ten Gallons, or till the Faints begin to rise, with a pretty brisk Fire.

This is an excellent Composition, and good in all Diseases of the Stomach arising either from Wind or a bad Digestion. It is greatly in use in some Parts of *England*, but comes too dear for the common Sort of People; on which account a Cordial Water is often sold under the Title of *the greater Composition of Wormwood Water*; drawn from the Leaves of Wormwood, Orange and Lemon-peel, Calamus Aromaticus, Pimento, and the Seeds of Anise and Caraway; which being all cheap Ingredients,
the

the Composition may be sold at a moderate Price. A Water drawn in this manner is a good Carminative ; but far inferior to that made by the above Recipe.

C H A P. XIX.

Of Antiscorbutic Water.

THE Scurvy being a Disease very common in *England*, this Antiscorbutic Water will be of great use.

Recipe for making ten Gallons of Antiscorbutic Water.

Take of the Leaves of Water-creffes, Garden and Sea Scurvy-grass, and Brooklime of each twenty Handfuls ; of Pine-tops, Germander, Horehound, and the lesser Centory, of each sixteen Handfuls ; of the Roots of Briony and sharp-pointed Dock, of each six Pounds ; of Mustard-seed one Pound and a Half. Digest the whole in ten Gallons of Proof Spirit, and two Gallons of Water, and draw of by a gentle Fire.

This is a good Water for the Purposes expressed in the Title, *viz.* against Scorbutic Disorders. It is also good in Tremblings and Disorders of the Nerves.

N

CHAP.

C H A P. XX.

Of compound Horfe-radish Water.

THERE are several Methods of making this compound Water ; but the three following Recipes are the best that has hitherto appeared.

Recipe for making ten Gallons of compound Horfe-radish Water.

Take of the Leaves of fresh Garden Scurvy-grass sixteen Pounds ; of fresh Horfe-radish Root, and the yellow Part of Seville Orange-peel of each eight Pounds ; of Nutmegs two Pounds. Cut and bruise these Ingredients, and digest them twenty-four Hours in ten Gallons of Proof Spirit and two Gallons of Water ; after which draw off ten Gallons with a gentle Fire.

Or,

Take of the fresh Roots of Horfe-radish nine Pounds ; of the Leaves of Water-creffes and of Garden Scurvy-grass, of each six Pounds ; of the outward, or yellow Peel of Oranges and Lemons, of each nine Ounces ; of *Winter's* Bark twelve Ounces ; of Nutmegs three Ounces. Cut, bruise and digest the Ingredients in ten Gallons of Proof

Proof Spirit, and two Gallons of Water, and draw off ten Gallons as before.

Or,

Take of the Leaves of Garden and Sea Scurvy-grass fresh gathered in the Spring, of each seven Pounds; Brook-lime, Water-cresses, and Horse-radish-root of each ten Pounds; of *Winter's* Bark and Nutmegs of each ten Ounces; of the outer Peel of Lemons one Pound; of Arum-root fresh gathered two Pounds; Proof Spirit ten Gallons, Water two Gallons. Bruise and slice the Ingredients; digest the whole, and draw off ten Gallons as before.

Either of the above Receipts will produce an excellent Water, against all Obstructions of the Kidnies and other Viscera. It is also of great Service in the Jaundice, Cachexies and Dropsies; and in all Scorbatic Cases, it is equal to any Medicine; as it opens the minute Passages, promotes Transpiration, and cleanses the Skin, and other small Glands, which are filled with gross Particles to the Detriment of their proper Offices,

C H A P. XXI.

Of Treacle Water.

THIS Water is made in a very different Manner, by different Persons; but the following Recipes are the best for this Purpose.

Recipe for making ten Gallons of Treacle Water.

Take of the fresh and green Husks of Walnuts four Pounds and a Half; of the Roots of Butter-burr, three Pounds; of Angelica and Master-wort, of each one Pound and a Half; of Zedoary twelve Ounces; of the Leaves of Rue and Scordium of each eighteen Ounces; of Venice Treacle three Pounds; digest them together four Days in twelve Gallons of Proof Spirit, and two Gallons of Water; after which draw off ten Gallons, to which add a Gallon and a Half of distilled Vinegar.

Or,

Take of the Rind of green Walnuts five Pounds; of Rue four Pounds; of Carduus, Marigold, and Balm, of each three Pounds; of fresh gathered Butter-burr-roots two Pounds and a Half; of Burdock-root one Pound and a Half; of green Scordium twelve Handfuls; of Venice Treacle and Mithridate

Mithridate of each two Pounds and a Half; Proof Spirit twelve Gallons; and Water two Gallons. Digest, and draw off ten Gallons, as before; to which add a Gallon and a Half of distilled Vinegar,

Some instead of distilled Vinegar unadvisedly add a proportional Quantity of Spirit of Vitriol, or other Mineral Acid, to their Treacle Water; but this Practice is very pernicious; and intirely alters the Nature of the Medicine. Vinegar is an Acid made by a double Fermentation, and therefore of a different Nature from the acid Juices of Vegetables, whether Oranges, Lemons, Citrons, Limes, Crabs, Barberies, &c, as also from those of Minerals, whether Vitriol, Sulphur, &c. It is indeed, like them, acid on the Tongue; but then it liquifies the Blood, is antipestilential, suddenly cures Drunkenness, Surfeits, the Plague, and does a thousand Things both as a Menstruum and Medicine, which they will not. This is an admirable and sprightly Alexipharmic and Sudorific, to which the Vinegar added, greatly contributes, and therefore good in Fevers, the Small-pox, Measles, and other pestilential Disorders.

C H A P. XXII.

Compound Camomile-flower Water.

TH E Camomile-flowers generally used are the double Sort, consisting wholly of Petals or Flower Leaves, without any Appearance of Stamina or Pistil, or the other Parts of Fructification, which in the single Flowers shew themselves in the Middle in Form of yellow Threads. But tho' the double Flowers are the Sort commonly used, they are not the best, or those which ought to be chosen. The single Flowers, or those which consist of only a single Series of Leaves, or Petals, in Form of Rays, surrounding a Cluster of yellow Threads or Stamina, have much more Virtue. It is indeed in these Stamina and their Apices, that great Part of the Virtue of the Flower resides, and these are wanting in the double Flowers.

Recipe for making ten Gallons of compound Camomile-flower Water.

Take of dried Camomile-flowers, five Pounds; of the outer Peel of Oranges, ten Ounces; of the Leaves of common Wormwood, and Penny-royal, of each twenty Handfuls; of the Seeds of Anise, Cum-
min,

of Distillation.

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min, and sweet Fennel, the Berries of Bay and Juniper, of each five Ounces. Digest these Ingredients two Days in ten Gallons of Proof Spirit, and three Gallons of Water, and draw off ten Gallons with a gentle Fire.

This is a very good Carminative and Stomachic; good in all Cholics and other Disorders of the Bowels from Wind. It also provokes the Appetite and promotes a good Digestion. Its Virtues as a Stomachic will not be less when made from the double Flowers; but if intended as a Carminative it should be made with the single Flowers.

C H A P. XXIII.

Imperial Water.

THIS Cordial Water has its Name from the great Opinion conceived of it by its first Inventors; and tho' their Opinion was, perhaps, justly founded, yet it is not at present so much in use as formerly.

Receipe for making ten Gallons of Imperial Water.

Take of the dried Peels of Citrons and Oranges, of Nutmegs, Cloves, and Cinnamon, of each one Pound; of the Roots of

N 4

Cypress,

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Cypress, Florintine Orrice, Calamus Aromaticus, of each eight Ounces ; of Zedoary, Galangal and Ginger, of each four Ounces ; of the Tops of Lavender and Rosemary, of each sixteen Handfuls ; of the Leaves of Marjoram, Mint, and Thyme, of each eight Handfuls ; of the Leaves of white and damask Roses, of each twelve Handfuls. Digest the whole two Days in ten Gallons of Proof Spirit, and four Gallons of damask Rose Water ; after which draw off ten Gallons.

All the Ingredients in Composition coincide in one intention, and are such as will give their Virtues by Distillation ; Circumstances that cannot be said of many other compound Waters. It is a very good Cephalic, and of great use in all nervous Cases. It is also a very pleasant Dram, especially if dulcified with fine Sugar, and good upon any sudden Sickness of the Stomach.

C H A P. XXIV.

Of Compound Piony Water.

THE Piony, from whence this Compound Water takes its Name, is a Plant divided into Male and Female ; but the former is the Sort intended to be used in this Composition. The Male and Female
Plants

Plants are distinguished both by their Roots and Leaves. The Male has a shining blackish Leaf, from which the Female differs by being lighter coloured. The Root of the Male kind is more bulbous, shorter, and branched than that of the Female, whose Shoots are much longer and thinner.

Recipe for making ten Gallons of compound Piony Water.

Take of the Roots of Male Piony, twelve Ounces ; of those wild of Valerian, nine Ounces ; and of those of white Dittany six Ounces ; of Piony-seed four Ounces and a Half ; of the fresh Flowers of Lilly of the Valley, one Pound and a Half ; of those of Lavender, Arabian Stæchas, and Rosemary, of each nine Ounces ; of the Tops of Betony, Marjoram, Rue and Sage, of each six Ounces ; slice and bruise the Ingredients, and digest them four Days in ten Gallons of Proof Spirit and two Gallons of Water ; after which draw off ten Gallons.

Or,

Take of the Flowers of Lillies of the Valley fresh gathered, and Male Piony-root, of each two Pounds ; of Cinnamon and Cubebs, of each eight Ounces ; of Rosemary and Lavender Flowers, of each two Hand-

Handfuls; of damask Rose Water two Gallons. Digest these four Days in ten Gallons of Proof Spirit, and draw off ten Gallons as before.

This is an excellent Cordial, and can be exceeded by nothing in all Nervous Cases, both in Children and grown Persons.

C H A P. XXV.

Of Nutmeg Water.

THE Nutmeg is a Kernel of a large Fruit not unlike the Peach, and is separated from that and its investient Coat the Mace, before it is sent over to us; except when the whole Fruit is sent over in Preserve, by way of Sweet-meat, or as a Curiosity. There are two Kinds of Nutmegs, the one called by Authors the Male, and the other the Female. The Female is the Kind in common use, and is of the Shape of an Olive: The Male is long and cylindric, and has less of the fine aromatic Flavour than the other, so that it is much less esteemed, and People who trade largely in Nutmegs will seldom buy it. Besides this oblong kind of Nutmeg we sometimes meet with others of very irregular Figures; but these are mere *Lusus Naturæ*, being produced by the same Tree. The long or Male Nutmeg, as we term it, is, by the *Dutch*, called

called the wild Nutmeg. It is always distinguishable from the others as well by its want of Fragrancy as by its Shape: It is very subject to be worm-eaten, and is strictly forbid by the *Dutch* to be packed up among the other, because it will be the means of their being worm-eaten also by the Insects getting from it into them, and breeding in all Parts of the Parcel. The largest, heaviest, and most unctuous of the Nutmegs are to be chosen, such as are of the Shape of an Olive, and of the most fragrant Smell.

Recipe for making ten Gallons of Nutmeg Water.

Take of Nutmegs bruised one Pound; Proof Spirit ten Gallons; Water two Gallons. Digest them two Days, and then draw off ten Gallons with a brisk Fire. You may either dulcify it or not as occasion offers.

Or,

Take of Nutmegs bruised one Pound; Orange-peel two Ounces; Spirit ten Gallons; Water two Gallons. Digest, and distil as before.

This is an excellent Cephalic and Cordial Water; agreeable to the Palate, comfortable to the Stomach, and grateful to the Nerves. It powerfully discusses Wind and Vapours from

the Stomach and Bowels, and is therefore of great Service in the Cholic, and Gripping of the Bowels.

C H A P. XXVI.

Of Compound Bryony Water.

THE white Bryony-root, from whence this Water takes its Name, is one of the largest Roots we are acquainted with. It is of an oblong Shape, and is frequently met with of the Thickness of a Man's Arm, sometimes of twice or three times that Bigness. Its Texture is somewhat lax and spongy; considerably heavy, but so soft that the thickest Pieces are easily cut through with one Stroke of a Knife: It is very juicy, and is externally of a brownish or yellowish white Colour, and of a pure white within: It is of a disagreeable Smell, and an acrid and nauseous Taste.

Recipe for ten Gallons of Compound Bryony Water.

Take of the Roots of Bryony four Pounds; wild Valerian-root one Pound; of Pennyroyal and Rue, of each two Pounds; of the Flowers of Fever-few, and Tops of Savin, of each four Ounces; of the Rind of fresh Orange-peel, and Lovage-seeds,
of

of each Half a Pound : Cut or bruise these Ingredients and infuse them in eleven Gallons of Proof Spirit, and two Gallons of Water, and draw off ten Gallons with a gentle Heat.

Or,

Take of fresh Bryony-root four Pounds ; of the Leaves of Rue and Mugwort, of each four Pounds ; of the Tops of Savin six Handfuls ; of Fever-few, Catmint and Pennyroyal, of each four Handfuls ; of Orange-peel eight Ounces ; of Myrrh four Ounces ; of *Russia* Castor, two Ounces ; Proof Spirit eleven Gallons, Water two Gallons. Digest, and distil as before.

This Composition is very unpalatable, but excellently adapted to the Intention of an Hysteric, in which Cases it is used with Success. It is very forcing upon the Uterus, and therefore given to promote Delivery, and forward the proper Cleansings afterwards ; as also to open Menstrual Obstructions, and in abundance of other Female Complaints. It is also good against Convulsions in Children, and of service in all nervous Complaints in either Sex.

It may not be amiss to observe here, that the oily Parts of the Ingredients will often render the Water foul and milky. If therefore the Distiller desires to have it fine and

and transparent, the Receiver must be removed as soon as the Liquor at the Worm appears the least turbid, which will be long before the Faints begin to rise. The Water, however, is not the worse for being milky, with regard to its Medicinal Virtue. Some, when the Liquor is milky, throw in a little burnt Allum to fine it; but this should never be done, because it spoils the Medicine.

C H A P. XXVII.

*Of compound Baum Water, commonly called
Eau de Carmes.*

THIS has its Name (*Eau de Carmes*) from the Carmelite Friars who were the Inventors of it. The great Profit accruing to these Fathers, from the Sale of this Cordial, induced them to keep the Method of making it a Secret; but notwithstanding all their Care the Secret has at last been discovered, and the following is the Method by which they prepare it.

Recipe for two Gallons of Eau de Carmes.

Take of the fresh Leaves of Baum four Pounds; of the yellow Peel, or Rind of Lemons, two Pounds; of Nutmegs and Coriander-seeds of each one Pound; of
Cloves,

Cloves, Cinnamon, and Angelica-root, of each Half a Pound. Pound the Leaves, bruise the other Ingredients, and put them with two Gallons of fine Proof Spirit into a large Glass Alembic (the Figure of which with its Head is represented on the Plate, Fig. 7.) stop the Mouth, and place it in a Bath-Heat to digest two or three Days. Then open the Mouth of the Alembic, and add a Gallon of Baum Water, and shake the whole well together. After this place the Alembic in Balneum Mariæ, and distil till the Ingredients are almost dry; and preserve the Water thus obtained in Bottles well stopped.

This Water has been long famous both at *London* and *Paris*, and carried thence to most Parts of *Europe*. It is a very elegant Cordial, and very extraordinary Virtues are attributed to it; for it is esteemed very efficacious not only in Lowness of Spirits, but even in Apoplexies; and is greatly commended in Cases of the Gout in the Stomach.

C H A P. XXVIII.

Of Ladies Water.

THIS Water has its Name from its Dearness, being much fitter for the Closet than to be sold in a Shop ; but as it is an excellent Cordial, I could not omit giving it a Place here.

Recipe for one Quart of Ladies Water.

Take of Sugar-candy one Pound ; of Canary Wine six Ounces, Rose Water four Ounces ; boil them into a Syrup, and mix with it of Heavenly Water (described Chap. xxx.) one Quart ; of Ambergrise and Musk of each eighteen Grains ; of Saffron fifteen Grains ; yellow Saunders two Drams. Digest the whole three Days in a Vessel close stoped, and decant the clear for use.

This is an extraordinary Cordial where the Perfumes are not offensive. It is too rich to be drank alone, and therefore should be mixed with Water, or some other Liquid.

CHAP.

C H A P. XVII.

Of Cephalic Water.

THIS Water has its Name from its use, being one of the best Cephalic Waters known.

Recipe for ten Gallons of Cephalic Water.

Take of Male Piony-root twelve Ounces ; of Angelica and Valerian of each four Ounces and a Half ; of the Leaves of Rosemary, Marjoram and Baum ; of the Flowers of Lavender, Betony, Piony, Marigolds, Sage, Rosemary, Lilies of the Valley, and of the Lime Tree, of each three Handfuls ; of Stæchus, or *French* Lavender, four Ounces and a Half ; of red Roses and Cowslips, of each six Handfuls ; of Rhodium Wood and yellow Saunders, of each two Ounces and a Half ; of Nutmegs four Ounces and a Half ; of Galangals, an Ounce and a Half ; of Cardamoms and Cubebs, of each one Ounce. Bruise these Ingredients, and digest them ten Days in eleven Gallons of Proof Spirit, and two Gallons of Water, after which add three Pounds of Cinnamon, and digest two Days more ; and then draw off ten Gallons with a pretty brisk
O Fire

Fire, and dulcify it to your Palate with fine Sugar.

This is an excellent Cordial, of great Use in Faintings or Sinking of the Spirits, and to remove any sudden Nauseas or Sickneſs at the Stomach.

C H A P. XXX.

Of Heavenly Water ; or, Aqua Cœleſtis.

THIS Water has its Name from the great opinion its Inventors had of it ; but at preſent it is not ſo much called for as formerly.

Recipe for ten Gallons of Heavenly Water.

Take of Cinnamon, Mace, and Cubebs, of each three Ounces ; Ginger one Ounce and a Half ; Cloves, Galangal, Nutmegs, and Cardamoms, of each one Ounce ; Zedoary one Ounce and a Half ; Fennel-ſeeds one Ounce ; of the Seeds of Aniſe, wild Carrot and Baſil, of each Half an Ounce ; Roots of Angelica, Valerian, Calamus Aromaticus, Leaves of Thyme, Calamint, Penniroyal, Mint, Mother of Thyme, and Marjoram, of each an Ounce ; Flowers of red Roſes, Sage, Rosemary, and Stæchas, of each fix Drums ; Citron-peel an Ounce : bruise all theſe Ingredients and diſteſt them three Days in eleven Gallons of Proof Spirit, and four Gallons of Water ; after which draw off ten Gallons,

lons, with a pretty brisk Fire; and dulcify the Goods with fine Sugar, adding Ambergrise and Musk of each three Scruples.

The Perfumes ordered to be added with the Sugar, rendering the Medicine offensive to some People, they may be omitted at pleasure. It is esteemed very efficacious in all nervous Complaints, particularly Palfies, Loss of Memory, and the like. In all Decays of Age, and languishing Constitutions, it is exceeded by nothing in suddenly raising the Spirits, and warming the Blood.

C H A P. XXXI.

Of Spirituous Penniroyal Water.

THE Plant from whence this Water has its Name has been already described Chap. xiii. Part II.

Recipe for ten Gallons of Spirituous Penniroyal Water.

Take of the Leaves of Penniroyal dried fifteen Pounds; Proof Spirit ten Gallons; Water two Gallons: draw off ten Gallons with a gentle Fire.

This is a good Carminative, of use in Cholics and Gripings of the Bowels; also in Plurisies and the Jaundice: It is of known

Efficacy in promoting the Menfes and other Disorders of the Female Sex.

C H A P. XXXII.

Of Compound Parsley Water.

THIS Plant from whence the Water is denominated is the common Parsley of our Gardens, an Herb too well known to need Description.

Recipe for ten Gallons of compound Parsley Water.

Take of Parsley-root, one Pound and a Quarter ; fresh Horse-radish Root, and Juniper-berries, of each fifteen Ounces ; the Tops of St. John's-wort, biting Arsmart, and Elder-flowers, of each ten Ounces ; the Seeds of wild Carrot, sweet Fennel, and Parsley, of each seven Ounces and a Half ; slice and bruise the Ingredients, and digest them four Days in eleven Gallons of Spirit, and two Gallons of Water ; after which draw off ten Gallons.

This is a very good Diuretic, frees the Kidnies from Sand and other Matter, which often forms Gravel and Stones. It is also good in cholic Pains arising from a Stone in the

the Bladder, and drains off all ill Humours by Urine.

C H A P. XXXIII.

Of Carminative Water.

THIS Water has its Name from its use, being an excellent Carminative.

Recipe for ten Gallons of Carminative Water.

Take of fresh Camomile-flowers, four Pounds; Dill-seed two Pounds and a Half; Leaves of Baum, Origany, and Thyme, of each one Pound; Seeds of Anise and Fennel of each six Ounces; Cummin-seed four Ounces; Peels of Oranges and Citrons, of each eight Ounces; Juniper and Bayberries, of each six Ounces; Cinnamon eight Ounces; Mace four Ounces. Digest these Ingredients, bruised in eleven Gallons of Proof Spirit, and two Gallons of Water; after which draw off ten Gallons; and dulcify it with fine Sugar.

This is an admirable Carminative, and therefore good in all Cholicky Pains and Gripings of the Bowels; and to remove Sicknefs and Nauseas from the Stomach.

C H A P. XXXIV.

Of Gout Water.

THIS Water also has its Name from its use, being of great Service in that Distemper.

Recipe for ten Gallons of Gout Water.

Take of the Flowers of Camomile, Leaves of Penniroyal, Lavender, Marjoram, Rosemary, Sage, and Ground-pine, of each eight Ounces; Myrrh four Ounces; Cloves and Cinnamon of each one Ounce; Roots of Piony two Ounces; Pellitory of *Spain*, and Cypress Orrice, of each one Ounce; the lesser Cardamoms and Cubebs, of each Half an Ounce; Nutmegs, two Ounces: Cut and bruise these Ingredients and digest them four Days in eleven Gallons of Proof Spirit and two Gallons of Water; then draw off ten Gallons, and dulcify with fine Sugar,

This is a very good Water in all nervous Cases; and a continued moderate Use of it will comfort and fortify the Fibres, so as to prevent the Discharge of such Juices upon the Joints as cause arthritic Pains and Swellings. It is also of excellent Use in Passies, Epilepsies, and Loss of Memory; particularly when these Distempers proceed from

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of Distillation,

from old Age, or when the principal Springs
of Life begin to decay.

C H A P. XXXV.

Of Anbalt Water.

THIS Water is supposed to have been
invented by a celebrated Physician of
Anbalt, a Province of the Circle of Upper
Saxony.

*Recipe for making ten Gallons of Anbalt
Water.*

Take of the best Turpentine a Pound and
a Half; Olibanum three Ounces; Aloes-
wood powdered one Ounce; Grains of
Mastick, Cloves, July-flowers, or Rosemary-
flowers, Nutmegs and Cinnamon of each
two Ounces and a Half; Saffron one Ounce;
powder the whole and digest them six Days
in eleven Gallons of Spirit of Wine; add-
ing two Scruples of Musk tied up in a
Rag; and draw off in Balneum Mariæ till
it begins to run foul.

This Water is a high aromatic Cordial,
invigorates the Intestines, and thereby pro-
motes Digestion and dispels Flatulencies.
It is also in great Repute as a sovereign
Remedy for Catarrhs and Pains arising from
O 4. Colds;

Colds ; as also in Palsies, Epilepsies, Apoplexies and Lethargies, the Parts affected being well rubbed with it.

C H A P. XXXVI.

Of Vulnerary Water, or, Eau d'Arquebuse.

THIS Vulnerary Water is greatly esteemed abroad ; and if properly tried, there is no doubt of its obtaining the same Reputation here.

Recipe for five Gallons of Vulnerary Water.

Take of the Leaves, Flowers and Roots of Comfrey, Leaves of Mugwort, Sage, and Bugle, of each eight Handfuls ; Leaves of Betony, Sanicle, or Ox-eye Daisy, the greater Figwort, Plantain, Agrimony, Vervain, Wormwood, and Fennel, of each four Handfuls ; St. John's-wort, Birth-wort, Orpine, Paul's-betony, the lesser Centory, Yarrow, Tobacco, Mouse-ear, Mint, and Hyssop, of each two Handfuls : Cut them, bruise them well in a Mortar, and pour on them three Gallons of white Wine and two Gallons and a Half of Proof Spirit ; digest the whole six Days with a gentle Heat, in a Vessel close stopped : after which distil off with gentle Fire, about five Gallons,

ions, or till it begins to run milky from the Worm.

This Water is of excellent Service in Contusions, Tumors attending Dislocations, Fractures and Mortifications, the Part affected being bathed with it. Some also use it to deterge foul Ulcers, and incarn Wounds; from whence it was called Vulnerary Water.

C H A P. XXXVII.

Of Cedrat Water.

THE Fruit called *Cedrat* by the *French* is a Species of the Citron, called by Botanists *Citratum Florentinum*, *fructu Mucronato & recurvo*, *Cortice verrucoso Odo-ratissimo*, Florentine Citron, with a pointed Fruit, which is recurved, and a warted sweet smelling Rind. This Fruit is in so great Esteem, that they have been sold at *Florence* for two Shillings each, and are often sent as Presents to the Courts of Princes. It is only found in Perfection, in the Plain between *Pisa* and *Leghorn*; and tho' the Trees which produce this Fruit have been transplanted into other Parts of *Italy*, yet they are found to lose much of that excellent Taste with which they abound in those Parts.

Recipe

Recipe for a Gallon of Cedrat Water.

Take the yellow Rinds of five Cedrats, a Gallon of fine Proof Spirit, and two Quarts of Water : Digest the whole twenty-four Hours in a Vessel close stoped ; after which draw off one Gallon in Balneum Mariæ, and dulcify with fine Loaf Sugar.

This is esteemed the finest Cordial yet known ; but as it is very difficult to procure the Fruit here, I shall give the Method of making this celebrated Cordial, with the Essence or essential Oil of the Cedrat, which is often imported from *Italy*.

Recipe for a Gallon of Cedrat Water with the Essence of the Fruit.

Take of the finest Loaf Sugar reduced to Powder a Quarter of a Pound, put into it a Glass Mortar, with one Hundred and twenty Drops of the Essence of Cedrat, rub them together with a Glass Pestle, put them into a Glass Alembic with a Gallon of fine Proof Spirits and a Quart of Water. Place the Alembic in Balneum Mariæ, and draw off one Gallon, or till the Faints begin to rise ; and dulcify with fine Sugar.

You

You may make this Water without Distillation, by mixing the Essence with the Sugar, as before directed, and dissolving it in the Spirit and Water directed as above. But the Water will be foul and milky, and therefore you must filtrate it thro' Paper, which will restore its Brightness and Transparency.

But whatever Method is used, you must be very careful that the Spirit been tirely freed from its essential Oil; and therefore if your Spirit be not very cleanly rectified, it will be adviseable to use *French Brandy*, lest the fine Flavour so highly esteemed in this Cordial be destroyed by the Spirit.

C H A P. XXXVIII.

Of Bergamot Water.

THE Bergamot is a Species of the Citron, produced at first casually by an *Italian's* Grafting a Citron on the Stock of a Burgamot Pear Tree, whence the Fruit produced by this Union participated both of the Citron Tree and Pear Tree. The Inventor is said to have kept the Discovery a long Time a Secret, and enriched himself by it.

The

The Bergamot is a very fine Fruit both in Taste and Smell; and its Essence or essential Oil highly esteemed.

Recipe for a Gallon of Bergamot Water.

Take the outer Rind of three Bergamots, a Gallon of Proof Spirit, and two Quarts of Water. Draw off one Gallon in Balneum Mariæ, and dulcify with fine Sugar.

If you make your Bergamot Water from the Essence or essential Oil, observe the same Directions as given in the preceeding Chapter for making Cedrat Water. One Hundred and sixty Drops of the Essence will be sufficient for a Gallon of Spirit, and so in Proportion for a greater or smaller Quantity.

C H A P. XXXIX.

Of Orange Cordial Water; or, Eau de Bigarade.

THE Orange called by the French Bigarade, is called by Botanists *Aurantium maximum, verrucoso Cortice*, the large worted Orange.

It

It is a large and beautiful Fruit, and greatly esteemed for the Fragrancy of its Effence. It is common in diverse Parts of *Italy, Spain, and Provence in France.*

Recipe for making a Gallon of Orange Cordial.

Take of the outer or yellow Part of the Rinds of fourteen Bigarades ; Half an Ounce of Nutmegs ; a Quarter of an Ounce of Mace, a Gallon of fine Proof Spirit, and two Quarts of Water. Digest all these together two Days in a close Vessel ; after which draw off a Gallon with a gentle Fire, and dulcify with fine Sugar.

This Cordial is greatly esteemed abroad, and would be the same here if sufficiently known.

If the Orange Peels are not easily procured, one Hundred and forty Drops of the Effence may be used in their stead, and the Water will be nearly equal to that made from the Peels.

CHAP.

C H A P. XL.

Of Jasmine Water.

THERE are several Species of Jasmine, but that Sort intended here is what the Gardeners call, *Spanish White* or *Catalonian Jasmine*. This is one of the most Beautiful of all the Species of Jasmine; the Flowers much larger than any of the others, of a red Colour on the Outside, and extremely fragrant. But if the Flowers of this Species cannot be procured, those of the common Sort may be used, but the Quantity must be considerably augmented.

Recipe for a Gallon of Jasmine Water.

Take of *Spanish Jasmine-flowers* twelve Ounces; *Essence of Florentine Citron*, or *Burgamot*, eight Drops; fine Proof Spirit a Gallon, Water two Quarts. Digest two Days in a close Vessel, after which draw off one Gallon, and dulcify with fine Loaf Sugar.

This is a most excellent Cordial, and deserves to be more known here than it is at present.

CHAP.

C H A P. XLI.

Of the Cordial Water of Montpellier.

THIS Water has its Name from the Place where it was first made, and what is now brought from thence is still in great Reputation.

Recipe for a Gallon of the Cordial Water of Montpellier.

Take of the yellow Rinds of two Bergamots, or fifty Drops of the Essence of that Fruit ; Cloves and Mace of each Half an Ounce ; Proof Spirit a Gallon ; Water one Quart : Digest two Days in a close Vessel, draw off a Gallon, and dulcify with fine Sugar.

C H A P. XLII.

Of Father Andrew's Water.

THIS Water has its Name from its Inventor ; and is greatly esteemed in France.

Recipe

Recipe for a Gallon of Father Andrew's Water.

Take of white Lilly-flowers eight Handfuls ; Orange-flowers four Ounces ; Rose Water a Quart ; Proof Spirit a Gallon ; Water a Quart : Draw off a Gallon in Balneum Mariæ, and dulcify with fine Sugar.

C H A P. XLIII.

Of the Water of Father Barnabas.

THIS Water has also its Name from its Inventor, a Jesuit of *Paris*.

Recipe for a Gallon of the Water of Father Barnabas.

Take of the Roots of Angelica four Ounces ; of Cinnamon and Orris-root, of each Half an Ounce ; bruise these Ingredients in a Mortar ; put them into an Alembic, with a Gallon of Proof Spirit and two Quarts of Water ; draw off a Gallon with a pretty brisk Fire.

C H A P.

C H A P. XLIV.

Of the Water of the four Fruits.

THIS Water has its Name from the four Fruits in its Composition, namely the Cedrat or Florentine Citron, the Bergamot, the common Citron, and the *Portugal* Orange.

Recipe for a Gallon of the Water of the four Fruits.

Take of the Essence of Cedrat, fifty Drops; of the Essence of Bergamot thirty-six Drops; of the Essence of Citron sixty Drops; and of the Essence of *Portugal* Orange sixty-four Drops; fine Proof Spirit one Gallon; Water two Quarts; draw off with a pretty brisk Fire till the Faints begin to rise, and dulcify with fine Sugar.

This is a very pleasant and odoriferous Cordial, and in great Esteem in *France*.

C H A P. XLV.

Of the Water of the four Spices.

THIS Water also derives its Name from the four Spices from whence it is drawn, viz. Cloves, Mace, Nutmegs, and Cinnamon.

Recipe for a Gallon of the Water of the four Spices.

Take of Cinnamon two Ounces; Nutmegs and Cloves, of each three Drams; Mace six Drams; bruise the Spices in a Mortar, and add Proof Spirit a Gallon, and Water two Quarts. Digest twenty-four Hours in a close Vessel, and distil with a brisk Fire till the Faints begin to rise; and dulcify with fine Sugar.

This is an excellent Stomachic, good in all Depressions of the Spirits and paralytic Disorders.

C H A P.

C H A P. XLVI.

Of the Water of the four Seeds.

THIS Water has its Name from the four Seeds from whence is it drawn, viz. the Seeds of sweet Fennel, Coriander, Angelica, and Anise.

Recipe for ten Gallons of the Water of the four Seeds.

Take of sweet Fennel-seed seven Ounces ; Coriander-seed nine Ounces ; of the Seeds of Angelica and Anise, of each three Ounces ; bruise all these in a Mortar, and put them into the Still with ten Gallons and a Half of Proof Spirits, and two Gallons of Water ; draw off with a gentle Fire till the Faints begin to rise, and dulcify with fine Sugar.

This Water is a very good Carminative, good in Cholics, Nauseas of the Stomach, and Gripings of the Bowels.

C H A P. XLVII.

Of the Divine Water.

THIS is one of those Waters whose Names have rendered them famous. The Basis of this Water is Orange-flowers, the other Ingredients being added to diversify the Flavour, and render it more agreeable.

Recipe for a Gallon of Divine Water.

Take of Orange-flowers fresh gathered two Pounds; Coriander-seed three Ounces; Nutmegs Half an Ounce; bruise the Nutmegs and Coriander-seeds; and put them, together with the Orange-flowers, into an Alembic with a Gallon of Proof Spirit and two Quarts of Water; draw off the Liquor with a gentle Fire, till the Faints begin to rise, and dulcify with fine Sugar.

This is a very pleasant Cordial, both with regard to its Smell and Taste; and on that Account in great Esteem abroad.

C H A P.

C H A P. XLVIII.

Of Roman Water.

THIS Water has its Name from its being made first at *Rome*; and from whence great Quantities are still exported to different Parts of *Europe*.

Recipe for a Gallon of Roman Water.

Take the outer, or yellow Peels of six Citrons; two Drams of Mace bruised; a Gallon of Proof Spirit, and two Quarts of Water: Draw off with a gentle Fire till the Faints begin to rise, and dulcify with fine Sugar.

This Water is generally of a red or purple Colour, the former of which may be easily given by infusing in it a few Grains of Cochineal, or the red Parts of Clove Gilly-flowers; and the latter by adding to the above a few Violets. When the Colour is extracted, run the Liquor through the filtrating Bag, and it will be very bright and clear.

C H A P. XLIX.

Of Barbadoes Water.

TH E R E is a great Variety of Waters called by this Name, made by foreign Distillers; but the following Recipes will be sufficient to shew the Distiller the Method of making them, and how to vary the Flavour of his Waters, so as to adapt them to the Taste of his Customers.

Recipe for a Gallon of rectified Barbadoes Water.

Take the outer Rind of eight large Florentine Citrons; Half an Ounce of Cinnamon bruised, and a Gallon of rectified Spirit. Distil to a Dryness in Balneum Mariæ. Then dissolve two Pounds of Sugar in a Quart of Water, and mix it with the distilled Liquor, and run it thro' the filtrating Bag, which will render it bright and fine.

Recipe for making a Gallon of amber-coloured Barbadoes Water.

Take of the yellow Rinds of six Bergamots, Half an Ounce of Cinnamon, and two Drams of Cloves, Bruise the Spices and digest the whole six Days in a Gallon
of

of rectified Spirit ; and then add a Dram of Saffron, and let the whole stand six Days longer in Digestion ; dissolve two Pounds of fine Sugar in a Quart of Water, add it to the Tincture, and run it thro' the filtrating Bag.

After the same manner may be made *Barbadoes Waters* of different Kinds, by adding Lemon, or Orange-peels instead of those of Citron or Bergamot ; or, by varying the Spices.

C H A P. L.

Of Ros Solis.

THE Ros Solis or Sun-dew, from whence this Cordial Water has its Name, is a small low Plant, with a fibrous Root, from whence spring small round hollowish Leaves, on Foot stalks about an Inch long, covered and fringed with short red Hairs, which give a red Cast to the whole Leaf. It grows in champaign and mossy Grounds, in a pale red Moss, and flowers in *May*.

Recipe for ten Gallons of Ros Solis.

Take of Ros Solis picked clean, four Pounds ; Cinnamon, Cloves, and Nutmegs, of each, three Ounces and a Half ; Mari-

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gold-

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gold-flowers one Pound ; Caraway-seeds ten Ounces ; Proof Spirit ten Gallons, and of Water three Gallons. Distil with a pretty brisk Fire, till the Faints begin to rise. Then take of Liquorice Root sliced Half a Pound ; Raisins stoned two Pounds ; red Saunders Half a Pound ; digest these three Days in two Quarts of Water, and strain out the clear Liquor, in which dissolve three Pounds of fine Sugar, and mix it with the Spirit drawn by Distillation.

Recipe for making ten Gallons of Ros Solis by Digestion.

Take Ros Solis clean picked three Pounds ; Nutmegs, Mace, Cloves, and Cinnamon, the Seeds of Caraway and Coriander of each three Ounces ; Ginger, the lesser Cardamom, Zedoary, and Calamus Aromaticus, of each one Ounce ; Cubebs and yellow Saunders of each Half and Ounce ; red Saunders three Ounces ; red Rose Leaves dried three Handfuls ; Proof Spirit ten Gallons ; digest the whole six Days in a Vessel close stoped, and then strain off the clear Liquor, and dulcify it with fine Sugar.

Or,

Take Ros Solis, picked three Pounds ; Cinnamon and Nutmegs, Caraway and Coriander-seeds, of each three Ounces ; Cloves, Mace,

Mace, and Ginger, of each one Ounce and a Quarter ; Cubebs, Cardamoms, Zedoary, and Calamus Aromaticus, of each Half an Ounce ; red Roses dried three Ounces ; Liquorice Root sliced, six Ounces ; Raisins stoned one Pound and a Half ; Cochineal and Saffron, of each three Drams ; digest the whole eight Days in ten Gallons of Proof Spirits ; strain off, and dulcify as before.

Recipe for ten Gallons of Turin Ros Solis.

Take of damask Roses, Orange-flowers, Lilies of the Valley, and Jasmine-flowers, of each two Pounds and a Half ; Cinnamon five Ounces ; Cloves three Drams : Put these Ingredients into an Alembic, with four Gallons and a Half of Water, and draw off three Gallons, with a moderate Fire ; to this Water add seven Gallons of Proof Spirit, in which a Dram of Cochineal and two Drams of Saffron has been infused ; dulcify with fine Sugar, and run the whole through the filtrating Bag.

All these different Kinds of Ros Solis are excellent Cordials, good in all Depressions of the Spirits, Nauseas, and paralytic Disorders.

C H A P. LI.

Of Usquebaugh.

USQUEBAUGH is a very celebrated Cordial, the Basis of which is Saffron. There are different Ways of making this famous Compound; but the following are equal to any I have seen.

Recipe for ten Gallons of common Usquebaugh.

Take of Nutmegs, Cloves, and Cinnamon, of each two Ounces; of the Seeds of Anise, Caraway and Coriander, of each four Ounces; Liquorice Root sliced Half a Pound; bruise the Seeds and Spices, and put them together with the Liquorice into the Still with eleven Gallons of Proof Spirits, and two Gallons of Water; distil with a pretty brisk Fire till the Faints begin to rise. But as soon as your Still begins to work, fasten to the Nose of the Worm two Ounces of *English* Saffron tied up in a Cloth, that the Liquor may run thro' it, and extract all its Tincture, and in order to this you should often press the Saffron with your Fingers. When the Operation is finished, dulcify your Goods with fine Sugar.

Recipe

*Recipe for making ten Gallons of Royal
Usquebaugh.*

Take of Cinnamon, Ginger, and Coriander-seed, of each three Ounces; Nutmegs four Ounces and a Half; Mace, Cloves and Cubebs, of each one Ounce and a Half. Bruise these Ingredients, and put them into an Alembic with eleven Gallons of Proof Spirit, and two Gallons of Water; and distil till the Faints begin to rise; fastening four Ounces and a Half of *English* Saffron tied in a Cloth to the End of the Worm, as directed in the preceding Recipe. Take Raisins stoned four Pounds and a Half; Dates three Pounds, Liquorice Root sliced two Pounds; digest these twelve Hours in two Gallons of Water; strain out the clear Liquor, add it to that obtained by Distillation, and dulcify the whole with fine Sugar.

*Recipe for ten Gallons of Usquebaugh by
Digestion.*

Take of Raisins stoned five Pounds; Figs sliced one Pound and a Half; Cinnamon Half a Pound; Nutmegs three Ounces; Cloves and Mace, of each one Ounce and a Half; Liquorice two Pounds; Saffron four Ounces; bruise the Spices, slice the Liquorice,

quorice, and pull the Saffron in pieces; digest these Ingredients eight Days in ten Gallons of Proof Spirit, in a Vessel close stoped; then filter the Liquor, and add to it two Gallons of *Canary Wine*, and Half an Ounce of the Tincture of Ambergrease.

*Recipe for making ten Gallons of French
Ufquebaugh.*

Take of Saffron three Ounces, of the essential Oil or Effence of Florentine Citron, Bergamot, *Portugal Orange*, and Lemon, of each a Hundred Drops; Angelica-seed, Vanellos and Mace, of each one Ounce and a Half; Cloves and Coriander-seed of each three Quarters of an Ounce; bruise the Seeds and Spices, and put all into an Alembic with eleven Gallons of Proof Spirit, and two Gallons of Water; and draw off with a gentle Fire till the Faints begin to rise, fastening to the Nose of the Worm four Ounces of Saffron in a Cloth. When the Operation is finished dulcify the Goods with fine Sugar.

These Waters are excellent Cephalic Cordials, and Alexipharmics; and are excelled by nothing in suddenly reviving the Spirits when depressed by Sicknes, &c.

C H A P. LII.

Of Ratafia.

RATAFIA is a Liquor in great Esteem, and most Persons are acquainted with it; tho' the true Method of making it is known only to a few. There are various Kinds of Ratafia made from different Fruits. I shall give Recipes for making those which are at present in most Esteem; which may serve as Instances for making these Goods from any other Kinds of Fruit.

I. Of red Ratafia.

There are three Sorts of Ratafia drawn from red Fruits, distinguished by the Epithets *fine*, *dry*, and *common*.

The Fruits most proper to make the red Ratafia are the black Heart Cherry, the common red Cherry, the black Cherry, the Merry or Honey Cherry, the Strawberry, the Raspberry, the red Gooseberry, and the Mulberry.

These Fruits should be gathered in the Height of their respective Seasons, and the largest and most beautiful of them chosen for the purpose.

Thus

Thus with regard to the Heart Cherry, it should be large, fleshy, and thorough but not over ripe; for then a Part of its Juice will be evaporated on the Tree: Care must be also taken, that its Colour be not decayed; but clear and almost transparent, and well tasted.

The black Cherry, or as it is often called, the black Arvon, must be extremely ripe, because it is used to colour the Ratafia when that of the other fails. The Criterion of judging when it is thoroughly ripe is its Blackness; for, when in Perfection, it is perfectly black. It should also be remembered that this Fruit is better and more profitable in Proportion to its Sweetness; as the Flavour of the Ratafia will be rendered more agreeable, and a less Quantity of Sugar necessary.

As the Gooseberry is an acid Fruit, it must be chosen as ripe as possible. The Fruit large, and the Skin and Husk so transparent as to see the Seeds through it. The Gooseberry should be used immediately after its being gathered; for it is very liable to ferment, which will inevitably spoil the Ratafia. Gooseberries are chiefly used to render the Ratafia dry or sharp, and consequently less soft; and therefore their Quantity should always be proportioned to that Intention. The

The Merry to be good should be small, black, the Skin transparent, full of Liquor of deep black Purple Colour. The greatest Care should be taken, that it be fresh gathered, and not rotten. It corrects the acid Juices of the other Fruits by its Sweetness, softens the Composition; and is of great Service in colouring the Ratafia.

The Mulberry is of the greatest Service in colouring the Ratafia. It should be chosen large, and fully ripe, at which time it is of a black Purple Colour. Its Taste also greatly contributes to render the Ratafia of a pleasant and agreeable Flavour.

The Strawberry greatly contributes to increase the rich Flavour of the Ratafia; but it must be chosen ripe, and large; fresh gathered and not bruised. Another Caution necessary to this Fruit is, that they are gathered in dry warm Weather; for if gathered in rainy Weather they will want that fine Taste, for which they are so greatly valued.

The Raspberry is also added to augment the Richness of the Liquor, to which its elegant perfumy Taste greatly contributes; by its agreeable Acidity it renders the Flavour more brisk

brisk and agreeable. It must be fresh gathered, full ripe, and free from Spots and Mouldness, which this Fruit is particularly subject to.

Having thus concisely enumerated the Qualities requisite in the several Fruits, to render the Ratafia of a rich and elegant Flavour, we shall proceed to give the best Methods for making Ratafia from them.

Recipe for making red Ratafia, fine and soft.

Take of the black Heart Cherries twenty-four Pounds; black Cherries four Pounds; Raspberries and Strawberries, of each three Pounds: Pick these Fruits from their Stalks, and bruise them, in which Condition let them continue twelve Hours; press out the Juice, and, to every Pint of it add a Quarter of a Pound of Sugar. When the Sugar is dissolved run the whole through the filtrating Bag, and add to it three Quarts of clean Proof Spirits. Then take of Cinnamon four Ounces; of Mace an Ounce; and of Cloves two Drains. Bruise these Spices, put them into an Alembic with a Gallon of clean Proof Spirits and two Quarts of Water, and draw off a Gallon with a brisk Fire. Add as much of this spicy Spirit to your Ratafia as will render it agreeable.

agreeable to your Palate ; about one fourth is the usual Proportion.

Ratafia made according to the above Recipe will be of a very rich Flavour, and elegant Colour. It may be rendered more or less of a spicy Flavour, by adding or diminishing the Quantity of Spirit distilled from the Spices.

Some in making Ratafia suffer the expressed Juices of their Fruits to ferment several Days; by this means the Vinosity of the Ratafia is increased; but, at the same time, the elegant Flavour of the Fruits greatly diminished. Wherefore if the Ratafia be desired stronger or more vinous, it may be done by adding more Spirits to the expressed Juice; by which means the Flavour of the Fruits may be preserved, as well as the Ratafia rendered stronger.

It is also a Method with some to tie the Spices in a Linen Rag, and suspend them in the Ratafia. But if this Method be taken it will be necessary to augment the Quantity of Spirit first added to the expressed Juice. There is no great Difference in the two Methods of adding the Spices, except that by suspending them in the Ratafia, the

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Liquor

Liquor is generally rendered less bright and transparent.

There is also another Method practised in making Ratafia, which is this : Take the Quantity of Fruit proposed, bruise it, and immediately pour the Spirit on the Pulp. After standing a Day or two express the Juice and Spirit, filtrate it, and add the Sugar and Spices as before. But this Method requires more Spirit than the former, as it will be impossible to press it all out of the Skins and other Parts of the Fruit remaining after the Juice is extracted.

2. Of making fine and dry Ratafia from red Fruit.

Tho' the Ratafia we have just mentioned will doubtless please the Palates of many People ; yet there are others who would prefer a different Sort ; it is therefore necessary to know how to make dry as well as sweet Ratafia, if we are desirous of pleasing all Sorts of Palates.

Dry Ratafia is prepared in the same manner as the preceding, but the Ingredients are different.

An equal Quantity of Cherries and Gooseberries are necessary in making dry or sharp Ratafia ;

Ratafia; because the Acidity of the Gooseberries gives the requisite Flavour to this Sort of Liquor. But, at the same time, care must be taken that the Gooseberries be fully ripe; for otherwise, tho' Gooseberries are more acid before they are ripe than afterwards; yet that Acidity is not the Flavour desired; it is acerb and rough, and will render the Flavour of the Ratafia disagreeable. The same Observation holds good also with regard to the Cherries; they must be fully ripe as in making the soft Ratafia.

Instead of black Cherries used in the Composition of the preceding Ratafia, Mulberries should be used in this: The reason for this Change is, that the Juice of the black Cherry is more sweet and glutinous than that of the Mulberry, and therefore less fit for making dry Ratafia. But the Mulberries must be the ripest and blackest possible, in order to give the better Colour to the Liquor.

More Spirit and less Sugar in proportion to the Juice of the Fruit, is also required in this Composition than in the foregoing; but with regard to the Spices, the same Quantity is generally added to both.

Recipe for making red Ratafia, fine and dry.

Take of Cherries and Gooseberries, of each thirty Pounds; Mulberries seven Pounds; Raspberries ten Pounds. Pick all these Fruits clean from their Stalks, &c. bruise them, and let them stand twelve Hours; but do not suffer them to ferment. Press out the Juice, and to every Pint add three Ounces of Sugar; when the Sugar is dissolved run it thro' the filtering Bag, and to every five Pints of Liquor add four Pints of clean Proof Spirit; together with the same Proportion of Spirit drawn from the Spices in the foregoing Composition.

But it may not be amiss to observe here, that different Distillers use different Quantities of the Spirit drawn from the Spices. The best Method therefore is to imitate the Flavour most universally approved of, which may be easily done by adding a greater or less Proportion of the spiced Spirit.

3. Of mixed Ratafia.

By mixed Ratafia is meant the Juices of Fruits prepared, and ready to be mixed with the Spirit when called for.

Recipe

Recipe for making mixed Ratafia.

Ratafia is composed of Cherries and Gooseberries; of these the best are to be chosen, bruised, and in that Condition suffered to remain some Days to ferment. The Juice is then to be strained off, the Quantity of Sugar and Brandy added, and the whole put into a Cask and close stopped. A Lee or Sediment will fall to the Bottom of the Cask, which Sediment will be of great Use in preserving the Ratafia.

The Proportion of black Cherries must be large in this Ratafia, because the Colour, which this is greatly valued for, chiefly comes from the Juice of that Fruit.

The Sugar must not be put in at once, because the Acidness of the Liquor would cause a considerable Effervescence, but by a little at a time.

These Instructions being observed, a Ratafia of this Kind may be easily made: And as the Spirit is not to be mixed with it, till the Ratafia is called for, a large Quantity of it may be made at a small Expence, when the Fruits are in Perfection, which cannot be done by the common Methods.

Recipe for making mixed Ratafia.

Take of common Cherries, thoroughly ripe, four Hundred and fifty Pounds; Gooseberries, large and ripe, two Hundred and twenty-five Pounds; black Cherries ripe and large, fifty Pounds. Bruise these Fruits, and in that Condition let them continue three or four Days to ferment. Then press out the Juice, and add one fifth Part of Spirit; that is, if you have two Hundred and fifty Pints of Juice you must add to it fifty Pints of Spirit. When your Spirit and Juice are mixed put them into a Cask, and for every Pint add three Ounces of Sugar. By this means your Ratafia will be always ready to mix with Spirit.

But as the Proportion of Spirit is but small, it will be necessary to taste your Ratafia at least every Month, lest it should ferment, and by that means lose both its Flavour and Colour. As soon therefore as you perceive the least Alteration in your Ratafia, more Spirit must be added to stop the Fermentation; and by this Method it may be kept the whole Year.

If you have any Ratafia remaining at the End of the Year, you must mix it with that just made, adding a large Proportion
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of black Cherries ; because the Colour in the old Ratafia will not be equal to that of the new. Or you may add to your old Ratafia a proper Quantity of the fresh Juice of black Cherries, which will restore its Colour, and, in a great Measure its Flavour too : So that if your Ratafia has been well preserved, it will, when mixed with fresh Juice of black Cherries, be but little inferior to the new.

4. *Of white Ratafia.*

As red Fruits are the Basis of that called red Ratafia, so, on the contrary, that made from the Juices of white Fruits is denominated white Ratafia.

There are various Kinds of Ratafia made from various Fruits ; but I shall only give Recipes for making three or four Sorts, which will be sufficient for all the rest, as the Method is nearly the same in all.

Recipe for making Ratafia from the Muscat, or white Frontiniac Grape.

The Berries of this Kind of Grape are large, and grow extremely close upon the Bunches, which are very long, and have commonly two Shoulders : The Fruit, when ripe, has a rich musky Flavour ;

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but it is commonly very late in Autumn, before these Grapes are in Perfection; and the Berries being so very close upon the Bunches, detain the Moisture in the Centre; so that they often perish: To prevent which some curious Persons look over their Vines, soon after the Grapes are formed, and, with a Pair of Scissars, cut out all small ones, so as to leave the others at a moderate Distance, whereby the Sun and Air are easily admitted, which dissipates the Moisture, and prevents their perishing. There is another Kind of this Grape, called by some the white Frontinac of *Alexandria*, and by others the *Jerusalem Muscat*, which is a very large Grape, and, when ripe, an excellent Fruit; but is rarely brought to Perfection in *England*. The Berries of the *Jerusalem Muscat*, are of an oval Shape, and very large. They grow very loose on the Bunches, are very fleshy and firm, and, when ripe, are of greenish white, and a delicate Flavour.

Either of these Kinds of Grapes will make very fine Ratafia; but which ever of them are chosen, they must be picked from the Stalks, and only the finest Berries made use of. The Stones must also be picked out; for if they are bruised with the Berries, the fine Flavour of the Juice will be greatly diminished.

When

When you have picked the Grapes from the Stalks, and taken out the Stones, press out the Juice, and filtrate it through a Flannel Bag. Then add the Quantity of Sugar and Spirit, and flavour it to your mind with a Spirit distilled from Spices, in the manner explained below.

The general Proportion of Sugar and Spirit, is, to twenty Pints of the Juice, five Pounds and a Half of Sugar, ten Pints of Spirit, and what Quantity you please of the spicy Spirit.

To make the spicy Spirit, take of Mace one Pound, Nutmegs four Ounces, Spirit three Gallons, and draw off the whole in Balneum Mariæ.

By the same Method you may make red Ratafia from the red Frontinac ; except that the Grapes, when bruised, must be suffered to ferment three or four Days, before the Juice is pressed out ; because the Colour, which resides principally in the Skins of the Grapes, will, by that means, be extracted.

The Berries of the red Muscat, or red Frontinac, are about the Size of those of the white ; but grow much thinner on the Bunches. This Grape, when thoroughly ripe,

ripe, has the richest and highest Flavour of any yet known ; but it must have a dry Soil and a South Aspect, otherwise it seldom ripens well in *England*. Besides the above Grape, there is another called by some red Muscat of *Alexandria*, and by others red *Jerusalem* Muscat. This is not quite so late in ripening as the white Muscat of *Alexandria* above described ; and for that reason more esteemed. The Berries of this Kind are not quite so large as those of the white, but of the same Form, and equal in Goodness.

5. Of Ratifia from Peaches.

The Ratifia made from the Peach is the finest and richest Flavour of any made from stoned Fruits. It is however necessary to gather the Peach when thoroughly ripe, but, at the same time not to suffer it to hang too long on the Tree : For as, on the one hand, it will not acquire its delicious Flavour and Smell till thoroughly ripe, so, on the other, it will lose both if suffered to hang on the Tree, after it has attained to a full Maturity. Another necessary Caution is, to gather it in fine warm Weather, and near the Middle of the Day ; because then both the Flavour and Smell are in the greatest Perfection.

It

It is also requisite to make Choice of the proper Sorts of Peaches; for there is a remarkable Difference in the Flavour of these Fruits. Gardeners reckon above thirty Sorts of Peaches, but not more than half that number are proper for making Ratafia. I shall therefore give a short Description of those that are most proper, that the young Distiller may not be disappointed in making Ratafia from Peaches.

1. The early Purple (called by the *French La Pourprée bâtive*.) This Tree hath smooth Leaves: The Flowers large, and open: The Fruit is large, round, and of a fine red Colour: The Flesh is white, but very red at the Stone; very full of Juice, which has a rich vinous Flavour. This Peach is ripe about the Middle of *August*.

2. The large, or *French Mignon*. The Leaves of this Tree are smooth, and the Flowers large and open. The Fruit is a little oblong, generally swelling out on one Side, and of a fine Colour. The Juice is very sweet, and of a high Flavour; the Flesh white, but very red at the Stone, which is small, and easily separates from the Flesh. This Peach is ripe in the middle of *August*.

3. The Chevreuse ; or, belle Chevreuse. This Tree hath smooth Leaves, and its Flowers are small and contracted. The Fruit is of a middling Size, a little oblong, and of an elegant Colour. The Flesh is white, but very red at the Stone, from which it separates ; full of a rich sugary Juice, and ripens towards the latter End of *August*.

4. The red Magdalen, called by the *French* about *Paris*, *Magdeleine de Courson*. The Leaves of this Tree are deeply sawed, and the Flowers large and open. The Fruit is large, round, and of a fine red Colour. The Flesh is white, but very red at the Stone, from which it separates. The Juice is very sugary, and of a rich Flavour. It is ripe the latter End of *August*.

5. *Smith's Newington*. This Tree hath sawed Leaves, and large open Flowers. The Fruit is of a middling Size, and of a fine red, next the Sun. The Flesh is very firm and white, but very red at the Stone, to which it closely adheres. It has a rich sugary Juice, and is ripe the latter End of *August*.

6. The Chancellor. The Leaves of this Tree are smooth, and the Flowers small
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and contracted. The Fruit is shaped somewhat like the Belle Chevreuse, but rounder. The Flesh is white and melting, and separates from the Stone, where it is of a fine red Colour. The Skin is very thin, and the Juice remarkably rich. It ripens about the End of *August*.

7. The Bellegarde; or, as the *French* call it, the *Gallande*. This Tree hath narrow Leaves, and small contracted Flowers. The Fruit is very large and round, and of a deep purple Colour, on the Side exposed to the Sun. The Flesh is white, melting, and separates from the Stone, where it is of a deep red Colour. The Juice is very rich. This Peach is ripe about the beginning of *September*.

8. The Bourdine. The Leaves of this Tree are smooth, and the Flowers small and contracted. The Fruit is large, round, and of a fine red Colour next the Sun. The Flesh is white, melting, and separates from the Stone; where it is of a fine red Colour. The Juice is vinous and rich. It is ripe the beginning of *September*, and greatly esteemed by the curious.

9. The Lisle; or, as the *French* call it, *la petite Violette Hâtive*. This Tree hath smooth Leaves, and small contracted Flowers.

ers. The Fruit is of a middle Size, and next the Sun of a fine violet Colour. The Flesh is of a pale yellow, melting, full of a rich vinous Juice; but adheres to the Stone, where it is very red. This Fruit is ripe the Beginning of *September*.

10. The old *Newington*. The Leaves of this Tree are sawed, and the Flowers large and open. The Fruit is fair, large, and of a beautiful red Colour next the Sun. The Flesh is white, melting, and closely adheres to the Stone, where it is of a deep red Colour. The Juice is very rich and vinous. It is ripe about the Middle of *September*.

11. The *Rambouillet*, commonly called the *Rambullion*. This Tree has smooth Leaves, and large open Flowers. The Fruit is of a middling Size, rather round than long, deeply divided by a Furrow in the Middle; of a fine red Colour next the Sun, but of a light Yellow next the Wall. The Flesh is melting, of a bright yellow Colour, except near the Stone, from which it separates, where it is of a deep red. The Juice is rich and of a vinous Flavour. This Fruit ripens about the Middle of *September*.

12. The Pourprée; or, as the *French* generally call it *Pourprée tardive*, the late Purple. The Leaves of this Tree are very large, and sawed, the Shoots strong, and the Flowers small and contracted. The Flesh, except near the Stone, from which it separates, and where it is red, is white, melting, and of a rich sugary Juice. It is not ripe till near the End of *September*.

13. The Nevette. The Leaves of this Tree are sawed, and the Flowers small and contracted. The Fruit is large, somewhat longer than round, of a bright red Colour next the Sun, and of a pale yellow on the other. The Flesh is melting, full of a rich Juice, and very red at the Stone, from which it separates. It ripens about the Middle of *September*, and is esteemed one of the best Peaches.

14. The Royal. This Tree hath smooth Leaves, and small contracted Flowers. The Fruit is large, round, and of a deep red on the Side exposed to the Sun, but of a pale yellow on the other. The Flesh is white, melting, and full of a rich Juice, of a white Colour, except near the Stone from which it separates, where it is of a deep red. This Fruit is ripe about the Middle of *September*.

15 The monstrous Pavy of Pomponne. The Leaves of this Tree are smooth ; the Flowers large and open. The Fruit is very large and round, many times fourteen Inches in Circumference. The Flesh is white, melting, and closely adheres to the Stone, where it is of a deep red Colour. The Side next the Sun is a beautiful red, and the other of a pale flesh Colour. It ripens about the End of *October*, and when the Autumn is warm, is an excellent Peach.

The above Description of the different Kinds of Peaches proper for making Ratafia, will be of use to the young Artist, as the fine Flavour of this Liquor in a great Measure depends on a proper Choice of the Fruits used in the Composition ; and if the Instructions relating to the Perfections and Ripeness of these Fruits are observed, an excellent Cordial may be easily made in the following manner.

Take your Peaches, bruise them, and instantly strain out their Juice thro' a Piece of strong Linen. In this Juice, without any Mixture of Water, dissolve your Sugar. And when the Sugar is melted, add the Quantity of Spirit. No Spices must be used in this Ratafia, the fine Flavour of the Peach being far preferable to all Spices in the
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the World. The Quantity of either the Sugar or Spirit may be augmented or lessened according to your own Judgment, or in Proportion to the Price of your Ratafia.

As soon as the Spirit is added to the dulcified Juice of the Peaches, the whole must be filtrated thro' a Flannel Bag, put into Bottles close stopped; for the fine Flavour of the Peach will soon be lost unless the Bottles are very well corked. Some also cover the Cork with Sealing-wax, which is not a bad Caution.

If you would have your Ratafia of a bright red Colour, you must let your bruised Peaches ferment a Day or two; by which means the Colour of the Skin, and that of the Flesh near the Stone, will be extracted, and give your Ratafia the Colour desired.

4. Of Orange-flower Ratafia.

The Orange-flower has been already described, Page 127. I shall therefore only add, that the Orange-flowers used in making Ratafia should be large, in their full Perfection, gathered before the Rising of the Sun, and carefully picked from their Stalks, &c. Some blanch the Orange-flowers, by putting them into a small Quantity of

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Water,

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Water, and boiling them a few Minutes over the Fire. But by this Method the most volatile Parts of the Flower are evaporated, by which the Ratafia will lose much of its delicate Flavour.

The best way therefore is to use the Orange-flowers without any previous Boiling.

Recipe for making ten Gallons of Orange-flower Ratafia.

Take of Orange-flowers fresh gathered, and clean picked from their Stalks, &c. five Pounds, and infuse them six Days in five Gallons of clean Proof Spirit. Dissolve fourteen Pounds of Sugar in five Gallons of Water; and after straining the Spirit from the Flowers, mix it with the Syrup, and filtrate the whole thro' a Flannel Bag.

Some instead of common Water use the Orange-flower Water; but it will be necessary in pursuing that Method to take care that the Water be fresh made, and very fragrant; for otherwise instead of improving, you will greatly injure the fine Flavour of your Ratafia.

The foreign Distillers keep two Sorts of Orange-flower Ratafia, one they call *single* and the other *double*. The former is made according

according to the above Recipe; but in making the latter they use double the Quantity of Orange-flowers, and considerably augment the Proportion of Sugar. It will be needless to give a Recipe for making that Sort of Ratafia, which they call *double*, as the Process is exactly the same.

5. *Ratafia of Portugal Orange.*

Ratafia may be made from any Sort of Orange; but that of the *Portugal* Orange is reckoned the best.

The Oranges must be chosen fair, large, and ripe; and the outer or yellow Peel be carefully taken off. The Juice of the Oranges must be then pressed out, dulcified with Sugar, and mixed with the Spirit: after which the outer Rinds are to be added, and after a proper Infusion, the whole filtrated through a Flannel Bag.

Recipe for making three Gallons of Portugal Orange Ratafia.

Take of the Juice of *Portugal* Oranges two Gallons; clean rectified Spirit one Gallon; four Pounds of Sugar; and the outer Peel of ten Oranges. Let the whole infuse a Fortnight, and then filter the Liquor through a Flannel Bag.

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Some instead of infusing the Peel as directed in the above Recipe, put the Peel into the Spirit, and distil it in Balneum Mariæ; after which they add the Spirit to the dulcified Orange-juice, and filtrate as before.

The foregoing Recipes for making Ratafia from different Fruits, &c. will be sufficient to instruct the young Distiller in the Method necessary to be pursued for making Cordials of this Kind; for it would be tedious to give Formula's for making all the Kinds of Ratafia kept by different Distillers. The Method in all is nearly the same; and the Proportion of Sugar and Spirit may be easily discovered by a few Experiments. I shall therefore conclude this Chapter with giving a Recipe for making what is called by our *English* Distillers Ratafia, tho' a very bad Composition.

Recipe for making ten Gallons of common Ratafia.

Take of Nutmegs eight Ounces; bitter Almonds ten Pounds; *Lisbon* Sugar eight Pounds; Ambergrise ten Grains: Infuse these Ingredients three Days in ten Gallons of clean Proof Spirit, and filtre thro' a Flannel Bag for use.

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The Nutmegs and bitter Almonds must be bruised; and the Ambergrise rubbed with the *Lisbon* Sugar in a Marble Mortar, before they are infused in the Spirit.

C H A P. LIII.

Of Gold Cordial.

THIS Cordial has its Name from Leaf Gold being formerly used in its Composition; but as later Experiments have abundantly demonstrated that Gold can add nothing to its Virtues, it is now generally omitted.

Recipe for making ten Gallons of Gold Cordial.

Take of the Roots of Angelica, four Pounds; Raisins stoned, two Pounds; Coriander-seeds, Half a Pound; Caraway-seeds and Cinnamon, of each Half a Pound; Cloves two Ounces; Figs and Liquorice-root, of each one Pound; Proof Spirit eleven Gallons; Water two Gallons: The Angelica, Liquorice, and Figs must be sliced, before they are added. Digest two Days, and draw off by gentle Heat, till the Faints begin to rise, hanging in a Piece of Linen fastened to the Mouth of the Worm an

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Ounce of *English Saffron*. Then dissolve eight Pounds of Sugar in three Quarts of Rose Water, and add it to the distilled Liquor. Some Distillers instead of Saffron colour their Goods with burnt Sugar, but by this means the Cordial is greatly impaired in its Virtues.

Or,

Take of the Juice of Alchermes five Ounces; Cloves two Ounces and a Half; Musk and Ambergrise, of each Half a Dram; Loaf Sugar ten Pounds; Proof Spirit eleven Gallons; digest the whole a Fortnight in a close Vessel, and filter thro' a Flannel Bag for use. Some add thirty Leaves of Gold; but the Medicine is not at all the better for it.

Either of the above Recipes will produce an excellent Cordial; good in Tremblings, Faintings, and Lowness of Spirits, &c. Also in Nauseas and Gripping Pains of the Stomach and Bowels.

CHAP.

C H A P. LIV.

Of Cardamum, or All-fours.

THIS Water has its Name from the four Ingredients in its Composition; and in some Countries is greatly used by the poorer Sort of People.

Recipe for making ten Gallons of Cardamum.

Take of Pimento, Caraway and Coriander-seeds, and Lemon-peel, of each three Pounds; of Malt Spirits eleven Gallons; Water three Gallons. Draw off with a gentle Fire, dulcify with ordinary Sugar, and make up the Goods, to the Strength you desire with clean Water.

This is rarely called for unless by the poor Sort of People, who are induced to use it from its Cheapness; tho' it is a better Cordial than many drawn from dearer Ingredients. It is an excellent Carminative, and is often sold for Aqua Mirabilis.

C H A P. LV.

Of Geneva.

THERE was formerly kept in the Apothecaries Shops a distilled spirituous Water of Juniper; but the Vulgar being fond of it as a Dram, the Distillers supplanted the Apothecaries, and sold it under the Name of Geneva. The common Sort however is not made from Juniper-berries as it ought to be, but from Oil of Turpentine; the Method of which we shall give in the Sequel of this Chapter.

Juniper-berries are, a roundish Fruit, of the Size of a Pea. They wither and wrinkle in the drying, and we meet with them variously corrugated, and usually covered with a bluish resinous Dust when fresh. They should be chosen fresh, plump, full of Pulp, and of a strong Taste and Smell. They are usually imported from *Germany*, tho' we have plenty of the Trees in *England*. It is but small with us, rarely rising to more than three or four Feet in Height, and scarce ever exceeding five or six. Some of the Juniper Shrubs are Males, some Females of the same Species; the Male Shrubs produce in *April* or *May* a small Kind of Juli with Apices on them very large.

large, and full of Farina; the Females produce none of these Juli but only the Berries, which do not ripen till the second Year, and then do not immediately fall off, so that it is no uncommon thing to see three Sets of Berries, or the Berries of three different Years at once on the same Tree.

If you make use of *English* Berries, let them be fully ripe before they are gathered; and in order to preserve them, spread them very thin on a boarded Floor, leaving the Windows and Doors open, and turn them once a Day till they are dry; after which pack them up in Barrels, so that no Air may come to them, and they will keep good all the Year. Some, when they are dry, throw them altogether in a Heap in a Corner of the Room, where they continue till wanted for use; but the Berries will not keep so well by this Method as by being packed in Casks; they are subject to contract a Mouldiness, which will give a Taste to the Goods, greatly to their Disadvantage.

Some Distillers as soon as their Berries are gathered, put them into Casks, and cover them with Spirits of Wine; by this Method the Berries are indeed well preserved, without any Danger of contracting an ill Smell, which they are very apt to do

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do by the other Methods unless the greatest Care be taken ; but then it must be remembered, that the Spirit will extract great Part of their essential Oil, in which their Virtues consist, and consequently the Berries themselves will be rendered of little Value. If, therefore, you preserve your Berries in this manner, you should put into each Cask or Jarr, only the Quantity you use for one Charge of your Still ; and when you have occasion to use them, put both the Spirits and Berries into your Alembic.

Thus your Berries will be finely preserved, without any Loss either of their essential Oil, or the Spirits made use of to preserve.

Recipe for making ten Gallons of Geneva.

Take of Juniper-berries three Pounds, Proof Spirit ten Gallons; Water four Gallons. Draw off by a gentle Fire till the Faints begin to rise, and make up your Goods to the Strength required with clean Water.

The Distillers generally call those Goods which are made up Proof by the Name of Royal Geneva; for the common Sort is much below Proof, ten Gallons of Spirit being

being sufficient for fifteen Gallons of Geneva. Nay, what is generally sold at the common Alehouses is made in the following manner.

Take of the ordinary Malt Spirits ten Gallons; Oil of Turpentine two Ounces, Bay Salt three Handfuls. Draw off by a gentle Fire till the Faints begin to rise, and make up your Goods to the Strength required with clean Water.

In this manner is the common Geneva made, and it is suprizing that People should accustom themselves to drink it for pleasure.

There is a Sort of this Liquor called *Holland's Geneva*, from it being imported from *Holland*, which is greatly esteemed.

The Ingredients used by the *Dutch* are; however, the same as those given in the first Recipe of this Chapter, only instead of Malt Spirit they use *French Brandy*. In the first Part of this Treatise we have sufficiently shewn the Nature of *French Brandy*, and in what its Excellence consists; and, also, that by the Help of a clean Spirit, Cordial Waters may be made with the same Goodness as those drawn with *French*

a mistake, *Hollands Gin is Rectified from* Brandy.
Malt spirits - - - - -

Brandy. If therefore the Distiller be careful in distilling and rectifying his Malt Spirit, he may make Geneva equal to that of the *Dutch*, provided it be kept to a proper Age; for all spirituous Liquors contract a Softness and Mellowness by Age, impossible to be imitated by Art.

C H A P. LVI.

Of Cherry Brandy.

THIS Liquor is greatly called for in the Country; and is made different ways. Some press out the Juice of the Cherries, and having dulcified it with Sugar, add as much Spirit to it as the Goods will bear, or the Price it is intended to be sold for. But the common Method is to put the Cherries clean picked into a Cask, with a proper Quantity of Proof Spirit, and after standing eighteen or twenty Days, the Goods are drawn off into another Cask for Sale, and about two thirds of the first Quantity of Spirits poured into the Cask upon the Cherries. This is suffered to stand about a Month to extract the whole Virtue from the Cherries, after which it is drawn off as before; and the Cherries pressed to take out the Spirit they had absorbed. The Proportion of Cherries and Spirit is not very nicely observed; the general Rule is

is to let the Cask be about half filled with Cherries, and then filled up with Proof Spirits. Some add to every twenty Gallons of Spirit half an Ounce of Cinnamon, an Ounce of Cloves, and about three Pounds of Sugar, by which the Flavour of the Goods is considerably increased. But in order to save Expences, not only the Spices and Sugar are generally omitted, but also great Part of the Cherries, and the Deficiency supplied by the Juice of Elder-berries. Your own Reason therefore, and the Price you can sell your Goods for, must direct you in the Choice of your Ingredients.

By the same Method you may make Raspberry Brandy; and if the Colour of the Goods be not deep enough, it may be improved by an Addition of Cherry Brandy.

C H A P. LVII.

Of Honey Water.

THIS Water has its Name from the Honey in its Composition; tho' that Ingredient is but of very little Service to the Water, if made according to the usual Method.

Recipe

Recipe for making a Gallon of Honey Water.

Take of the best Honey and Coriander-seeds, of each one Pound; Cloves, one Ounce and a Half; Nutmegs and Gum Benjamin, of each an Ounce; Vanilloes Number four. The yellow Rind of three large Lemons: Bruise the Cloves, Nutmegs, Coriander-seed, and Benjamin; cut the Vanilloes in pieces, and put all into a Glass Alembic, with one Gallon of clean rectified Spirit, and after digesting forty eight Hours, draw off the Spirit in Balneum Mariæ. To a Gallon of the above Spirit, add of damask Rose Water and Orange-flower Water, of each a Pound and a Half; Musk and Ambergrise of each five Grains. Grind the Musk and Ambergrise with some of the Water in a Glass Mortar, and afterwards put all together into a digesting Vessel, shaking them well together, and let them circulate three Days and three Nights in a gentle Heat: Then let all cool; filter and keep the Water in Bottles well stopped for use.

This Water was first made by that faithful Chemist Mr. *George Wilson*, for King *James II.* It is an Antiparaliuc, smooths the Skin, and gives one of the most agreeable Scents imaginable. Forty or sixty Drops

Drops put into a Pint of clean Water, are sufficient for washing the Hands and Face; and the same Proportion to Punch, or any Cordial Water, gives a very agreeable Flavour.

C H A P. LVIII.

Of Unequalled Water, generally sold by the French Name l'Eau sans Pareille.

There are two Sorts of this Water, one drawn considerably below Proof, and rendered fine by Filtration, and the other without the Faints, the Receiver being removed as soon as they begin to rise. The latter is much the best, tho' dearer than the former.

Recipe for making a Gallon of the common Eau sans Pareille.

Take the outer Peels of twelve Citrons, three Quarts of fine Proof Spirit, and a Quart of Water. Put all into a Glass Alembic, and distil to a Dryness in Balneum Mariæ; filtre the Water, and put it into Bottles well stopped.

This is the common Sort, and what is generally sold here under the Name *Eau sans Pareille*.

Recipe

*Recipe for making a Gallon of the best Sort
of Eau sans Parcille.*

Take of the Effence of Cedrat, Bergamot, Orange, and Lemon, of each two Drams; rectified Spirit a Gallon; Water two Quarts. Put all into a Glafs Alembic, and distil in Balneum Mariæ till the Faints begin to rise, when the Receiver must be immediately removed.

Some to save the Trouble and Expence of Distillation, mix the Essences with the Spirit of Wine, in the manner before mentioned in the Chapter for making Hungary Water; but this is greatly inferior to that made by Distillation.

C H A P. LIX.

Of the Water of Bouquet.

THIS Water has its Name from its Inventor, and is greatly esteemed abroad for its Smell. It is indeed drawn from the most odoriferous Flowers, and therefore it is no wonder that it is held in great Esteem.

Recipe

Recipe for making a Gallon of Bouquet's Water.

Take of the Flowers of white Lillies, and *Spanish* Jessamin, of each Half a Pound; Orange-flowers and those of the Jonquil and Pink of each four Ounces; damask Roses one Pound. Let those be fresh gathered, and immediately put into a Glass Alembic with a Gallon of clean Proof Spirit, and two Quarts of Water. Place the Alembic in *Balneum Mariæ*, draw off till the Faints begin to rise. You may use Spirit of Wine, instead of Proof Spirit; but it will be absolutely necessary that it be entirely inodorous; for otherwise your Water will fall short of the desired Perfection.

C H A P. LX.

Of Cyprus Water.

THIS Water is only a dilute Tincture of Ambergrise; but as it is used by those who are fond of that Perfume, and known by the Name of Cyprus Water, or *Eau de Chypre*, I would not omit giving the Recipe here, intending to give a full Account of Ambergrise in a succeeding Chapter.

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Recipe

Recipe for making a Gallon of Cyprus Water.

Take of the Effence of Ambergrise Half an Ounce; put it into a Glafs Alembic with a Gallon of Spirit of Wine and two Quarts of Water. Place the Alembic in Balneum Mariæ, and draw off till the Faints begin to rise.

C H A P. LXI.

Of Vestal Water, or Eau de Vestale.

THIS is a very agreeable Water, and has been long in use in several Parts of *Europe*.

Recipe for a Gallon of Vestal Water.

Take of the Seeds of Daucus Creticus or Candy Carrots, two Ounces; Spirit of Wine a Gallon; Water two Quarts. Distil in Balneum Mariæ till the Faints begin to rise. Then add to the Spirit drawn over an Ounce of the Effence of Lemons, and four Drops of the Effence of Ambergrise; redistil in Balneum Mariæ, and keep the Water in Bottles well stoped for use.

C H A P.

C H A P. LXII.

Of Beauty Water, or Eau de Beauté.

THIS Water has its Name from its use in washing the Face, and giving an agreeable Smell. It is drawn from Thyme and Marjoram, which gives it a very elegant Odour.

Recipe for making a Gallon of Beauty Water.

Take of the flowery Tops of Thyme and Marjoram, of each one Pound; Proof Spirits five Quarts; Water one Quart. Draw off in Balneum Mariæ, till the Faints begin to rise, and keep it close stopped for use.

C H A P. LXIII.

Of Royal Water.

THIS Water has its Name from being considered as the most excellent of all scented Waters. It is compounded of the Cedrat, Nutmegs and Mace, from whence the most elegant Smell is produced; and no Water is at present thought equal to this. There are two Sorts of Royal Water, one produced by a single Distillation, and the other by a double Distillation,

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and thence called rectified, or double distilled Royal Water.

Recipe for a Gallon of Royal Water.

Take of Mace one Ounce ; Nutmegs Half an Ounce ; Essence of Cedrat, or Bergamot two Drams : Put these into a Glass Alembic (after bruising the Spices) with five Quarts of fine Proof Spirit, and draw off one Gallon in Balneum Mariæ.

Recipe for making a Gallon of double distilled Royal Water.

Take of Mace one Ounce ; Nutmegs Half an Ounce ; bruise them, and put them into an Alembic with six Quarts of fine Proof Spirit, and draw off five Quarts with a gentle Fire. Then take the Spirit drawn off and put it into a Glass Alembic, with two Drams of the Essence of Cedrat, or Bergamot, and draw off a Gallon in Balneum Mariæ.

Either of these Recipes will produce an elegant Water ; but the latter greatly exceeds the former.

CHAP.

C H A P. LXIV.

*Of the Tincture, or Essence of Ambergrise,
Musk and Civet.*

1. **A**UTHORS have been long divided with regard to the Origin of Ambergrise; some taking it for a vegetable Juice, which either dropped into the Water from the Trunks or Branches of some Trees growing on the Sea-coast, or exudated from their Roots which ran out of the Earth into the Sea; some for an animal Production, and formed either by a secret Process from Honey-combs, or the Dung of Birds; and others have very circumstantially recorded that it is produced in the Whale. These Opinions are however now looked upon as false; Ambergrise being universally allowed to be a Mineral Production, of the Number of Bitumens. It is a light and frothy Substance, which generally bubbles up out of the Earth in a fluid Form, principally under Water, where it is by Degrees hardened into the Masses we see it in.

Ambergrise in its natural, or common Form is a lax and coarse Substance of an irregular Structure, friable, and so light as to swim upon Water. It is of a pale gray Colour, with a faint Tinge of brown in it; but Pieces perfectly and uniformly of this

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Colour are rare, what we usually meet with is composed of whitish, yellowish, and blackish Granules; and in Proportion as there is more or less of this whitish Matter in these Masses, it is more or less scented and valuable. It is found in Pieces of perfectly irregular Figures, and from the Bigness of a Pea to those of ten, twenty, or more Pounds; nay there have been Masses found of more than two Hundred Weight.

It should be chosen in clean and not over friable Pieces, of a pale grey Colour, and as uniform as possible in its Structure, with small black Specks within.

There are two Sorts of Essences made from this Perfume; one without Addition of any other odoriferous Substance, and the other from Ambergrise compounded with Musk and Civit.

Recipe for making the Essence of Ambergrise.

Take of Ambergrise and white Sugar-candy, of each three Drams; grind them well together in a Glass Mortar, adding to them by slow Degrees, five Ounces of rectified Spirit of Wine, digest the whole in a Martrass (represented Fig. 8.) well stopped for four Days, and then separate the clear Tincture or Essence, which keep in a Bottle well stoped for use.

Re-

Recipe for making the compound Essence of Ambergrise.

Take of Ambergrise and white Sugar-candy of each two Drams; Musk twelve Grains; Civet two Grains; grind all these well together in a Glass Mortar, adding by Degrees four Ounces of rectified Spirit of Wine; digest and separate the clear Essence for use, as in the preceding Recipe.

2. Musk is a dry, light, and friable Substance; of a dark blackish Colour, with some Tinge of a purplish or blood Colour in it. It is soft, and somewhat smooth and unctuous to the Touch, and of a highly perfumed Smell. It is brought to us sewed up in a Kind of Bladders or Cases of Skin, covered with a brownish Hair, which are the real Bags in which the Musk is lodged while on the Animal. Musk should be chosen of a very strong Scent, and in dry sound Bladders; and must be kept close shut down in a Leaden Box, by which means it will retain its Smell, and not grow too dry.

Recipe for making the Essence of Musk.

Take of Musk and white Sugar-candy of each one Dram; rub them well together
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in

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in a Marble Mortar, adding by Degrees during the rubbing five Ounces of rectified Spirit of Wine : Put the whole into a Matraass, digest three Days in a gentle Heat, and pour off the clear Essence, which keep in a Bottle well stoppt for use. Some add a few Grains of Civet to their Essence of Musk, which considerably augments the Fineness of the Perfume.

3. Civet is produced, like Musk, in Bags growing to the lower Part of the Belly of an Animal. It is of different Colours from a pure lively whitish, to a black ; but the nearer it approaches to the white the better it is ; of an extremely strong Smell, and a bitterish pungent Taste.

The Essence of Civet is rarely used alone, but of great service in making Additions to other odoriferous Waters, and therefore I shall here give the Method of making it.

Recipe for making the Essence of Civet.

Take of Civet and double refined Sugar, of each two Drams ; rub them well together in a Glass Mortar, adding by Degrees five Ounces of rectified Spirit of Wine ; Put the whole into a Matraass, digest three Days in a gentle Heat, and pour off the clear Essence for use. Tho' the Essences
in

in this Chapter are, properly speaking, Chemical Preparations, and therefore foreign to the Business of the Distiller ; yet as they are often added to perfumed Waters, and easily made, I thought the above Recipes would not be unacceptable to the Reader.

C H A P. LXV.

Of Faints, and the Uses they may be applied to.

IN many of the preceding Recipes I have ordered the Receiver to be removed as soon as the Faints begin to rise ; because otherwise the Goods would contract a disagreeable Taste and Smell. It is not however to be understood that these Faints are to be thrown away, nor the Working of the Still immediately stopped ; for they are far from being of no Value, notwithstanding they would be of great Disadvantage if suffered to run among the more spirituous Parts of the Goods before drawn off. As soon therefore as you find the clear Colour of the Goods begins to change of a bluish or whitish Colour, remove the Receiver, place another under the Nose of the Worm, and continue the Distillation as long as the Liquor running from the Worm is spirituous, which may be known by pouring a little of
it

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it on the Still Head, and applying a lighted Candle to it; for if it is spirituous it will burn, but otherwise not. When the Faints will no longer burn on the Still Head, put out the Fire, and pour the Faints in a Cask for that Purpose; and when, from repeated Distillations, you have procured a sufficient Quantity of these Faints, let the Still be charged with them almost to the Top. Then throw into the Still three or four Pounds of Salt, and draw off as you would any other Charge as long as the Spirit extracted is of a sufficient Strength; after which the Receiver is to be removed, and the Faints saved by themselves as before.

The Spirits thus extracted from the Faints will serve in several Compositions as well as fresh; but they are generally used in Aniseed Water, because the predominant Taste of the Aniseeds will entirely cover that they had before acquired from other Ingredients.

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